



Business Plan

McGill iGEM 2022-2023
Finance & Entrepreneurship

McGill iGEM 2022-2023 Finance & Entrepreneurship Team

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1. Executive Summary

1.1 McGill iGEM and Our Mission

McGill iGEM strives to develop novel synthetic biology approaches to solve globally relevant issues. Our team prioritizes educational initiatives to educate youth, young adults and postsecondary students on interdisciplinary STEM topics, connecting the many facets of Canada's bioeconomy as well as internationally. We aim to provide students the opportunity to experience the full life cycle of a research project from ideation to market launch, cultivating transferable skills, such as leadership, entrepreneurship, critical thinking, and collaboration. Our group prides itself on fostering the brightest minds of tomorrow and democratizing access to science to pave the path for a better future.

McGill iGEM is determined to make research opportunities more accessible to students of all backgrounds in order to increase youth participation in STEM. Our work begins in elementary and secondary schools as well as summer camps and indigenous education centers around Montreal, where we host information sessions, presentations, and hands-on workshops to familiarize students with current areas of exploration in synthetic biology. The goal is to educate students on how to get involved with research and innovation. We have connected with several CÉGEPs (colleges) throughout Quebec, SHAD Canada, the First Nations Regional Adult Education Centre in Kahnawà:ke, among others for these collaborative education and workshop events. We hope to break down existing barriers to entry to provide equitable opportunities for youth to contribute to science in general.

Every fall, we recruit dozens of new undergraduate members for our team, providing hands-on, student-guided training on synthetic biology techniques. By the end of our yearly project, all of our new members have developed the essential skills required to perform independent research, an opportunity seldom offered elsewhere.

McGill iGEM makes a positive impact on the wider undergraduate community year round by hosting student-led educational sessions and research workshops through our SynBio Collective. Any and all students may participate in these sessions (including non-McGill students). This collective affected around 200 students in the Greater Montreal Area, though we aim to expand our impact as our infrastructure grows. McGill iGEM additionally hosts its own podcast, STEMCast, where we disseminate novel advances in research, making science more accessible and understandable by bringing in researchers, industry leaders, and bioethicists to discuss a range of topics in STEM. Our team also creates blog posts and social media posts to increase youth interest and accessibility to synthetic biology.

Finally, our team hosts an annual conference, In Vitro, where we bring together all facets of the bioeconomy to advance the biotech community in Canada and internationally. This year, our conference had over 200 participants from all over North America, enabling startups, students, iGEM teams, researchers, VCs, industry professionals, big pharma, and many other stakeholders to connect, exchange ideas, and contribute to collaborative innovation in biotechnology.

iGEM is more than an undergraduate research program, but rather a collaborative network that connects the best and brightest scientific minds from around the globe. At McGill iGEM, we aim to lead by example in demonstrating the meaningful contribution of undergraduate research to academia, entrepreneurship, and society. This report outlines our plan for commercializing our project as a modular pancreatic cancer therapeutic.

1.2 Sustainability in Science

As an internationally-focused student-led research organization, McGill iGEM applies high standards in evaluation of all our scientific endeavors and therapeutics in development. We recognize that our team and entrepreneurial pursuits have a unique role in making a real difference in society, not only through development of novel therapeutics, but also through the route we take to generate them. McGill iGEM uses the environmental, social, and corporate governance (ESG) framework to contribute to sustainable development of our team and *PROTEUS* in order to create a positive impact today as well as for future generations. All members of our team are stewards of the ESG mission as well as the United Nations Sustainable Development Goals (SDGs), bringing these commitments to fruition for a better tomorrow.

With input from all stakeholders and mentors, our team ensures our research, innovations, and collaborations prioritize communities and the conservation of the world around us. McGill iGEM evaluates our strategies and plans of action by cross-referencing external frameworks such as the United Nations SDGs, the S&P Global Corporate Sustainability Assessment, and the Carbon Disclosure Project. In the coming years, our organization may begin to commercialize, developing manufacturing labs and expanding our infrastructure. These frameworks will be essential to ensure we maintain corporate responsibility and sustainability moving forward.

1.3 Overview of this Report

This report is broken down into 10 sections. The Table of Contents provides page numbers for quick reference to each subsection. The report aims to describe McGill iGEM 2022-2023's commercialization strategy for our project, *PROTEUS*. Section I provides an executive summary of McGill iGEM's initiatives and the *PROTEUS* project, after which the following sections provide in-depth detail on all aspects of the project and its strategy for market entry. This includes, but is not limited to, a description of the team, our project, market and patent research for the cancer therapeutics space, several possible business models, and our plan for development and implementation. While *PROTEUS* is in a pre-seed stage of development, backbone financial projections have been included to contextualize our team's objectives and timeline for the *PROTEUS* project.

1.4 *PROTEUS* as a Marketable Product

Delineating the commercial potential of a cancer therapeutic requires both an evaluation of the possible socioeconomic and health benefits worldwide of applying this therapeutic for pancreatic cancer treatment as well as the intellectual capital associated with the modularity of the proof of concept.

As will be discussed in the Market Research section, pancreatic cancer is one of the most common cancers worldwide and the third leading cause of cancer-related death in the United States (National Institutes of Health, 2023). Furthermore, it is the deadliest of all major cancers with a 12%, 5-year relative survival rate in the US (American Cancer Society, 2023); according to the Canadian Cancer Society, this rate drops to 2% for metastatic cases (Canadian Cancer Society, n.d.). Two major factors in this incredibly low survival rate are the current lack of consistent, curative methods of treatment for victims of pancreatic cancer as well as the difficulty in early-stage detection of the cancer (Oberstein & Olive, 2013). While the 5-year survival rate has increased in recent years, it remains abysmally low compared to progress made with other cancers (Roser & Ritchie, 2019). Furthermore, current pancreatic cancer care and treatment is exorbitantly expensive; patients with pancreatic cancer accrue more than 15 times the annual healthcare costs of control groups, making professional treatment financially infeasible for many in nations with primarily private healthcare coverage (Samaan et al., 2023). Furthermore, in the next 40 years, the clinical, societal and economic burden of pancreatic cancer is predicted to double (Lippi & Mattiuzzi, 2020), representing devastating downstream effects to society at large. These facts demonstrate a substantial gap and opportunity in the current pancreatic cancer therapeutic market.

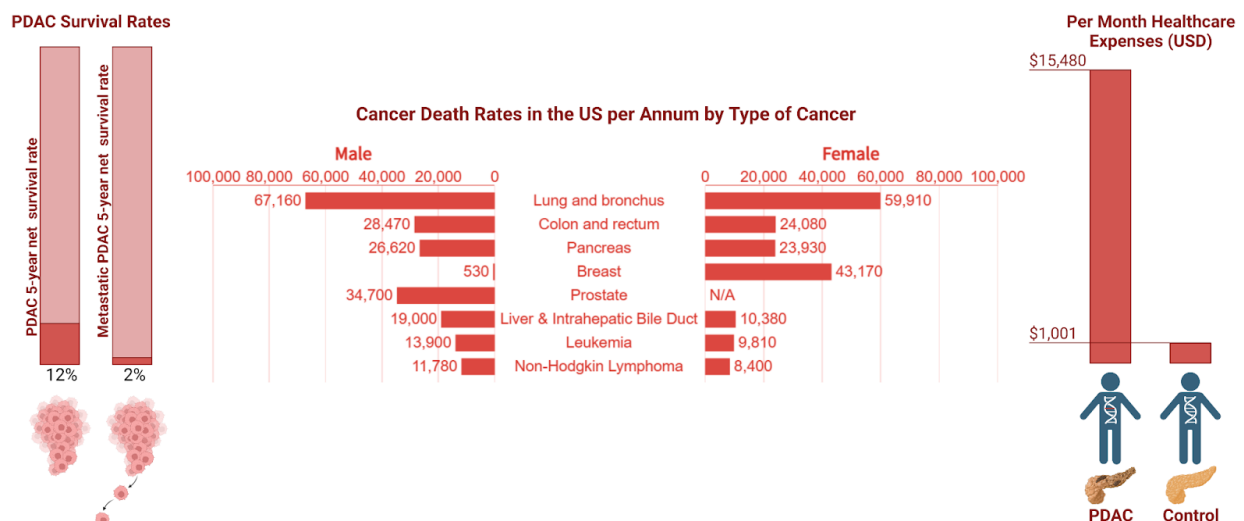


Figure 1. Critical pancreatic cancer facts and figures. The left hand side shows PDAC survival rates at the time of diagnosis for general PDAC cases and specifically metastatic cases. The centre chart shows the most deadly common cancers based on annual death rates in the US. Lung and bronchus cancer has the highest death toll, followed by colon and rectal cancers, then pancreatic cancers. Pancreatic cancer affects women and men at similar rates. The rightmost figure shows monthly healthcare expenses for PDAC patients versus a control group. PDAC

patients tend to spend approximately \$15500 USD per month on healthcare, compared to the approximate \$1000 USD spent by the control group. This is approximately 15 times more than the control group. (Canadian Cancer Society, n.d.; American Cancer Society, 2023, Cipora et al., 2023).

As a treatment targeting cancer at its genetic roots, our therapeutic has the potential to effectively prevent and cure the over 531 000 deaths per annum caused by pancreatic cancer worldwide (Lin et al., 2022). This represents an immeasurable benefit to society and the economy, with hundreds of thousands of families recovering their loved ones and hundreds of thousands of workers now able to contribute throughout their working lives.

Beyond a treatment as a pancreatic cancer therapeutic, *PROTEUS* has the potential to be re-engineered for fundamentally any other cancer (or any disease arising from genetic deficiencies in fact) due to its modularity as a general CRISPR-guided genetic tool. Ownership of this mechanism through provisional patenting represents massive intellectual capital, assuming the product is functional *in vivo*. The market potential for other therapeutics harnessing the same Craspase mechanism designed by the McGill iGEM 2022-2023 team are substantial; noteworthy future applications could include treatment of viral diseases such as AIDS, retardation of senescence, and CRISPR-guided small-molecule detection.

1.5 Market Opportunity and Target Audience

In 2023, an estimated 64,050 pancreatic cancer cases will be diagnosed, and an estimated 50,550 individuals will die in the US alone (American Cancer Society, 2023), though pancreatic cancer is an international dilemma affecting all sexes, genders, and races (Moshayedi et al., 2022). Current treatment options include; surgery, radiation therapy, immunotherapy, and chemotherapy. These methods often extend survival or relieve symptoms, but seldom produce a cure (American Cancer Society, 2023). Furthermore, these therapies are often only applicable for small percentages of patients; fewer than 20% of patients are candidates for surgery, and even fewer cases are eligible for immunotherapy (American Society of Clinical Oncology, 2022). The most common treatment for those who *are* eligible for surgery is adjuvant treatment with chemotherapy and sometimes radiation, though these methods still rarely cure the disease (American Society of Clinical Oncology, 2022). In short, pancreatic cancer treatment is a fledgling field with extensive opportunity to improve the lives of the hundreds of thousands of victims of the disease.

The primary target market for a pancreatic cancer therapeutic is patients diagnosed with pancreatic cancer. Additionally, given the preventative nature of our treatment, which will be further discussed in the Product Description section, we aim to have secondary targets that are especially susceptible to disease development. Common targets include direct family of pancreatic cancer patients, those with family history of KRAS-driven cancers, the elderly population (given that most cancers, and especially pancreatic cancer, disproportionately affect the elderly in terms of quantity of cases and severity on health (Lin et al., 2022), and those with strong risk factors of pancreatic cancer, such as chronic diabetics, smokers, and

those with poor access to oral hygiene (Tsai & Chang, 2019). Risk factors will be discussed in additional detail in the Product Description section. Finally, *PROTEUS* will be available, upon a medical professional's approval, to the general public; in order to prevent any spontaneous development of pancreatic cancer, individuals can be pre-treated with the system through injection without any major risk factors. This is highly recommended due to the fact that Ras mutations are highly expressed in pancreas cells during early or even pre-development stages of pancreatic cancer tumours (Luo, 2021). The body often accumulates Ras mutations long before extensive tumour development, and KRAS mutations are seldom capable of causing tumorigenesis alone; in fact, longstanding KRAS mutations were observed in healthy populations without negative effects on quality of life (Badheeb et al., 2022). This is problematic given KRAS' function as a driver mutation; once Ras mutations exist, subsequent mutations result in carcinogenesis. In essence, accumulation of Ras mutations acts as a timebomb for pancreatic cancer. *PROTEUS* effectively clears mutated cells before they can induce carcinogenesis by effectively targeting high risk regions in the body, such as the pancreas.

In terms of the supply chain, *PROTEUS* would most likely be provided to wholesalers then shipped to pharmacies, or directly shipped to pharmacies from manufacturing facilities. Depending on the jurisdiction, patients would then request *PROTEUS* from clinicians, or clinicians would prescribe the medication to their patients. Pharmacies would ship *PROTEUS* to medical professionals for injection of the therapy.

Beyond a treatment as a pancreatic cancer therapeutic, *PROTEUS* has the potential to be re-engineered for fundamentally any other cancer (or any disease arising from genetic deficiencies in fact) due to its modularity as a general CRISPR-guided genetic tool. Ownership of this mechanism through provisional patenting represents substantial intellectual capital, assuming the product is functional *in vivo*. The market potential for other therapeutics harnessing the same Caspase mechanism designed by the McGill iGEM 2022-2023 team is extensive; noteworthy future applications could include treatment of viral diseases such as AIDS, retardation of senescence, and CRISPR-guided small-molecule detection.

1.6 Commercialization Strategy

The CRISPR therapeutics market is a rapidly developing space with a predicted compound annual growth rate of 22.3% between 2022 and 2027 (BCC Research Team, 2023). The cancer therapeutics submarket represents a significant segment of this space, making it highly competitive and fast-growing. Current gene editing technologies are highly expensive, making them inaccessible to large portions of the population; analysis by the Institute for Clinical and Economic Review suggests the average cost of a gene therapy is between \$1 million and \$2 million per dose (Sterling, 2023). To date, no CRISPR cancer therapeutic has been effectively commercialized, representing a significant market opportunity to provide a new therapeutic modality to combat cancer. This section will outline McGill iGEM's strategy for commercializing *PROTEUS*. An in-depth discussion of our commercialization strategy is provided in the Business Model Section.

CRISPR technology has shown great promise in cancer research, offering precise and customizable gene-editing capabilities. The *PROTEUS* CRISPR cancer therapeutic has the potential to disrupt the market by providing targeted and personalized treatment options. However, it is essential to navigate regulatory challenges, demonstrate safety and efficacy, secure intellectual property rights, and establish strong partnerships to successfully commercialize *PROTEUS*. Our commercialization strategy relies on a robust product development plan that encompasses the following key aspects: research and development (R&D), manufacturing and scale-up, regulatory strategy, marketing and sales, and intellectual property and partnerships.

- a. R&D: We will continue investing in cutting-edge research and development to enhance our understanding of our CRISPR technology and its application in cancer therapeutics. This includes verifying our system in mammalian cell lines, developing and optimizing the adeno-associated virus delivery system, and conducting preclinical and clinical trials to establish safety and efficacy.
- b. Manufacturing and Scale-up: To ensure scalability and cost-effectiveness, we will establish strategic partnerships with contract manufacturing organizations (CMOs) specializing in biologics production. This will allow us to meet regulatory requirements, maintain quality standards, and streamline the manufacturing process for commercialization.
- c. Regulatory Strategy: Navigating regulatory pathways is crucial for successful commercialization. We will adopt a proactive approach to regulatory affairs:
 - i. Preclinical and Clinical Trials: Conducting rigorous preclinical studies to establish safety and efficacy profiles for *PROTEUS*. We will then initiate Phase 1, 2, and 3 clinical trials to evaluate the therapeutic potential, optimize dosing, and monitor long-term effects. Collaboration with regulatory agencies will be sought to ensure compliance with guidelines.
 - ii. Regulatory Filings: We will compile comprehensive data from preclinical and clinical trials to support regulatory filings for marketing authorization. Close collaboration with regulatory bodies and proactive engagement will be maintained to expedite the review process and secure approvals in a timely manner.
- d. Marketing and Sales Strategy:
 - i. Target Market and Positioning:
 1. To effectively commercialize our CRISPR cancer therapeutics, we will identify specific target patient populations and disease indications (as mentioned previously). By leveraging the personalized and targeted nature of our therapies, we will position ourselves as a leading provider of precision medicine for cancer treatment.

- ii. Marketing and Sales:
 1. Key Opinion Leader (KOL) Engagement: Over the course of this year's iGEM project, our team members have engaged with renowned oncologists, researchers, and industry leaders in the field of cancer treatment to build advocacy and credibility for our CRISPR-based therapies. KOL endorsements and testimonials will be leveraged in marketing campaigns to influence healthcare professionals' prescribing decisions.
 2. Medical Education and Awareness: We will continue to invest our team's time and resources in scientific and medical education programs and initiatives to increase awareness of CRISPR and CRISPR-based therapeutics. This has involved organizing scientific conferences, workshops, youth education programs, regular blog posts, and podcast episodes on our platform, STEMCast. Our extensive educational initiatives are all designed to increase youth participation in STEM and to increase awareness and acceptance of CRISPR and its many applications in oncology and disease treatment in general. We aim to continue our education activity to disseminate knowledge and clinical evidence of the success of *PROTEUS* as a cancer therapeutic.
 3. Market Access and Reimbursement: To ensure widespread adoption, we will work closely with payers, including insurance companies and government agencies, to secure favorable reimbursement policies for our CRISPR cancer therapeutics. Economic and clinical data will be generated to demonstrate cost-effectiveness and patient outcomes.
 4. Sales Force and Distribution: A dedicated sales force will be trained to effectively communicate the value proposition of our therapies to healthcare professionals. We will also establish strategic distribution partnerships to ensure efficient product availability and reach.
- e. Intellectual Property and Partnerships:
 - i. To protect our inventions and maintain a competitive advantage, we will prioritize intellectual property (IP) strategies. This includes filing for provisional patenting to safeguard our CRISPR technology and therapeutic applications. Additionally, we will seek collaborations and partnerships with academic institutions, pharmaceutical companies, and research organizations to leverage complementary expertise, access resources, and expedite development and commercialization.

This commercialization strategy provides a comprehensive roadmap for successfully bringing *PROTEUS* to market. By focusing on product development, regulatory compliance, targeted marketing, and strategic partnerships, we aim to revolutionize cancer treatment and improve patient outcomes while establishing our company as a leader in the field of CRISPR-based therapeutics. Further discussion on commercialization is provided in the Business Plan section.

2. Firm Description

2.1 Background and Founding Team

McGill iGEM was founded in September 2021 by driven undergraduate students from McGill University in Montréal, Canada. Our team is a student-led organization that offers unmatched research experiences for students working towards a project to be presented at the annual International Genetically Engineered Machine (iGEM) competition, the largest global synthetic biology competition. McGill iGEM recruits dozens of undergraduate students each year and provides them with knowledge and hands-on research experience in synthetic biology, all while exposing them to the entrepreneurial space by developing consequential projects from ideation to market launch.

McGill iGEM is one of the leading iGEM teams worldwide at a preeminent Canadian institution. Our team is composed of 22 undergraduate students from McGill University's faculties of Medicine and Health (23%), Arts and Science (9%), Science (18%), and Engineering (50%). The diverse background of the team provides the team with a broad range of expertise in the field of synthetic biology, allowing for well-rounded considerations of the project's implications. The team is divided into the Wet Lab team, those of which achieve proof of concept through experimentation, and the Dry Lab team, who further the project through software development and computational modeling. Each member of the team is also heavily involved in McGill iGEM's various subteams, including Human Practices, Media & Outreach, and Finance. The Human Practices team coordinates the various educational initiatives that McGill iGEM runs and analyzes the feasibility of the year's project by engaging with expert clinicians and researchers in the field. The Media subteam works to present McGill iGEM's core values, not only by putting together the team Wiki for the annual iGEM Grand Jamboree, but also by fostering a presence with the general public via several social media outlets. The Finance team collaborates with sponsors, grant issuers, and potential investors to ensure the team has sufficient funding to support McGill iGEM's various endeavours. The Team and Organizational Structure Section contains further details regarding the specific responsibilities of each corresponding role.



Figure 2. McGill iGEM 2022-2023 team members and their primary roles and responsibilities.

McGill iGEM also has an extensive network of support with our graduate advisory board of eight members and two Principal Investigators, Dr. Lisa Münter from the Department of Pharmacology & Therapeutics and Dr. Codruta Ignea from the Department of Bioengineering. Biweekly meetings are held with the graduate advisors for guidance towards both the Wet Lab and Dry Lab project advancements.

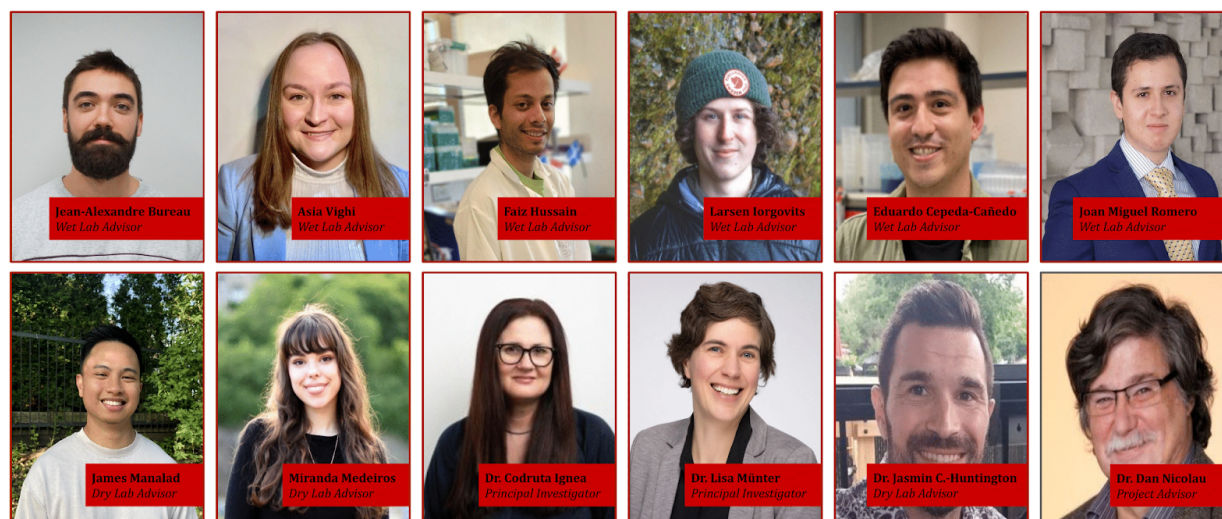


Figure 3. McGill iGEM 2022-2023 graduate advisory board, instructors, and principal investigators.

In its founding year, McGill iGEM competed at the 2022 iGEM Grand Jamboree in Paris, France where we presented CoBiota, a cholesterol-lowering probiotic that converts cholesterol to coprostanol in the gut as a preventative against cardiovascular disease. With CoBiota, McGill iGEM placed in the Top 10 overall, was awarded with a Gold Medal, and received

nominations for Best Wiki and Best Presentation at the competition. McGill iGEM sees great market potential in the CoBiota product. Since the competition, the returning members have continued to validate the probiotic's metabolic pathway with the ultimate goal of following through with CoBiota's commercialization. More recently, the 2022-2023 McGill iGEM team founded *PROTEUS*, a CRISPR-mediated pancreatic cancer therapeutic, which is to be presented at the 2023 iGEM Grand Jamboree. McGill iGEM aims to continue with *PROTEUS* beyond the competition and pursue its commercialization, as we see great potential for it to have a profound impact on cancer patients worldwide.

2.2 Intellectual Property and Technological Assets

2.2.1 Established IP: CoBiota



Figure 4. Cobiota's logo design; Cobiota was McGill iGEM 2021-2022's top ten iGEM project.

McGill iGEM is a multi-year organization with experience in intellectual property (IP) protection through previous years' projects. Our team has been meticulous in past iterations to maintain privacy over intellectual property (IP) and to request signed non-disclosure agreements (NDAs) from parties to which we must disclose pertinent elements of our IP. For McGill iGEM's 2021-2022 project, CoBiota, the team spoke with the McGill Research and Innovation Office of Innovation and Partnerships to discuss the logistics of filing a patent as a McGill-affiliated organization. Our team conducted an extensive literature search to protect against existing patent infringement, and one team member eventually branched off to develop the product into a company of the same name. Returning team members from the 2021-2022 team continue to perform experiments to further validate the product. After more robust results have been collected, the 2021-2022 team plans to file a patent application in Canada through the Canadian Intellectual Property Office (CIPO). Currently, the team is pursuing potential partnership opportunities with global probiotics and food and beverage companies, most notably Lallemand, in preparation for IP protection, scale-up, and eventual product launch.

By late August 2023, the 2021-2022 team aims to finalize the experimental validation of the product, after which the focus will be on filing a Canadian patent and establishing official partnerships with companies capable of scaling up the product. Until a patent is officially filed with the CIPO, the team is vulnerable to potential replication of the product. Thus, after completing experimental validation, patent filing is the priority. After filing, the CIPO provides the applicant 18 months following the filing date before the application is made available to the public via the Canadian Patents Database. At this point, competitors and generic producers could replicate the invention; thus, in this timeframe, our team aims to scale up production through partnerships to establish a dominant production of the probiotic. After a CIPO

examination and payment of maintenance and final fees, CoBiota would gain patent protection for 20 years in Canada following the filing date. This serves as an ideal outline of the IP strategy for our existing intellectual material, as well as a basis for the path our team aims to take with regards to our 2022-2023 project.

2.2.2 Future Outlook

This year's team is building upon the successes and setbacks of previous years to construct a sound IP strategy– taking into consideration our current IP portfolio, the requirements for differentiating our product, the timeline for patenting in several jurisdictions, and the strategy for disclosure of information prior to patent protection. As a Canadian organization, we aim to focus our efforts on establishing IP protection in Canada. This is for the convenience of proximity, the presence of a large potential market (Canada has a crude pancreatic cancer incidence rate approximately 247% higher than the worldwide rate), and the potential for a natural expansion to the country with the second highest incidence count worldwide: the USA (World Health Organization, n.d.).

To date, the team has validated the system in both *Saccharomyces Cerevisiae* (strain BY4741) and is working towards mammalian validation in HKP1 mouse lung cancer cell lines. Furthermore, after filing a patent, we plan to further validate in mouse models, then potentially in human trials. The innovative aspect of the product is derived from the method of attack against cancerous cells; instead of targeting *hallmarks of cancer*, our system targets the expressed cancer mutation itself. Using the novel CRISPR-Cas system 'Craspase', our team developed fusion proteins that are proteolytically cleaved upon recognition of specific mutant mRNAs by the gRAMP machinery of Craspase; after cleavage, one half of the protein translocates to the cell membrane where it forms pores that cause cell death through pyroptosis. This effectively prevents carcinogenesis at its root by forcing the cell to undergo pyroptosis before it can become phenotypically cancerous. Usage of Craspase as a therapeutic is entirely novel; to date, no other group has engineered the gRAMP machinery for treating genetic diseases, nor for any other therapeutic applications. The advantage of targeting mRNA rather than hallmarks of cancer is the efficacy of cancer removal; targeting hallmarks of cancer may attempt to limit angiogenesis, poison cells, or induce DNA damage in cancer cells. Unfortunately, these solutions often only kill fractions of cancer cells, which allows fitter mutants to survive and then proliferate due to natural selection. Furthermore, these treatments may result in the destruction of healthy tissue in the process. Oftentimes, especially for pancreatic cancer, current treatment options are only palliative, not curative. On the other hand, targeting mutant mRNA itself enables for specific killing with minimal off-target effects; the result is exceedingly rapid and high efficiency destruction of mutant cells. Based on *in vitro* success, our team is optimistic towards the future outlook of experimentation with the system, though *in vivo* validation will be the true testament to the utility and commercial potential of the system.

Based on the results of eventual *in vivo* experimentation, our team has the opportunity to bring a highly effective pancreatic cancer therapeutic to market, which could impact millions globally (Lin et al, 2022). The product would address fundamental needs in pancreatic cancer treatment given the disease's abysmal survival rate and underdeveloped curative modalities as compared to other cancers. Beyond the application as a pancreatic cancer therapeutic, our system has the potential to unlock a new avenue for genetic disease treatment in general; the inherent modularity of the machinery would allow others in the synthetic biology field to investigate diseases that could be targeted and treated by the system.

2.2.3 Overview of IP Strategy

This year, the McGill iGEM team has decided to pursue product patenting. Our team investigated other potential avenues, such as maintaining our product as a trade secret, solely pursuing the publication of a research paper, or changing our focus to a new project related to the system we developed. After weighing the feasibility, utility, and monetizability of the project, our team predicts that patenting the system and pursuing the product commercially would be the most beneficial strategy. The metrics used to guide this decision will be expanded on in the Intellectual Property Strategy section of this report.

In order to commercialize, our team would seek to incorporate as well as establish partnerships with major manufacturing corporations to develop a functional scale-up. The main benefits of incorporating include:

- Limiting personal liability due to the company becoming a separate legal entity
- Greater access to capital through venture capitals and angel investors
- Corporate tax rates on any revenues
- Establishing an official strategy for company equity/shares

Incorporation requires filing with Corporations Canada, which involves the preparation and submission of Articles of Incorporation and the Registered Office and Directors form, as well as a \$200 federal incorporation fee plus provincial registration fees. Our team has prepared the required documents and plans to officially incorporate upon rudimentary *in vivo* validation. The Basic Incorporation option in Canada allows for a maximum of 10 directors as well as a maximum of two share classes (voting and non-voting). Our team would elect 5 members from our founding team and 5 members of our graduate advisory board (most likely our two principal investigators and three PhD students that had a central role in supporting the development of the project) to maintain a balance of interest. We would select two share classes, Class A (voting), and Class B (non-voting) to provide the company the ability to grant equity without forfeiting control over its organizational structure.

After incorporation, the company would have reason to focus on expanding technological assets and filing additional patents in order to bolster IP and pursue scale-up to become a profitable venture. The modularity of our proof of concept (POC) system would

provide opportunities for the team to expand its IP portfolio by investigating other applications, such as in lung, colorectal, and brain cancers. After establishing an *in vivo* POC with pancreatic cancer, this would be one method of expanding new technology assets in the pre-seed to seed stages of the company.

Due to our affiliation and funding from McGill University, the company would need to minimally develop our POC patent in parallel with the university. Internally, McGill has its own process for pursuing patenting and takes a 40% royalty on all revenues generated from licensing. Luckily, this comes with the benefit of interest-free financial coverage of the legal expenses associated with maintaining the patent up until the date of the license (at which point there is a payback period to the university for patent expenses incurred). Further patent applications for future investigations would be solely regulated by McGill iGEM, so long as future research and development as well as recurring patent fees can be independently covered by revenues, low-interest debts, and other investment instruments.

All plans for future IP development rely on a foundation of security with regards to the intellectual assets developed by the team; to remain as the sole beneficiaries of our idea, it is necessary that the technologies we develop are not disclosed in their entirety. Thus, our team has developed and refined non-disclosure agreements as well as standard internal policies and strategies for minimizing external exposure to our intellectual assets. This allows our team to have a uniform approach to all disclosures that limits the liability of sharing information on our project; we have taken all necessary precautions to see the fruits of our labour. A more in-depth discussion of our team's IP strategy and IP roadmap are provided in the Intellectual Property Strategy section.

2.3 Vision and Long-Term Goals

McGill iGEM strives to ideate novel projects within the vast field of synthetic biology, such that our work can be brought to market launch, posing a solution to pertinent modern day problems. No other undergraduate research program at McGill allows for participation throughout an entire project lifecycle, from ideation to proof of concept to monetization.

We envision the *PROTEUS* therapeutic system trailblazing a future of innovative cancer therapies, utilizing a novel treatment modality that targets the cancer transcript itself, rather than hallmarks of cancer. Our program's goal is to use synthetic biology as a way to change the world. Currently, individuals who receive late diagnosis of pancreatic cancer have little to no chance of extended survival (Canadian Cancer Society, n.d.); surgical resection is often the most successful palliative treatment (Perinel & Adham, 2019). Our idea is to use *PROTEUS* in tandem with other immunomodulators such as checkpoint blockade inhibitors (i.e. pembrolizumab, ipilimumab, atezolizumab, etc.).

We have sought out numerous professional perspectives on our pyroptosis-inducing therapeutic system to guide our implementation strategy for developing a safe-to-use therapeutic. One of our primary concerns was the potential for hyper-inflammation caused by pyroptosis. Based on expert advice, we aim to have *PROTEUS* function as a "clean up"

mechanism for residual cancer cells following the use of another therapeutic. We are also in the process of selecting an ideal viral delivery system for *PROTEUS* that minimizes unwanted side effects and host cytotoxicity.

Following the global iGEM competition, our team will continue to validate and perfect the *PROTEUS* system. To ensure patient perspectives are considered in the design process, we have connected with pancreatic cancer patient advocacy groups and have conducted patient surveys to get their insight. Following the proof of concept in yeast and mammalian cells presented at the Grand Jamboree, our team aims to validate *PROTEUS* in mouse models, then eventually move forward to clinical trials.

Currently, *PROTEUS* proof of concept is being validated using *S. cerevisiae* strain BY4741. *PROTEUS* is also being validated in mammalian cell lines such as HKP1 mouse lung cancer cells for more accurate modelling. Following *in vitro* validation, the team plans to use syngeneic mouse models to more accurately represent the human immune system and inflammatory responses. Animal trials of *PROTEUS* will also allow us to measure and monitor optimal dosage, adverse side effects and safety concerns prior to moving forward with the regulatory processes.

In addition to ensuring its safety profile and efficiency, it is important to demonstrate that *PROTEUS* will have a positive real-world impact as exemplified through a variety of the sustainable development goals. The 17 Sustainable Development Goals (SDGs) were adopted in 2015 by all UN Member states in order “to promote prosperity while protecting the planet” (iGEM, 2021). Our team strives to focus on Goal 3 (“Good Health and Well-being”), given *PROTEUS*’ potential to reduce mortality rates from pancreatic cancer. Long-term, we hope that, within a five-year timeline, *PROTEUS* will enter the stage of clinical trials, after which it will eventually be approved by Health Canada for patient administration in Canada. Good Health and Well being via *PROTEUS* will ideally be applied to a variety of cancers and/or RNA-diseases. McGill iGEM prioritizes modularity throughout the completion of the *PROTEUS* design scheme, especially through the mechanism of which the Craspase system acts. Furthermore, McGill iGEM has explored various literature to gain inspiration for potential BioBricks, one of our special award categories. There are many possible applications of the Csx30 cleavage site, i.e. cleavage of any pro-protein/transcription factor for nuclear localization, functional activation, etc. In order to validate successful modularity of our BioBricks, we also tag the protein of interest with a fluorescent reporter, such as GFP. Thus, we can validate that our system can facilitate the translocation/activation of various proteins.

Within the next two decades, our team aspires to successfully complete clinical trials so that our therapeutic can be brought to market. Through experimental trials, we will need to establish adequate dosage of *PROTEUS*, optimal administration techniques (i.e. gutless adenoviral delivery vectors) and foreseeable adverse side effects. Ultimately, we hope that *PROTEUS* can move to stage four of clinical trials, deemed market-surveillance for an optimistic future of RNA-targeted cancer therapies. Specific to cancer, McGill iGEM believes that *PROTEUS* can serve as a wide-demographic cancer therapy, not limited to PDAC. Cancer

is a disease notoriously caused by a variety of mutations, however some cancers are more homogeneous than others. Through modification of *PROTEUS* components, we strongly believe our Craspase system has potential to target any oncogenic mutation, a lucrative quality for RNA-based therapies.

Our team has immense dedication to the pursuit and continuous improvement of our novel therapeutic project; we believe it has real potential to make change in the lives of those affected by pancreatic cancer. We are committed to staying at the forefront of emerging techniques and therapeutic systems within the field of biotechnology.

3. Corporate Organizational Structure

3.1 Roles and Responsibilities

3.1.1 Current Structure

Principal Investigator

The Primary and Secondary Principal Investigators of McGill iGEM must be assistant, associate, or full professors at McGill University. The Principal Investigators are responsible for overseeing the proper functioning of the Club and competitive research team, such as ensuring all safety and operational guidelines are met within the lab space. They participate in the selection process of Executive Directors and Club team members, as well as processing any formal complaint made against any team member(s) from within and outside of the organization. In addition, they act as liaisons in managing any McGill funds and administrative work associated with the Club team. For more information, refer to section 3.2.

Graduate Advisor

The Graduate Advisors of McGill iGEM consists of graduate students, PhD candidates, and professors of McGill University. They provide formal training and guidance on all operations in the wet lab and dry lab team. For more information, refer to section 3.2.

Executive Committee

The Executive Committee is composed of the following portfolios:

- Executive Directors (2)
- Operations Director
- Communications Director
- Media Director
- Finance Director
- Human Practices Director
- Wet Lab Director(s)
- Dry Lab Director(s)

The Executive Committee is chaired by the Executive Directors. Communication between the Executive Committee and the Students' Society of McGill University, like the bi-annual financial audit, falls under the responsibility of the Operations Director.

Executive Directors

The Executive Directors are chief spokesperson of the team. They are the main point of contact to the Principal Investigators, and the SSMU VP Student Life. In general, the Executive Directors oversee all aspects of the club, such as co-signing all financial transactions with the Finance Director, and co-signing all minutes with the Operations Director. In addition, they ensure that the team is abiding by the Constitution, By-laws and Policies of Students' Society of McGill University and McGill University.

Operations Director

The Operations Director oversees and coordinates the programming activities and events of the club, including its weekly meetings, annual conference and the iGEM Competition. The Operations Director maintains an updated contact list of all members of the Club with first and last names, email address, student ID, position, date of joining and date of leaving. They maintain communications between all subteams and the Executive Committee, while working closely with Wet Lab Leads to ensure the smooth operation of the research team.

Communications Director

The Communications Director is responsible for developing healthy relations with students and community organization inside and outside of McGill University. They act as representatives of the Club to the community, promoting the image of the team and overseeing all media output like the podcast, posters, and social media posts.

Media Director

The Media Director determines the design language pertaining to the presentation of the iGEM project, which affects the production of the promotional video, wiki, and the slide deck. They design multimedia resources for various social media channels, such as Twitter, LinkedIn, Instagram, and Facebook. In addition, they help assemble the promotional material for the Finance team, such as the sponsorship package, the crowdfunding advertisement, and the pitch deck.

Finance Director

The Finance Director coordinates funding for the programming activities and events of the club. A critical role for the Finance Director is to keep track of all financial transactions and receipts, in preparation for the biyearly audits for the SSMU and ensuring the financial stability of the research team. All grant applications must be approved by the Finance Director. In addition, they are responsible for overseeing the entrepreneurial aspect of the project.

Human Practices and Education Director

The Human Practices Director organizes and participates in the outreach of key stakeholders, experts, and individuals directly affected by the problem our team is hoping to solve. Subsequently, the Human Practice Director is responsible for documenting these interactions for display on the iGEM Wiki. They prepare and submit any necessary ethics documents to the appropriate Research Ethics Board for any human studies, including interviews, surveys, and experiments. In addition, they are responsible for organizing and delegating any education initiatives within and outside of McGill University.

Wet lab Director(s)

The Wet Lab Director(s) are responsible for overseeing all laboratory operations and organizing all meetings related to wet lab work. Before the start of the summer research period, they are responsible for formulating the experimental outline of the project. They direct tasks to members of Wet Lab, keep a detailed track for all lines of experimentations, order reagents and lab material, and ensure adherence to lab safety guidelines. In addition, they are the point person for organizing wet lab workshops sponsored by McGill iGEM.

Dry Lab Director(s)

The Dry Lab Director(s) oversee all computational and bioinformatic experiments related to the Club's research project. They are responsible for laying out the experimental outline on the dry lab side of the research project and organizing all dry lab meetings. They keep up to date on all lines of experimentation and document all dry lab activities.

Chief Electoral Officer

The Executive Committee elects a Chief Electoral Officer (CEO) to oversee the running of elections via a majority vote. The elected CEO must be part of the McGill iGEM Graduate Advisory board. The primary and secondary PI informs their choice of executive director to the CEO and the CEO calls for a vote for all returning members to ratify the two chosen executive directors. In addition, the CEO collects all nominations for the Executive Committee and once the Executive Directors has chosen the director for the subteam, the CEO holds a vote within the team to confirm the candidate.

Human Resource

The Human Resource is responsible for processing complaints made by any members of McGill iGEM against any members of the organizations. Human Resource does so through monitoring of an anonymous form accessible to all members of McGill iGEM. Once Human Resource has collected the responses, they shall bring the concerns to the Executive Committee or in private.

Teams & Subteams

All members of McGill iGEM has to participate in either one, or both, of the following teams, headed by their respective director in the Executive Committee:

- Wet Lab
- Dry Lab

In addition, all members of McGill iGEM has to participate in one or more of the following subteams, headed by their respective director in the Executive Committee:

- Human Practice
- Education
- Finance and Entrepreneurship
- Media and Outreach

Wet Lab Team

The Wet Lab Team plays a pivotal role in bringing ideation to life. They are responsible for carrying out all of the hands-on lines of experimentation and participating in activities directly relevant to the wet lab, such as literature review, documentation of lab work, experimental design, and maintenance of the lab space.

Dry Lab Team

The Dry Lab Team is responsible for the computational and data-driven arm of the research project. They harness the power of computer science, bioinformatics, and modeling to make the research project more comprehensive. They work in collaboration with the Wet Lab team by providing data analysis for the results produced by the Wet lab, as well as developing important tools that complement the development of the research project .

Human Practice Subteam

The Human Practice Subteam reaches out to external organizations, stakeholders, and leading scientists to discover ways to collaborate and improve the ethics and implementation of the research project. In addition, they are in charge of the composition of the Bioethics report.

Education Subteam

The Education Subteam collaborates with schools of all levels, summer camps, and other organizations to educate students on the power of research and synthetic biology. The subteam is responsible for hosting info sessions, workshops, and seminars that focus on a hands-on approach to teaching.

Finance and Entrepreneurship Subteam

The Finance and Entrepreneurship subteam secures funding for the operations of the team and develops the entrepreneurial aspect of McGill iGEM. Each member is responsible for finding

grants opportunities and writing grant applications. In addition, they are responsible for the composition of the manuscript and the business report.

Media and Outreach Subteam

The Media and Outreach Subteam is responsible for the production and publication of all forms of media communication, including, but not limited to, podcast, Substack newsletter, Linked-in, and Instagram Post. In addition, the subteam is involved in the design of all McGill iGEM related websites, including the STEMcast website, the main website, and the wiki websites.

3.2 Advisory Board and Support Network

As an organization under McGill University, McGill iGEM has an extensive support network over many different faculties, including the Faculty of Engineering, the Faculty of Science, and the Faculty of Medicine.

Our advisory board consists of two principal investigators along with several professors and graduate students with knowledge and expertise in the field of synthetic biology, biological engineering, bioinformatics, and biochemistry. We hold bi-weekly meetings whereby our executive team provides the advisory board with updates on the project status. After the current state of the project is presented, the advisory board deliberates over the current issues and roadblocks that we are facing, and provides insights and suggestions to overcome these problems.

The principal investigators (PIs) support McGill iGEM not only with project-specific suggestions, but also in overseeing the entirety of the team and administrative affairs. They provide guidance in our yearly recruitment process, specifically in aiding our executive team in making final decisions regarding the appointment of new members. They oversee the proper functioning of McGill iGEM and ensure all safety and operational guidelines are met within the lab space. The PIs are also our financial account holders and give final approval in the purchasing of lab equipment and reagents.

The Wet Lab graduate advisors provide ample guidance in the development of our experimental plan. They also share expertise they have obtained from their own research experiences with our executive team in order to aid in troubleshooting flaws in our protocols and techniques. The Dry Lab instructors and graduate advisors work directly with the Dry Lab team. Their knowledge and expertise in working with programming toolkits and large data sets aided our team with the development of the pipeline, and necessary troubleshooting methodologies. This well-assembled advisory board comes from a diverse educational background that well encapsulates the fields of synthetic biology, biological chemistry, and bioengineering, ultimately providing McGill iGEM with a broad range of expertise that ensures success in all aspects of *PROTEUS*.

As well, McGill iGEM was honoured to collaborate with the McGill Investment Club (MIC) and use their insight in developing our commercialization strategy. Their knowledge of

the financial sector and their expertise in the investment industry greatly aided our efforts in realizing the marketable potential of *PROTEUS*.

3.3 Skill Gap Analysis

A skill gap analysis is an essential tool for determining the strongest skills provided by the current team, as well as those that should be developed and sought after in recruitment. Our team developed a skill gap analysis table, breaking down the skills into Technical Expertise, Regulatory and IP Handling, Business Management and Finances, and Human Resources. While this analysis did not go into depth with traditional 'Soft Skills', we believe that by developing a strong Human Resources team in the coming months and years, these skills will naturally be considered in the recruitment process, as well as in employee development strategies and seminars.

Given that McGill iGEM is currently a pre-seed organization, our highest focus should be placed on fleshing out the technical aspects of our research and development. Without a sound proposal, it will be impossible to successfully garner financial support for *PROTEUS*, let alone turn a profit upon market launch. Thus, our team needs to develop technical skills in immunology, oncology, synthetic biology (with a particular focus on CRISPR-Cas systems and genetic-level therapeutic technologies), molecular modeling, and big data (given the need to handle thousands of patient reads in the Dry Lab pipeline). These skills are considered high priority given their immediate relevance to the health of the venture.

Another major concern is regulatory compliance and IP protection. Given our team's significant skew towards STEM, we will need to recruit expertise in the biotechnology regulatory space. Having a dedicated legal team may be difficult and excessive this early on, though seeking guidance to minimize disclosures, protect our IP, and avoid encroaching on previously patented materials would be the most financially responsible course of action. Given our current skill gap in the Regulatory and IP Handling category, this will be an essential focus for our early recruitment strategy.

Business development is another area where our team currently lacks some essential proficiency. Given the extensive regulatory hurdles for both cancer therapeutics and gene editing tools, market launch is a distant concern. Thus, marketing, accounting, and customer service have lower general priority. However, garnering financial support by developing relations with large oncological corporations, suppliers, and producers will be essential for pre-seed and seed funding once we arrive at initial regulatory milestones. This makes business management generally a medium priority in terms of immediate relevance to the success of the venture.

Finally, Human Resources is a low to medium concern; eventually, our team will aim to establish a full division for Human Resources. However, until our team can garner substantial funding to scale up, dedicating an entire division to recruitment and employee relations is nonessential and a non-lucrative use of labour. Early on, we will be making minimal recruitments to the team both to preserve our IP protection and to limit financial liability. We will

focus on a small team of regulatory experts, a few key business development strategists, and perhaps a few technical experts to finalize our early-stage technological assets. Large-scale recruitments and team development will become more essential considerations following our initial pre-seed developments.

Given the considerations discussed above, our team ranked our current skills, and our most critical shortfalls. The Skill Gap Analysis table below describes the importance of each hard skill considered (mostly in terms of immediate relevance to the venture's success), then the current level of expertise (on a scale of 1 to 5), and finally the required level for our early-stage success (also on a scale of 1 to 5).

	Skill	Importance	Current Level	Required Level
Technical	Immunology Expertise	Med/High	2	4-5
	CRISPR-Cas / Synbio Expertise	High	3	5
	Protein Engineering Expertise	High	2	4-5
	In Silico Biomolecule Modeling	High	2	4
	Data Science and Data Management	Med	3	4-5
Reg. / IP	Regulatory Compliance Knowledge	High	1	5
	IP Portfolio Management	Low	1	3
Business Mgmt / Finances	Business Development Expertise	High	2	4-5
	Sales and Marketing	Low	1	4
	Administration and Accounting	Med/High	1	3
	Customer Service and Partnerships	Low/Med	2	3-4
HR	Human Resources Management	Low	2	3

Figure 5. Skill Gap Analysis table broken down by category of skill, then ranked in terms of immediate relevance, as well as our current team's level of expertise. The most essential skills to develop based on the analysis are highlighted in grey, while the less pressing skills are left as white. Evidently, our focus in early-stage team development and recruitment will be on rounding out our technical skills, as well as working on regulatory considerations and business development.

3.4 Partnerships with academic institutions or industry collaborators

McGill iGEM's primary partnership is with McGill University. Having a close relationship with a prominent academic institution such as McGill University brings numerous benefits to our team and has greatly aided in the development of *PROTEUS*.

As mentioned previously, our team has an extensive support network of mentors and administrators within the university. In addition to regularly meeting with our advisory board,

being affiliated with McGill University provides us access to a variety of experts who can give insight into different aspects of our project. This is the case both for technical consultations, given the variety of researchers whose work is tangential to ours in some facet, as well as for consultations relating to other aspects of the project, such as ethical considerations and the commercialization process. Furthermore, in cases where few McGill professionals are fit to advise on a specific part of the project, many of the faculty have connections to external researchers and other institutions who may be able to provide support.

Our team's partnership with McGill University has also been a great privilege in the administrative realm. Being an organization under the university allows us to take advantage of the centralized banking system, giving us interest-free, highly secure, administratively managed accounts. Although administrators from McGill handle the finer details and relevant procedures, we maintain autonomy in the purchases we make and the contracts into which we enter. With regard to those very purchases, our affiliation with McGill allows us to obtain reagents, equipment, and other supplies for our experiments that we would not otherwise be able to obtain as a team of undergraduates.

In addition to helping us obtain the necessary materials for our research, McGill also provides two other key resources that would typically be inaccessible to a team such as ours. The first of these resources is money: the university has been very generous with their financial support of our project. Research is an expensive undertaking, and the development of *PROTEUS* would not be possible without contributions from the Faculties of Engineering and Medicine & Health Sciences, and the Departments of Bioengineering, Biology, and Biological & Biomedical Engineering. The second of these resources is space. Even within a well-funded university, there is a limited amount of laboratory space for faculty to conduct their own research. Despite this, McGill has allowed us to access multiple labs and to use the equipment and storage within them.

As the commercialization of the *PROTEUS* project progresses, our partnership with McGill University will evolve to include new components. The first case of this will be when the *in vivo* POC has been verified and we are ready to pursue patenting. As detailed in section 2.2.3, Overview of IP Strategy, we plan to collaborate with McGill in the patenting process. The development of the patent will occur in parallel with the university and their own internal patenting procedure. In exchange for a 40% royalty on revenue generated from the licensing, McGill will temporarily cover, interest-free, any legal fees associated with maintaining the patent until the date of the license. Following the patent process, we will be able to take advantage of McGill's nature as a large and selective institution when we start recruiting during scale-up. Although it would be unwise to limit ourselves to members of the university, it exists as a large pool of talent that we can easily reach and recruit from, serving as a sound starting point for our growth.

When that point has been reached, we will be at the later stages of our commercialization strategy. It will then also be necessary to expand our partnerships beyond just the university. Although our team at McGill iGEM has not yet decided what revenue

generation strategy or strategies (further described in section 8.2) we will be pursuing, the viable options are likely to lead to collaboration with a large biopharma company in some capacity. Specifically, the extent of this new partnership would depend on our own facility to develop and produce products using *PROTEUS* and *PROTEUS*-adjacent technology. Keeping these tasks in-house has a higher potential for profit, but outsourcing them to an industry partner would place much less burden on us. Beyond this, there is potential for a strong partnership with a large player in the biopharma industry to turn into an eventual acquisition. This is something that will be considered in detail by our team down the line depending on the state of the market at that time. While we have many leading strategies, we cannot yet finalize a single exit strategy (given exit will depend on the state of the venture perhaps decades down the line), but it is always important to keep all of our options open.

4. Market Research

4.1 Problem Statement

Cancer is a general term for diseases arising from random genetic mutations which ultimately have devastating effects on how proteins in the human body natively function (National Cancer Institutes, 2021; Pessoa et al., 2022).

Genes are the blueprints that dictate cell function and performance, such as when cells must grow, divide, and die through automated cell death, or apoptosis. Oncogenic mutations, however, lead to rapid cell proliferation and resistance to cell death mechanisms, known as immune evasion (Pessoa et al, 2022). In this way, they reroute nutrients essential for healthy cells, causing tissue necrosis and severely harming the host.

In general, cancer cells are recognizable through cancer hallmarks. However, cancers tend to be difficult to detect, enabling for metastasis, or detachment of tumor cells from the primary tumour. These cells then spread to other parts of the body, forming secondary tumours (Pessoa et al., 2022).

Cancer mutations are generally acquired after birth and are related to factors such as substance abuse, diet, exercise, and BMI (National Cancer Institutes, 2015; World Health Organization, 2022). In 2020, cancer accounted for nearly ten million deaths globally, with a predicted diagnosis in two of every five Canadians (World Health Organization, 2022). Despite recent scientific advancements, cancer incidence continues to climb without a matched improvement in patient prognoses. Between 2015 and 2030, the number of new cancer cases is expected to increase by 40% in Canada (Canadian Cancer Society, 2023).

Pancreatic cancer is the deadliest common cancer in terms of survival rate (American Cancer Society, 2023), characterized by its poor prognosis and post-diagnostic quality of life. This cancer is currently the third leading cause of cancer-related death in the United States (American Cancer Society, 2023). Incidence rates vary internationally, though higher incidence is especially common in developing and industrialized countries with increasing quality of life (World Cancer Research Fund International (WCRF), 2022). While the 5-year survival rate has

increased in recent years, it remains abysmally low compared to progress made with other cancers; between 1977 and 2013, 5-year survival increased by 16.7% for all cancers in the US, but only by 5.7% for pancreatic cancers (Roser & Ritchie, 2019). Pancreatic cancer is conquering the Western world with alarming growth in its incidence rate; between 1992 and 2017, its incidence rate increased by 55%, affecting women and men approximately equally (Lippi & Mattiuzzi, 2020; McGuigan et al., 2018). It is estimated that by 2030, pancreatic cancer will be the second most common cause of cancer-related death (American Cancer Society, 2023).

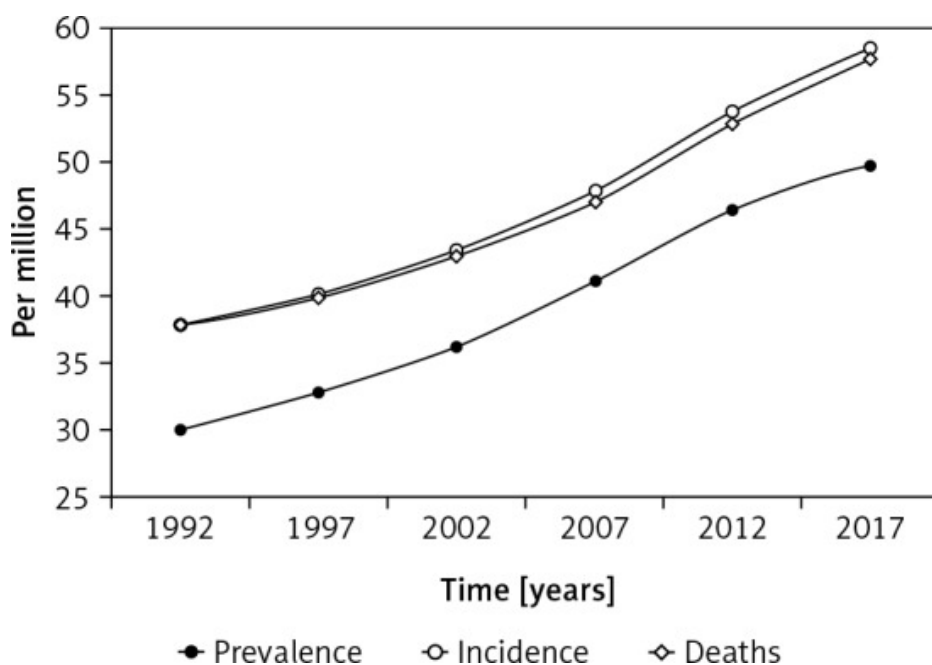


Figure 6. Pancreatic cancer prevalence, incidence, and mortality rates worldwide between 1992 and 2017. Evidently, nearly all victims succumb to the disease, and incidence is climbing steadily.

Risk factors of pancreatic cancer include lifestyle factors such as obesity, smoking, and alcohol consumption (National Cancer Institute, 2015; World Health Organization, n.d.). However, it may also result from non-modifiable risk factors like age, ethnicity, gut microbiota, and genetic susceptibility (Canadian Cancer Society, 2023). As obesity continues to infiltrate into the Western world, meta-analyses of studies indicate correlations between cancer and obesity. In fact, these studies suggest a 10% increased risk of pancreatic cancer for every 5 BMI unit increase (Jacobs et al., 2020).

The KRAS oncogene is a hallmark of pancreatic cancer, present throughout a wide patient demographic. KRAS is a key mediator of pancreatic tumour formation and maintenance, making it a leading target for therapeutic intervention (Luo, 2021). There is urgency surrounding the prognosis of pancreatic cancer, as cases are often late-diagnosed, with advanced metastatic challenges (Luo, 2021; Michael et al., 2019).

In addition, roughly 85% of tumors are non-resectable at the time of diagnosis. Surgical resection is often the only feasible treatment for pancreatic cancer, although recurrence is probable and long-term survival is seldom achieved (Michael et al., 2019).

The problem described is what inspired McGill iGEM to engineer a novel, personalized therapeutic technique. A personalized treatment modality restricts negative off-target effects, ideally killing cancerous cells before surgical resection becomes the only viable treatment. At McGill iGEM, we utilize the newly discovered genome editing tool, Craspase, to engineer forced pyroptotic cell death among cancerous cells (Hu et al., 2022). Craspase is a CRISPR RNA-guided, RNA activated protease, which mediates our personalized therapeutic approach (Hu et al., 2022). Our team takes advantage of the high homogeneity of KRAS mutations in pancreatic cancer to design a genetic-level therapeutic that targets the cancer at its root, rather than targeting *hallmarks* of cancer.

Our team prioritizes designing modular parts for our project in hopes of engineering a novel therapeutic for many different manifestations of cancer, and perhaps more generally any disease arising from mutations or changes to the host genome.

4.2 Target Market and Stakeholders

4.2.1 Target Market and Customer Segments

The primary target market for *PROTEUS* would evidently be those affected by or likely to be affected by pancreatic cancer. Given *PROTEUS*' potential as a preventative, it is important to consider the most significant customer segments for the therapeutic based on age, ethnicity, sex, BMI, and other factors. Pancreatic cancer increases in incidence with an increase in age, BMI, and tobacco use (American Cancer Society (ACS), 2020; Jacobs et al., 2020). There is a slight bias in incidence towards men instead of women, and towards people of African descent instead of caucasians (ACS, 2020). As with most cancers, family history and inherited genetic syndromes increase likelihood of pancreatic cancer as well (ACS, 2020). Creating a marketing strategy to reach all of these customer segments is difficult given the variety of factors affecting pancreatic cancer development. Nonetheless, the most important segments will be the elderly, the medically obese, smokers, and those with a likely genetic predisposition.

As a medical treatment of disease similar in principle to a drug, our team must consider the healthcare system of each jurisdiction we aim to target; in countries such as the US, there is a lack of universal healthcare (MIT Medical, n.d.), while in Canada, France, and Germany, for example, there is a certain level of governmental health coverage for most or all citizens (Brown, 2003). However, for jurisdictions with some form of universal healthcare, it is a separate question whether *PROTEUS* would be likely to meet the requirements to be covered by the state's healthcare plan. For instance, in Canada, gene therapies are considered an 'emerging technology' (Health Canada, 2005); as recently as 2018, no gene therapy technologies were approved in Canada, let alone covered by provincial or territorial health insurance plans (CADTH, 2019). Given the novelty of CRISPR-Cas therapeutics, as well as their

currently steep price, it is not reasonable to expect that *PROTEUS* will be adopted by government-subsidized or government-issued health coverage for at least a decade, regardless of the jurisdiction. Thus, at least in the coming years, the primary market segment will more specifically be those capable of covering the costs of the treatment on their personal account, or through health insurance. The most significant customer segments are summarized in the table below:

Customer Trait	Customer Segmentation
Affluence	• Demographic: ageing, educated, stable career, high annual income, secondary assets • Geographic: Europeans, North Americans, Oceanians; wealthier regions of the aforementioned (urban areas throughout: Catalonia, Île-de-France, London, Bavaria, Scandinavian nations; Bay Area, Boston, New York, Toronto, Vancouver; New South Wales and Victoria) along with major business hubs such as Singapore, Beijing, Shanghai, Shenzhen/Guangzhou/Hong Kong, Taiwan, and Tokyo • Behavioural: supportive of genetic modification and gene therapies, risk-taking, science or technology background (a STEM background would increase comfort with the product)
Elderly	• Demographic: elderly, taking on managerial roles, high income • Geographic: ageing economies (i.e. developed or developing nations), most notably Japan, Europe, and North America • Behavioural: educated, especially in sciences
High BMI	• Demographic: ageing, lower income, underprivileged • Geographic: primarily developing nations with high economic disparities; the US, the Arab Gulf States, Mexico • Behavioural: fast food consumers, high meat and fat consumers, often lower social status (especially in developed nations)
Tobacco Use	• Demographic: lower income, underprivileged, undereducated • Geographic: Europeans, especially Eastern Europe, China, Indonesia, and the Middle East • Behavioural: more 'European' perspective on smoking

Figure 7. Customer trait and segmentation analysis, summarizing key characteristics and their importance. These customer traits are predicted to be common among users of our therapeutic.

Based on the customer segment analysis, it is evident that the geographic focuses will be in North America and Europe. Given the proximity to McGill iGEM, Canada and the US will be the focus of our marketing and business development. Depending on initial success, we would broaden our geographic availability to Europe and the developing regions of Asia and Oceania. According to the World Cancer Research Fund, the countries with the highest incidence rates of pancreatic cancer are mostly concentrated in Eastern Europe (WCRF, 2022); hence, this will be a region we would expand to quickly following initial successes.

The traits of old age and affluence tend to conflict with high BMI and tobacco use; the latter two lend to more modest socioeconomic positions in most cases. Given the often high cost of gene therapies, we assume that *PROTEUS* will similarly not be a budget treatment. Thus, obesity and tobacco use are considered less lucrative customer traits compared to age and affluence.

As a personalized medicine, *PROTEUS* provides a unique therapeutic option compared to existing pancreatic cancer treatments; instead of targeting hallmarks of the disease, *PROTEUS* targets the cancer at the transcriptomic level, filling a current gap in the market that is essential for patient survival and long-term health. McGill iGEM believes that this will be a significant enough incentive to encourage consumers to use the product post-approval.

4.2.2 Stakeholders and Supply Chain

At a high level, drug and therapeutic manufacturers produce and distribute medications to various intermediaries, including wholesalers and pharmacies. In Canada, the government negotiates drug prices, while in the US, various entities, including Pharmacy Benefit Managers (PBMs) and insurers, negotiate these prices (U.S. GAO, 2007). Both countries have insurance systems—Canada's predominantly public and the US's a mix of private and public—that determine coverage and reimbursement rates for medications. The insurance providers then dictate the out-of-pocket costs or co-payments required from patients, which can vary based on the specifics of each individual's insurance plan. Patients, subsequently, obtain their medications from pharmacies or hospitals, paying the amount designated by their respective insurance coverage.

Given that McGill iGEM aims to target a niche market, we would likely partner with a cancer therapeutic wholesaler to better reach our target market. We would ideally partner with a top ten oncological pharmaceutical company based in North America, such as Johnson & Johnson (J&J), Celgene, Pfizer, or Bristol-Myers Squibb (BMS). Eventually, depending on success in North America, we may seek partnerships worldwide, such as with Astellas in Japan, Merck in Germany, AstraZeneca in the UK, and Roche or Novartis in Switzerland. Afterwards, individual physicians would request *PROTEUS*; the therapeutic would then be delivered to the hospital to be administered to the patient. This would be the standard process for the more common mutation targets, such as KRAS G12V. If we eventually expanded to target other mutations, the protocol would be the same.

Given that gene therapies tend to be expensive and new on the therapeutic stage, it is unlikely that federal, or state/provincial health insurance plans will cover *PROTEUS* in the near future. The much more likely scenario is that individuals pay out of pocket, or rely on their health insurance provider, at least in the early stages.

The Stakeholders and Supply Chain diagram below provides a visual overview of the general protocol from manufacturing to administration.

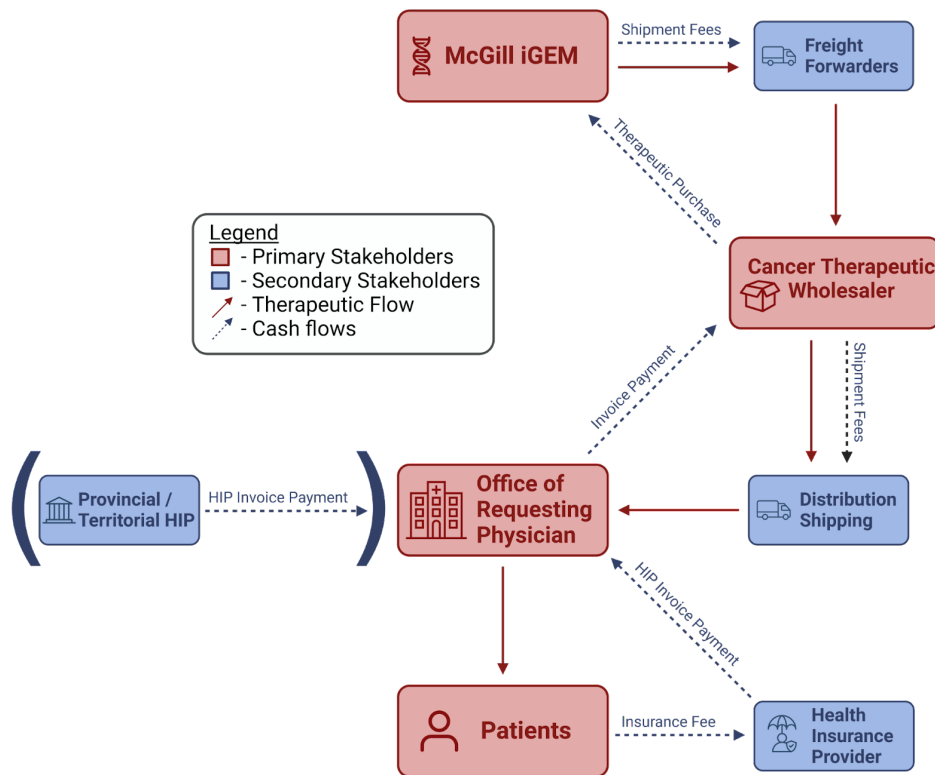


Figure 8. General protocol from manufacturing to delivery of *PROTEUS* with cash flows and flow of the therapeutic.

4.2.3 Consumer Segment for Clinical Trials

Following a discussion with Cameron Black at Repare Therapeutics, our team came to the conclusion that recruitment for clinical trials should focus on highly specific oncogenic mutations, such as KRAS G12V, rather than just pancreatic cancer in general. This would allow us to recruit a population with more homogeneous cancer developments and allow us to establish a niche market in which we excel. This would make our product more appealing to large pharmaceutical wholesalers given we can target one specific mutation with high efficacy.

Prior to any true market launch, our product needs to undergo rigorous validation through clinical trials. For cancer therapeutics, this is usually a decades-long journey. Given the lengthy timescale, it is essential that we meticulously define our customer segments for clinical trials by developing a clinical trial application (CTA). Our team aims to format this application according to recommendations provided by the Government of Canada (Health Canada, 2015).

4.3 Analysis of Market Size and Growth Potential

The global pancreatic cancer treatment market size was valued at \$2.15 billion in 2021 and is projected to grow from \$2.48 billion in 2022 to \$6.85 billion by 2029, exhibiting a soaring compound annual growth rate (CAGR) of 15.7% (Fortune Business Insights, 2021). Thus, by 2040, the absolute earliest feasible timing for *PROTEUS*' market launch, the market could reach a size as large as \$35 billion. The global pancreatic cancer market is experiencing rapid growth, driven by factors such as rising cancer prevalence, enhanced research initiatives, and population growth among the elderly. As of 2021, North America accounted for approximately half of the market share, followed by Europe, mainly led by Germany, France, and the UK (Fortune Business Insights, 2021). The Asia-Pacific region is anticipated to expand at the highest CAGR during the forecast period. The rising prevalence of pancreatic cancer, improvement in healthcare infrastructure, and increase in the number of pharmaceutical companies developing products for pancreatic cancer treatment are some of the factors propelling the growth in the region.

4.3.1 Total Addressable Market

Approved CRISPR-Cas therapies tend to cost upwards of \$1 million today (Nature Editorials, 2023). Assuming a total cost per treatment of \$1 million – between development of optimal gRNAs, plasmid construction, viral vector development, shipping, wholesaler fees, and hospital fees – and a rough estimate of around 815 000 new pancreatic cancer cases per annum by 2040 (Rawla, Sunkara, & Gaduputi, 2019), this provides a total addressable market (TAM) of around \$815 billion per year. Given the exceedingly low survival rate of pancreatic cancer (~12% (ACS, 2023)), we have assumed that all individuals with a new development of pancreatic cancer would have demand for the product, ignoring all geographical, financial, and religious barriers to treatment.

4.3.2 Potential Available Market

According to the American Cancer Society, up to 10% of pancreatic cancers are directly caused by inherited variants in the genome (ACS, 2020). This means that by 2040, approximately 80 000 individuals will develop pancreatic cancer yearly due to genetic inheritance (as opposed to environmental factors) assuming the 10% figure holds constant. If the usual victim of genetically inherited pancreatic cancer is a grandparent, and the average fertility rate is simplified to 2 (Roser, 2014), this suggests that 6 additional family members could have a genetic predisposition to the disease (the two children as well as the four grandchildren of the afflicted individual). Thus, for each case of genetically inherited pancreatic cancer, 6 additional individuals might logically seek preventative treatment. This brings the population of treatable individuals up to 1 304 000 (815 000 plus 81 500 times 6). Taking this value as the potential available consumers, the potential available market (PAM) can be estimated at ~\$1.3 trillion.

4.3.3 Serviceable Available Market

Out of the pool contributing to the TAM, only a fraction would be able to be served based on geographic restraints and production capacity. Assuming we initially launch our product with shipping to consumers throughout Canada and the US, this vastly decreases the total number of available consumers by eliminating Europeans, East and Southeast Asians. Based on this limitation, the number of potential consumers by 2040 drops to around 100 000 assuming similar long-term pancreatic cancer incidence rates between Canadians and Americans (Rahib et al., 2021). This equates to a serviceable available market (SAM) of ~\$100 billion by 2040.

4.3.3 Serviceable Obtainable Market

Out of the SAM, many consumers will not be able to afford the treatment (at least not considering government-sponsored subsidies or coverage, or support from private health insurance). Furthermore, many may be hesitant to seek treatment given the perceived risks. Lastly, some may have religious beliefs that interfere with seeking treatment through *PROTEUS*.

Financial obstacles are likely the easiest to quantify. In the US, 8.3% of citizens did not have any form of health insurance in 2021 (US Census Bureau, 2022). In one survey, 55% of Americans reported some form of medical debt, especially due to deductibles required by health insurers (Winters, 2022). Given the high cost of most CRISPR-Cas therapeutics, we assume those who can afford therapy are households with at least \$500 000 in total assets, which equates to the top 20% of American households (Loudenback 2022). Most individuals in this bracket have private or work-paid health insurance, covering most or all health-related expenses, allowing for coverage of treatments of greater cost than the household's total assets. Assuming 20% of Americans can afford the treatment out of pocket or via some form of insurance, this equates to 18 600 individuals with pancreatic cancer in the US who can afford treatment by 2040. While Canada does have public health care, it is highly unlikely that *PROTEUS* will be offered by provincial health insurance plans. Hence, using a similar rational progression, about 1850 Canadians with pancreatic cancer would be able to afford treatment. This limits the population to 20 450.

Consumer hesitance towards CRISPR-Cas therapies is difficult to quantify in today's world given the lack of these therapies on the market as of now. Given pancreatic cancer's exceedingly low survival rate, we have simplified the calculation of the serviceable obtainable market (SOM) by assuming consumer hesitance does not affect the number of individuals who would seek treatment through *PROTEUS*.

The final consideration in this analysis is religion. According to Statistics Canada, around 21% of Canadians reported attendance at religious activities at least monthly (Government of Canada, 2021). Out of this portion, we assume 10% would be strictly opposed to seeking medical care via CRISPR-Cas therapies, totalling at 2.1% of Canadians. While

strong religious affiliations are becoming increasingly rare in developed nations (Stastna, 2019; Government of Canada, 2021), if we hold the 2.1% estimate, this equates to ~1 million Canadians by 2040 (Government of Canada, 2022). Assuming an equal spread of pancreatic cancer cases among religious and non-religious individuals, this equates to ~210 individuals who will not seek our treatment in Canada due to religious concerns. Out of this 210, ~40 would have been able to afford treatment.

Using a similar logical progression, we can calculate the number in the US; 42% of Americans reported regular attendance at religious services (Ayvaci, 2016), of which 10% we assume would be opposed to CRISPR-Cas therapies for 4.2% of the population, or ~16 million Americans (Colby, 2015). This equates to ~4000 individuals, of which ~800 could have afforded treatment.

One final consideration is the potential for competing genetic-level pancreatic cancer therapeutics. While today, there are no direct competitors to this niche market given the novelty of the product and the specific consumer segment targeted, it may be the case that competition develops in the next 20 years during *PROTEUS*' validation. This would limit the number of serviceable obtainable consumers. Given the current lack of competition, we simplified the SOM calculation by assuming no early-stage competition in the specific niche carved out by *PROTEUS*. Following approval, it is likely that generic replication of the product would be conducted by competing firms in the oncology therapeutic space, though this is a long-term concern, and not immediately pressing to *PROTEUS*' success.

Given the factors considered here, we estimate that the serviceable obtainable population would be approximately 19 600 individuals for a SOM of ~\$20 billion.

4.3.4 Summary of Analysis

Below is a figure of the estimated SOM, SAM, TAM, and PAM for *PROTEUS*.

Serviceable Obtainable Market	
~19 600 patients	~\$20 billion CAD
Serviceable Available Market	
~100 000 patients	~\$100 billion CAD
Total Adressable Market	
~815 000 patients	~\$815 billion CAD
Potential Available Market	
~1.304 million patients	~\$1.3 trillion CAD

Figure 9. Estimated market size for *PROTEUS* using several levels of filtering.

Pancreatic cancer incidence rates are predicted to continue climbing far into the 21st century, representing substantial potential for growth in this niche market. Due to the requirements to make long-term predictions for *PROTEUS*, many of the aforementioned figures are subject to substantial change.

4.4 Competitive Landscape Analysis and SWOT Analysis

4.4.1 The Current Space and Competition

As of now, there are four FDA-approved drugs for treatment of pancreatic cancer (Grand View Research, n.d.): Abraxane®, (albumin-bound paclitaxel); Gemzar® (gemcitabine); 5-FU (fluorouracil); and onivyde® (irinotecan liposome injection). Combination therapies are often used for metastatic cases as well. While not in direct competition with *PROTEUS* – given that these are small molecule therapeutics that indiscriminately prevent cells from dividing (harming both healthy and cancer cells) – these drugs constitute the primary competition for our novel genetic-level therapeutic. Luckily, given the novelty of our approach, there are no therapeutics on the market that use a similar tactic against pancreatic cancers, providing *PROTEUS* an opportunity to fill a current gap in the pancreatic cancer treatment space.

The traditional initial treatment for PDAC is combination therapy using paclitaxel and gemcitabine. If the disease progresses after gemcitabine-based therapies, fluorouracil and irinotecan liposome injection combination therapies are often subsequently used. Unfortunately, these treatment modalities are potentially cytotoxic and indiscriminately target all dividing cells (both cancerous and healthy). For example, severe or life-threatening neutropenia (neutrophil deficiency) occurred in 20% of patients receiving irinotecan/5-FU/leucovorin (LV) combination therapy, one of the most common combination therapies following gemcitabine-based treatment (DailyMed, 2023). Furthermore, with current pancreatic cancer long-term survival rates still sitting at a low 12%, these treatments are not incredibly fruitful, providing a large gap in the market for safe, effective therapies.

The pancreatic cancer drug treatment field is constantly changing, though highly effective drugs have yet to be brought to market. Regardless, some of the key recent developments in the industry are reported below:

- In January 2022, the FDA gave orphan drug status to multitargeted T-cell therapy. This therapy has shown the potential to elicit sustained responses in individuals with advanced or metastatic pancreatic cancer when combined with chemotherapy as a first-line therapeutic option (Grand View Research, n.d.)
- In September 2021, Roche partnered with NovoCure to develop Tumor Treating Fields, electric fields that disrupt cancer cell division, in combination with atezolizumab for patients with metastatic PDAC (Novocure, 2021; Fortune Business Insights, 2021)
- In July 2021, Novartis received orphan drug status for NIS793, a novel antibody specific for Transforming Growth Factor Beta (TGFβ), a cytokine that has an important role in metastatic PDAC and other solid tumors (Novartis, 2021)

- In June 2020, Ipsen pharma received fast-track designation by the FDA for the use of irinotecan in combination with 5-FU/LV and oxaliplatin, known altogether as NALIRIFOX, for the treatment of patients suffering from locally advanced and metastatic PDAC (Fortune Business Insights, 2021)
- In December 2019, AstraZeneca and Merck received FDA approval for Lynparza (olaparib) for its use in the U.S. for the maintenance treatment of patients suffering from pancreatic cancer (Fortune Business Insights, 2021)

Pancreatic cancer treatment expenses are already high; chemoradiation often costs upwards of \$10 000 USD per month, depending on the stage (I through IV) and phase (initial, continuing, or terminal) that the patient is in (Tramontano, 2019). Stage indicates the extent of tumour development (lower stages correspond to less metastasis and infiltration into surrounding tissue), while phase describes the patient's general health progression (with 'initial' being the most healthy and 'terminal' being terminally ill). The table below offers insight into average monthly expenses by phase, stage, and treatment modality:

					Cancer-attributable cost predictors		
	N (%)	Total monthly cost (95% CI)	Monthly patient- liability cost (95% CI)	Monthly cancer- attributable cost (95% CI)	Year	Age	Year_Age
Initial Phase							
Stage I							
Best supportive care	229 (20.5)	\$5020 (\$4189–\$5850)	\$407 (\$325–\$489)	\$3544 (\$2634–\$5851)		+	
Surgery	212 (19.0)	\$3452 (\$2562–\$4343)	\$466 (\$328–\$604)	\$2232 (\$1278–\$3185)			
Chemoradiation	218 (19.5)	\$10,987 (\$9717–\$12,257)	\$1598 (\$1492–\$1704)	\$9860 (\$8554–\$11,167)	+		
Surgery, chemo, and radiation	133 (11.9)	\$7240 (\$5898–\$8583)	\$1228 (\$1127–\$1329)	\$6171 (\$4751–\$7591)	+		
Stage II							
Best supportive care	662 (14.8)	\$5662 (\$5128–\$6195)	\$431 (\$378–\$483)	\$4265 (\$3670–\$4861)			
Surgery	546 (12.2)	\$5305 (\$4543–\$6066)	\$687 (\$538–\$836)	\$4085 (\$3293–\$4877)			
Chemoradiation	851 (19.0)	\$10,618 (\$10,132–\$11,104)	\$1583 (\$1535–\$1630)	\$9560 (\$9058–\$10,063)	+	–	
Surgery, chemo, and radiation	1031 (23.0)	\$7450 (\$7169–\$7731)	\$1337 (\$1280–\$1394)	\$6342 (\$6017–\$6666)	+	–	
Stage III							
Best supportive care	159 (14.8)	\$7552 (\$5919–\$9186)	\$517 (\$368–\$666)	\$6503 (\$4834–\$8172)			
Radiation	130 (12.1)	\$9054 (\$7701–\$10,407)	\$1154 (\$973–\$1335)	\$8155 (\$6790–\$9519)		–	
Chemotherapy	234 (21.8)	\$8884 (\$7923–\$9843)	\$1294 (\$1171–\$1416)	\$7645 (\$6702–\$8588)			
Chemoradiation	445 (41.5)	\$11,183 (\$10,495–\$11,872)	\$1695 (\$1623–\$1768)	\$10,046 (\$9328–\$10,766)	+		
Stage IV							
Best supportive care	515 (14.2)	\$6182 (\$5526–\$6837)	\$513 (\$438–\$587)	\$4840 (\$4169–\$5511)			
Chemotherapy	1282 (35.3)	\$9070 (\$8712–\$9429)	\$1377 (\$1336–\$1419)	\$7766 (\$7369–\$8163)	+		
Chemoradiation	1455 (40.1)	\$8760 (\$8434–\$9098)	\$1695 (\$1623–\$1768)	\$7819 (\$7460–\$8194)			
Continuing Phase							
Stage I							
Best supportive care	160 (15.9)	\$4204 (\$3284–\$5123)	\$398 (\$314–\$483)	\$3075 (\$2110–\$4041)			
Surgery	281 (27.9)	\$2549 (\$2177–\$2921)	\$361 (\$300–\$423)	\$1503 (\$1079–\$1926)			
Chemoradiation	135 (13.4)	\$7007 (\$5945–\$8069)	\$878 (\$762–\$993)	\$6053 (\$4958–\$7149)			
Surgery, chemo, and radiation	169 (16.8)	\$3181 (\$2669–\$3693)	\$475 (\$392–\$557)	\$2187 (\$1602–\$2773)			
Stage II							
Best supportive care	308 (8.8)	\$4178 (\$3280–\$5074)	\$364 (\$304–\$423)	\$2803 (\$1820–\$3786)			
Surgery	500 (14.3)	\$3226 (\$2810–\$3641)	\$416 (\$368–\$464)	\$1659 (\$1106–\$2212)	+	+	–
Surgery and chemotherapy	421 (12.0)	\$4222 (\$3804–\$4641)	\$644 (\$588–\$701)	\$3141 (\$2437–\$3845)	+	+	–
Chemoradiation	515 (14.7)	\$6946 (\$6141–\$7750)	\$980 (\$874–\$1086)	\$5806 (\$4980–\$6632)	+		
Surgery, chemo, and radiation	1107 (31.6)	\$3898 (\$3632–\$4163)	\$578 (\$411–\$610)	\$2990 (\$2708–\$3273)	+	+	–
Stage III							
Best supportive care	60 (9.5)	\$5026 (\$2939–\$7112)	\$620 (\$321–\$919)	\$3675 (\$1478–\$5870)		+	
Radiation	74 (11.7)	\$7454 (\$4772–\$10,135)	\$702 (\$471–\$933)	\$6630 (\$3868–\$9393)			
Chemotherapy	124 (19.7)	\$6612 (\$5730–\$7493)	\$954 (\$835–\$1073)	\$5340 (\$4269–\$6411)			
Chemoradiation	290 (46.1)	\$6672 (\$5642–\$7703)	\$918 (\$824–\$1011)	\$5494 (\$4430–\$6561)		–	
Stage IV							
Best supportive care	111 (8.1)	\$3186 (\$2349–\$4023)	\$352 (\$254–\$450)	\$1811 (\$971–\$2652)			
Chemotherapy	464 (33.9)	\$7019 (\$6544–\$7493)	\$1087 (\$1025–\$1150)	\$5651 (\$5112–\$6189)	+	+	–
Chemoradiation	627 (45.8)	\$7657 (\$7224–\$8091)	\$1179 (\$1124–\$1235)	\$6500 (\$6048–\$6951)	–		

* A positive (+) symbol indicates that the covariate in the regression model has a parameter estimate greater than 0, while a negative (–) symbol indicates that the parameter estimate is less than 0. Except for best supportive care costs, treatment modality costs are not shown if less than 10% of patients within a stage received that treatment.
CI = Confidence Interval.

Figure 10. Mean monthly cost estimates by stage at diagnosis by treatment modality, and significant predictors of cancer-attributable costs during the initial and continuing phases (Tramontano, 2019).

Based on these reports, we can develop a cost comparison and treatment modality analysis to determine the market potential for *PROTEUS*.

	Chemotherapies	Radiation	Surgery	<i>PROTEUS</i>
Cost (USD): Stage I	\$10987 / month (chemoradiation)		\$3452 / month	TBD, one-time dose
Cost (USD): Stage II	\$10618 / month (chemoradiation)		\$5305 / month	TBD, one-time dose
Cost (USD): Stage III	\$8884 / month	\$9054 / month	N/A	TBD, one-time dose
Cost (USD): Stage IV	\$9070 / month	\$8760 / month (chemorad.)	N/A	TBD, one-time dose
Manufacturers / Personnel	Paclitaxel: BMS, others Gemcitabine: Eli Lilly, others Fluorouracil: many Irinotecan: originally Pharmacia & Upjohn (acquired by Pfizer), now sold in several formulations	Radiation oncologist, dosimetrist, radiation therapist	Surgical oncologists	McGill iGEM
Expected Side Effects	Low blood cell counts, anemia, nausea & vomiting, loss of appetite, taste changes, sore throat, diarrhea, constipation, flu-like symptoms, muscle and joint pain, fatigue, skin problems, hand-foot syndrome, peripheral nerve damage, hair loss	Diarrhea, fatigue, hair loss, nausea, vomiting, skin changes, urinary and bladder problems	Bleeding, blood clots, tissue / organ damage, infections	Hyperinflammation, others TBD
Expected Efficacy	Low	Low	Variable*	High
Strengths / Weaknesses	Median cost, prolongs life; many side effects, low efficacy	Median cost, prolongs life; side effects, low efficacy	Lower cost, effective early on; ineffective after stage II	High expected efficacy, minimal expected side effects; costly

Figure 11. Treatment modality analysis for pancreatic cancer treatment modalities (Tramontano, 2019; ACS, 2019; National Cancer Institute, 2022; Canadian Cancer Society, 2018).

Based on this analysis, *PROTEUS* has substantial promise in terms of establishing a unique market position as a high efficacy therapeutic against pancreatic cancer. Current treatment modalities lack curative capacities in general, besides perhaps Stage I surgical resection. The difficulty of the latter is that pancreatic cancer is rarely diagnosed in Stage I. Regardless, *PROTEUS* does have the weakness of being expected to be a more costly procedure. However, given the fact that a pancreatic cancer ‘cure’ has truly yet to be found, *PROTEUS* has a unique selling point that distinguishes it from competing treatment modalities offered by similar companies in the space.

In order to dig out a market share in the pancreatic cancer therapeutic space, McGill iGEM will need to put substantial effort into marketing *PROTEUS* as curative (assuming this is validated through clinical trials), compared to the palliative or low-efficacy curative treatments provided by competitors. Most existing drug competitors are licensed by big pharmaceutical companies, such as Eli Lilly (with their gemcitabine, Gemzar®), Roche (with their tablet form erlotinib, Tarceva®, which is used in combination with pancreatic cancer chemotherapies), Pfizer (with their irinotecan, Camptosar®), Celgene (with their albumin-bound paclitaxel, Abraxane®), Ipsen (with their irinotecan, Onivyde®), and others. While we will be in competition with these companies, we may seek partnerships with some down the line as wholesalers of our product to reach a wider target audience internationally. Establishing a market share while maintaining relations with these companies will be an essential balancing act for McGill iGEM in the coming years.

4.4.2 SWOT Analysis

Our team developed a SWOT (Strengths, Weaknesses, Opportunities, and Threats) Analysis for *PROTEUS* to delineate our key opportunities to succeed, as well as some concerns we should keep in mind to prevent failure of the venture. Below is our SWOT canvas.

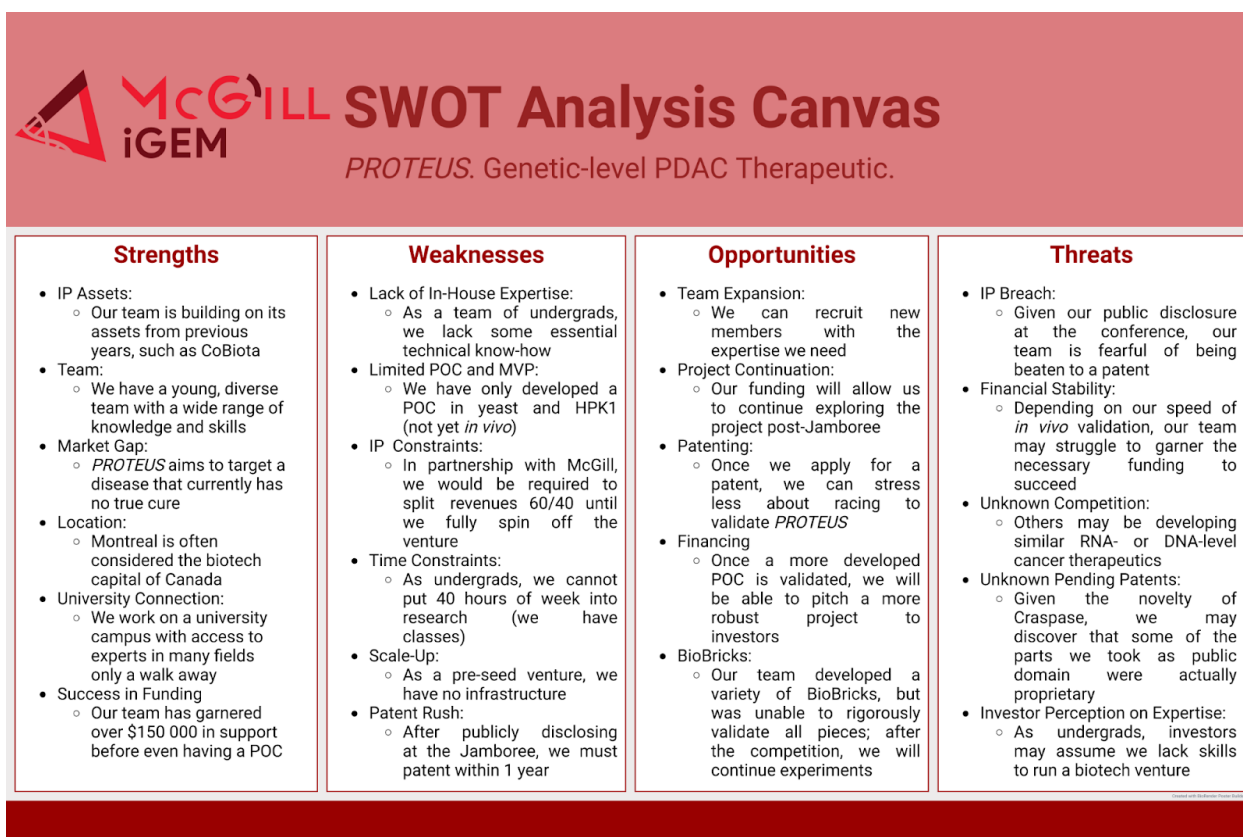


Figure 12. McGill iGEM SWOT Analysis as related to our *PROTEUS* project.

4.6 Macro-environment and Market Trends

In today's rapidly evolving global landscape, biotech startups operate against a backdrop of significant macroeconomic volatility. Characterized by fluctuating financial markets, shifting regulatory environments, and the unprecedented challenges of a globalized economy, the macro environment presents both opportunities and threats for emerging biotech ventures. These external forces, often beyond the control of individual enterprises, have profound implications on investment, research priorities, and commercialization strategies. We aim to provide a comprehensive overview of the current macroeconomic conditions, shedding light on the key factors that biotech startups must navigate to thrive amidst uncertainty.

In 2021, there was a significant boost in revenue for biotech, primarily driven by the market for COVID-19 vaccines, therapies, and testing. This growth stabilized in 2022 with public biotech companies in the US and Europe amassing revenues of \$215 billion USD, a 1% decrease from 2021 (Ernst & Young, 2023). Major biotech companies like BioNTech and Gilead Sciences saw revenue drops due to reduced demand for their COVID-19 related products. Despite this, the biotech industry's fundamental growth remained stable, and excluding the COVID-19 product revenue impact, the industry's growth was 3.7% in 2022, a small decline from 5.2% in 2021 (Ernst & Young, 2023).

One encouraging sign is the biopharma industry's continued innovation. Even though FDA approvals declined in 2022 due to staffing issues, there is a healthy pipeline of over 20 000 active drug candidates globally. New therapeutic modalities like gene therapies, mRNA platforms, radiopharmaceuticals, and antibody-drug conjugates are emerging. An example of industry confidence in these new platforms is Pfizer's acquisition of ADC specialist Seagen for US\$43 billion in 2023 (Mullard, 2023).

However, a concern for the biotech sector is financing. After a spike in funding in 2020 and 2021, where nearly US\$240 billion was raised, there was a 54% decline in 2022, with only US\$54.6 billion raised. This is the lowest since 2016, but it aligns with pre-pandemic expectations (Ernst & Young, 2023). The industry will now have to adapt to a post-pandemic financing landscape after two years of unusual conditions.

In 2022, the biotech sector underwent significant financial transformations. One of the prominent shifts was in debt financing, which experienced a 10% decline, largely due to rising interest rates (Ernst & Young, 2023). The biotech Initial Public Offering (IPO) market took a substantial hit, plummeting by 93% relative to the previous year. In 2021, biotechs had reaped about US\$21 billion in IPO financing. However, with the correction in biotech valuations, those companies that became public recently faced severe challenges. A staggering 91% of the firms that went public between 2020 and 2021 witnessed their market value nosedive, averaging a decline of over 50% (Ernst & Young, 2023).

Financial sustainability became a pressing concern, with 55% of emerging biotechs indicating they lacked the cash reserves to keep them afloat beyond two years (Ernst & Young, 2023). This figure was alarmingly higher than that of 2021. Moreover, 29% had a mere year's worth of cash left in their reserves. Venture funding, another crucial aspect of the industry, also declined by 29% in 2022. Despite this drop, the total, which amounted to US\$18.9 billion, surpassed the average of the past decade. A notable outlier was the high-profile startup, Altos Labs, which secured a commitment of US\$3 billion (Ernst & Young, 2023). Nonetheless, there was a palpable shift in investor sentiment, with many leaning towards ventures that presented fewer risks.

The Mergers and Acquisitions (M&A) landscape also morphed, becoming an essential exit strategy for biotechs in light of the constrained capital environment. However, the inclination for vast deals in 2022 waned compared to previous years. Remarkably, 56% of the deals in that year witnessed larger pharma entities acquiring biotech firms (Ernst & Young, 2023). The start of 2023 maintained this subdued M&A trend. The US regulatory environment added another layer of complexity, especially concerning antitrust regulations. A case in point was the US Federal Trade Commission's endeavor to halt Amgen's acquisition of Horizon Therapeutics.

Interestingly, instead of opting for outright acquisitions, the industry displayed a preference for establishing alliances and partnerships. This trend was evident in 2022 when life sciences firms embarked on alliance deals potentially worth US\$132.1 billion. In essence, the

challenges faced in 2022 reshaped the financial and regulatory landscape of the biotech industry, ushering in new practices and strategies.

Biotech companies are also increasingly confined by political and geographical challenges. During 2022, the biotech sector in China was experiencing a boom, with returning Chinese entrepreneurs benefiting from favorable policies and abundant funding. For instance, Suzhou Abogen Biosciences, spearheaded by an ex-Moderna scientist, raised over \$1 billion in a year, with significant investments pouring into Chinese healthcare, reaching \$38 billion in 2021 (Mak, 2022). International collaborations signify China's active participation in the biotech scene. Over 60% of the 500+ biotech collaborations in China in 2021 were international (Mak, 2022). However, this positive integration contrasts starkly with the financial disconnection happening in China's tech sector due to diplomatic conflicts. Market anxieties have intensified; a mere rumor of possible U.S. sanctions against the Chinese biotech sector led to significant market value losses for companies like Wuxi BioLogics and BeiGene (Mak, 2022). The sector's stability remains in question as companies face potential delisting from New York stock exchanges and restrictive foreign investment policies from Beijing. The industry seems poised for a geopolitical challenge soon.

Research-stage biotech companies, particularly those in the preclinical phase, face an array of challenges in the current financial landscape. Driven by the COVID-19 pandemic, investor interest in biotech startups surged, leading to record VC fundraising in 2020 and 2021. However, as of 2022, the rise in interest rates coupled with a subdued deal-making environment has caused a significant slowdown in investments. Furthermore, with late-stage investments being hit harder due to crossover investors finding better value opportunities in the public market, startup valuations are expected to decline. While a downturn in valuations might be advantageous for Big Pharma seeking acquisitions, a substantial uptick in M&A activity remains elusive. As a solution, many biotech startups and large pharmaceutical companies are increasingly exploring partnership deals, which can provide essential funding for specific trials. Preclinical biotechs, especially those yet to secure funding, need to strategically navigate this shifting terrain, exploring diverse financing options, and potentially leveraging partnerships to ensure their longevity and success. These market tendencies will play a key role in dictating McGill iGEM's future partnerships and collaborations in the biotech and Big Pharma ecosystems.

Based on these market trends, our team has developed a PEST (Political, Economic, Social, and Technological) analysis to assess the major external factors in the world economy that might affect our organization's initial commercial launch. In this analysis, we also discuss environmental and legal considerations for a more holistic analysis of the international market trends on *PROTEUS*.

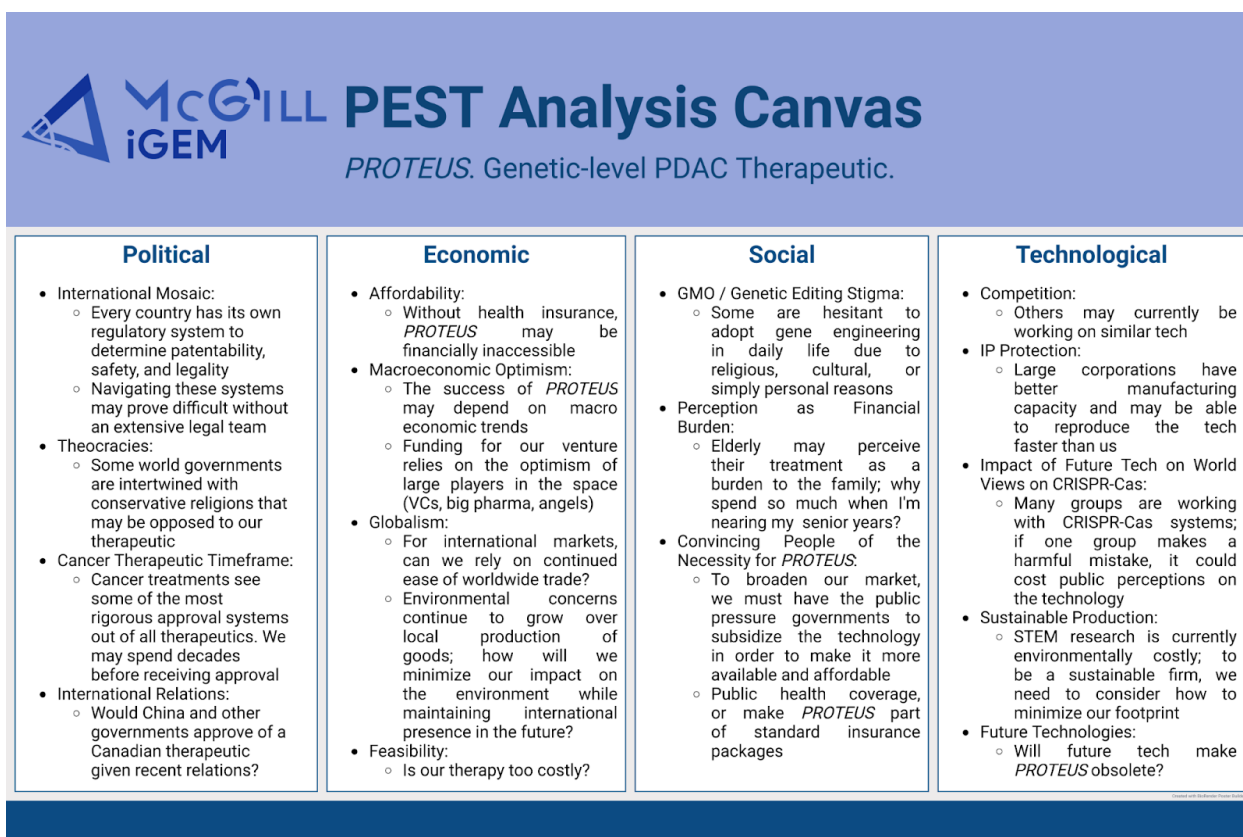


Figure 13. McGill iGEM PEST Analysis as related to our *PROTEUS* project.

5. Product Description

5.1 Project overview

PROTEUS is a CRISPR mediated pancreatic cancer therapeutic. Specifically, it makes use of Craspase, a novel CRISPR RNA-guided, RNA-activated protease, to target the KRAS G12V mutation. KRAS mutations are present in nearly 100% of all pancreatic ductal adenocarcinoma (PDAC) cases (Waters, 2018). After detecting the mutation in a cell, the system induces a pyroptotic suicide event to kill the cell. The modularity of this system allows for patient-specific treatment, making it applicable for targeting a wide variety of cancer manifestations.

Pyroptosis is an inflammatory programmed cell death mechanism that results in membrane perforation, cell swelling, and the release of cellular content (Wang et al., 2021). This process is mediated by pro-inflammatory molecules called inflammasomes that are active during the innate immune response (Bertheloot et al, 2021; Tan et al., 2021). A family of naturally-occurring proteins called gasdermins are used alongside inflammasomes to induce pyroptosis among infected cells. These proteins have a conserved structure, consisting of a cytotoxic N-terminal domain and an autoinhibitory C-terminal domain that are connected via a flexible linker region (Broz et al., 2020). When the linker region is cleaved by a family of

endonucleases called caspases, the N-terminal domain is activated and oligomerizes to form large pores in the membrane that induce swelling and ultimately membrane rupture (Kovacs & Miao, 2017).

The primary mechanism of *PROTEUS* is centered around the cleavage and subsequent activation of the pore-forming N-terminal domain of gasdermins that induce pyroptosis in tumor cells. Rather than cleavage with a caspase, *PROTEUS* relies on a caspase-like protease known as TPR-CHAT that associates with the type III-E effector gRAMP protein complex to specifically cleave an ancillary protein, Csx30 (Wang et al., 2022). Upon the binding of gRAMP guide RNA (gRNA) to its target RNA, TPR-CHAT undergoes a conformational change that exposes its active site protease domains, which can then target and cleave the endogenous protein Csx30 (Willson, 2022).

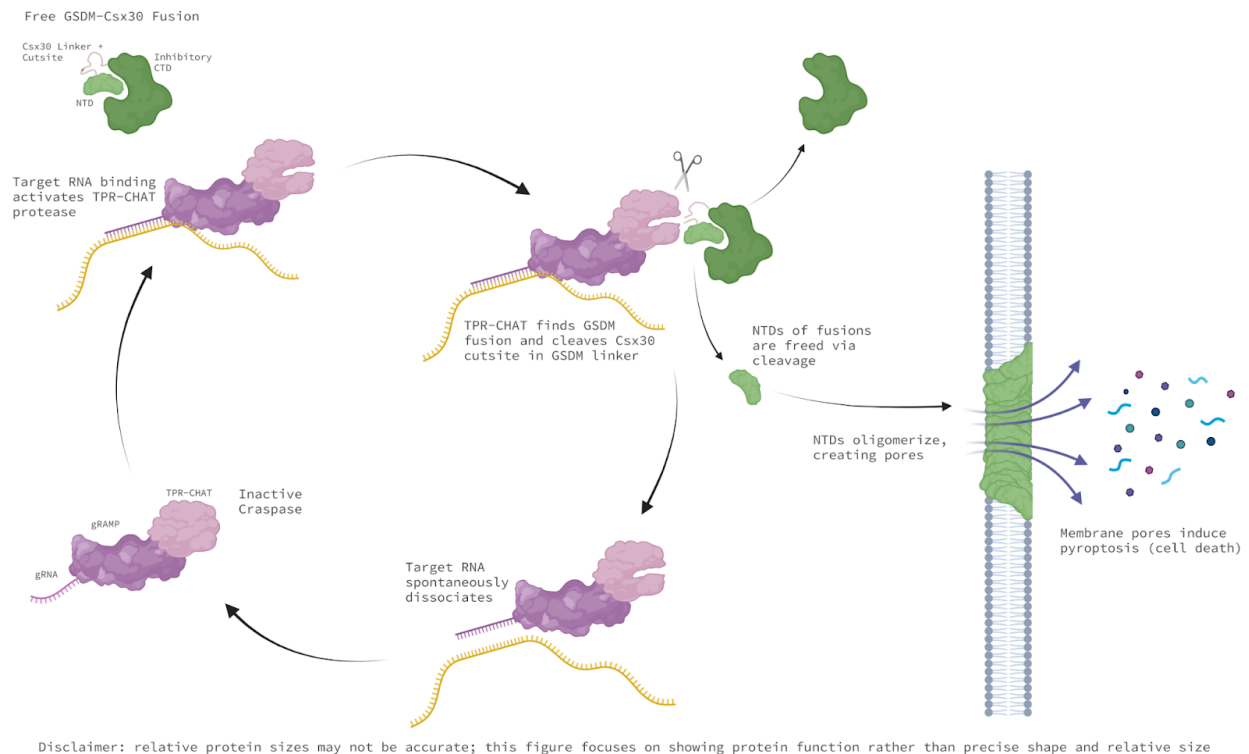


Figure 14. Craspase binding its target oncogenic mRNA, cleaving a GSDM-Csx30 fusion protein, creating membrane pores in the cell which induce pyroptosis.

McGill iGEM has designed a series of Gasdermin D (GSDMD) and Gasdermin E (GSDME) fusion proteins, inserting an optimally truncated version of the Csx30 protein into the linker region of both GSDMD and E. Insertion of Csx30 in this region was chosen given that the linker is intrinsically disordered and will not interfere with the native structure and function of the GSDMs. When TPR-CHAT is activated and cleaves Csx30 in the GSDM linker region, the N-terminal domain ruptures the cell membrane, thus killing the tumor cell. By this design, a personalized treatment approach is developed as we can design the gRNA to be perfectly

complementary to a patients' mutated KRAS G12V oncogenic transcripts. Using a high-specificity gRNA, *PROTEUS* reduces the likelihood of negative off-target effects given that pyroptosis is only induced upon Csx30 cleavage initiated through patient-specific gRNA binding with the mutant KRAS G12V. Therefore, healthy cells lacking the mutation are unaffected while tumor cells are effectively eliminated.

gRNAs can easily be modified to target other oncogenic mRNA sequences. Thus, beyond solely targeting KRAS G12V-induced PDAC, our system could target any oncogene expressed by a patient. First, we would perform whole exome sequencing on a patient, identifying the most highly expressed cancer transcripts in the process. Then, we design optimally efficient gRNAs to bind to these targets using the exact same strategy as our proof of concept with KRAS G12V. The modularity of the system allows *PROTEUS* to be applied broadly in many different manifestations of cancer.

After optimization and incorporation of the gRNA that binds to a desired target mRNA, the *PROTEUS* therapeutic will be delivered to patients via intravenous injection using a gutless adenovirus (GLAd) as the delivery method for the system's expression vectors. In this way, human cells will become infected with the necessary genetic information to constitutively produce the Craspase therapeutic and induce cell death solely in cancerous cells.

The figure below shows our current plan for in-patient treatment post-patenting and post-approval by the required regulatory agencies.

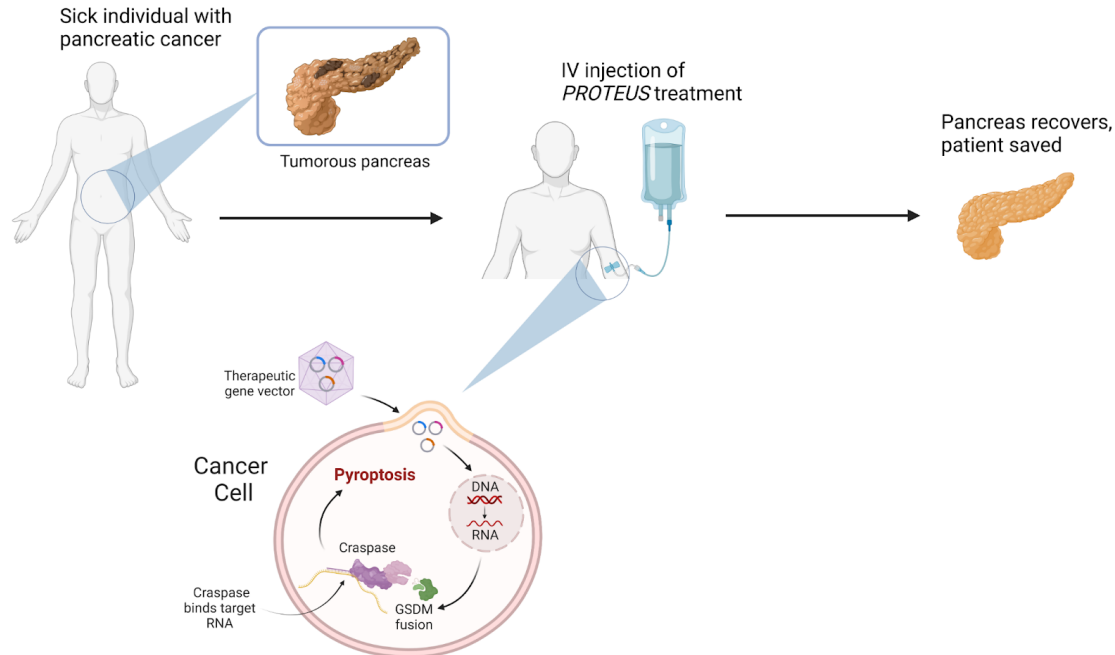


Figure 15. Overview of proposed in vivo treatment; a patient with pancreatic cancer receives intravenous injection of the therapeutic. On the cellular level, this involves a viral vector infecting each cancer cell, causing the cell to express Craspase and our proprietary gasdermin fusion protein. If the gRNA binds to oncogenic mRNA in the cell, the gasdermin fusion is cleaved, killing the cell. If the therapeutic is effective, this allows the patient to recover.

Currently, *PROTEUS* has been successfully validated in both yeast and mammalian cell lines. Compared to existing methods, this therapeutic system would offer both preventative and curative treatment capability; those at high risk of developing pancreatic cancer may seek this treatment in advance, and those with early-stage pancreatic cancer can similarly be treated by eliminating the cancer before it metastasizes. Later stage pancreatic cancer cases would first undergo surgical resection of major tumours before treatment with *PROTEUS*. This is to avoid the potential for over-inflammation from pyroptosis. We would recommend *PROTEUS* as a combination therapy with another established treatment modality, such as immunotherapy and/or surgical resection. We are not recommending the use of *PROTEUS* as a standalone treatment in a late-stage tumour given that over-activating pyroptosis may result in harmful hyperinflammatory responses (Bergsbaken et al., 2009).

5.2 Technology behind *PROTEUS*

5.2.1 External Technology

Craspase

Craspase (CRISPR-guided caspase) is a CRISPR-Cas type III-E RNA-targeting effector complex gRAMP/Cas7-11 that is associated with the caspase-like protein, TPR-CHAT. Hu et. al. 2022 elucidated the system's cleaving mechanism through cryo-electron microscopy (Hu et al., 2022).

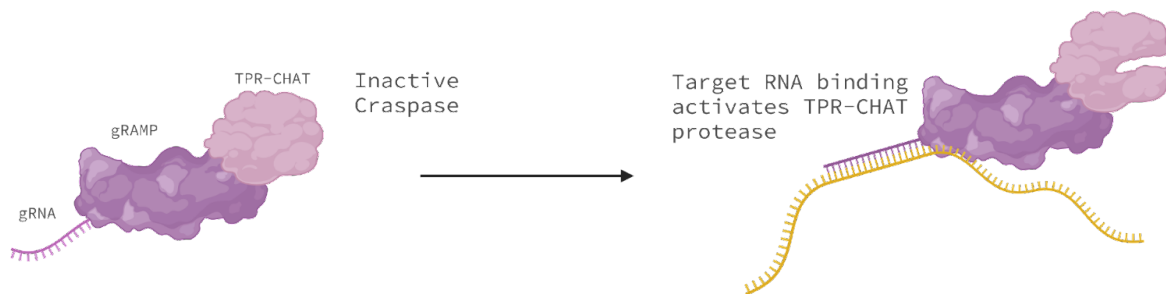


Figure 16. Inactive Craspase with its two complexed proteins, gRAMP and TPR-CHAT, and unhybridized gRNA; activated Craspase, where the gRNA hybridizes with its target mRNA transcript, activating the protease activity of TPR-CHAT through a conformational change.

Since the original publication, co-authors Sam P.B. Van Beljouw and Stan J. J. Brouns have gone on to patent the Craspase system. It is filed under WO 2022/255865 A1 and has been granted protection internationally, including in Canada, where we plan to launch our minimal viable product in a proof of business (van Beljouw, 2021). As we envision *PROTEUS* to be a competitive therapeutic tool, McGill iGEM will be seeking a patent licensing agreement to credit the Craspase system as the baseline technology.

Next-Generation Sequencing

PROTEUS is designed to be a precision medicine tool, tailored according to a patient's genomic data. With next-generation sequencing (NGS) becoming widely adopted within clinical practices, the design and optimization of gRNAs specific to a patient's cancer cells is instrumental in the implementation of the *PROTEUS* system. Next-generation sequencing is a high-throughput technology that can perform massive parallel sequencing of multiple DNA fragments (Behjati & Tarpey, 2013). It is an improvement on Sanger sequencing, and is able to capture a broader spectrum of mutations, including copy number variants and structural variants, which previously required additional assays to be identified (Behjati & Tarpey, 2013). NGS is used to perform either whole genome sequencing (WGS) or whole exome sequencing (WES), in which the non-coding regions of the genome are excluded, allowing for the identification of disease-causing variants.

Cancer mutations are often categorized as gain-of-function or loss-of-function mutations. True to their name, these mutations can alter a protein's native function, leading to many of the phenotypic consequences that surround cancer. For this reason, *PROTEUS* will primarily rely on patient data retrieved from WES. Protein-coding regions, or exons, account for a small fraction of the entire genome (~2%); however, they account for roughly 85% of known disease-related variants (Illumina, 2015). This makes WES a much more cost-effective and efficient method than WGS in the context of variant identification. Although a relatively new technology, there has been some progress on integrating WES into clinical practices. As such, we see NGS as an adjacent tool in the implementation of *PROTEUS*; genomic data will be collected and sent to clinics with the capacity to perform NGS. McGill iGEM is not claiming proprietorship of this technology, nor will it fall under the scope of our commercialization plan.

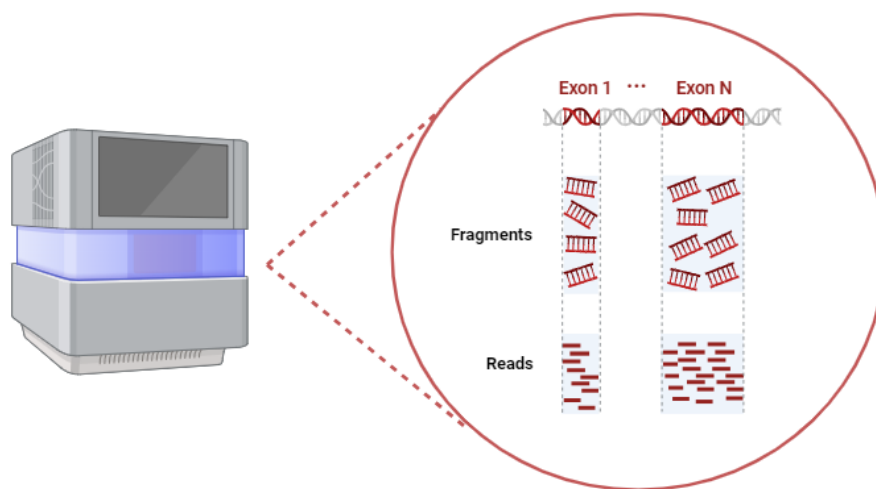


Figure 17. Visualization of whole exome sequencing. Exons are amplified and fragmented to create a DNA library prior to sequencing. The resulting sequences of inferred DNA base pairs are stored in a FASTQ file format to be used for further bioinformatics analyses.

However, it must be acknowledged that there are some barriers to using NGS in healthcare settings. Currently, WES is only available via referral through a physician or to patients participating in clinical trial studies (Yale Medicine, 2022). Within Canada, there also exist provincial guidelines that determine a patient's eligibility, and provincial funding for the clinical implementation of WES is limited (Weymann, 2022). Clinics performing WES will also be required to have the appropriate infrastructure to process and store this data (Behjati & Tarpey, 2013). Moving forward, as WES becomes more commonly adopted in clinics and the cost of sequencing decreases, we see a strong potential for WES to become a routine genetic test that will greatly complement the implementation of *PROTEUS*.

5.2.2 Internal Technology

Fusion Proteins

Underlying *PROTEUS* is a chimeric fusion protein. This complex is composed of the N-terminus and C-terminus domains of the GSDM proteins, connected by the native linker sequence which we have modified to enable cleavage from the Craspase complex. Csx30 is a 64 kDa protein and contains a binding domain recognized by TPR-CHAT. The Csx30 insertion can be truncated to this 19 kDa region, facilitating its insertion into the linker while also minimizing structural interference with the adjacent domains. We validated the recognition and cleavage of the modified linker through a series of specially designed GSDMD and GSDME fusion proteins. For each GSDM, we narrowed down four candidate regions along the linker sequence to determine the most optimal position for the Csx30-insertion:

- N-terminal end of linker
- C-terminal end of linker
- Native caspase cut site of GSDMs
- Centre of the linker, with an additional 15% deletion from either side of the insertion region

Additionally, in each construct where the native caspase cut site was not modified, the cut site was mutated via PCR to prevent spontaneous off-target cleavage by host caspases. In our proof-of-concept, several expression vectors were constructed to enable the cloning of these fusion proteins in yeast cells. FLAG tags were also added to the N-terminus of GSDMD fusions, or the C-terminus of E fusions, to enable detection and identification through immunological assays. Beyond our proof-of-concept, this also has implications in clinical settings, where physicians will be able to validate whether the fusion proteins were cut *in vivo*. The figure below demonstrates a general GSDM fusion protein with the Csx30 linker insertion.

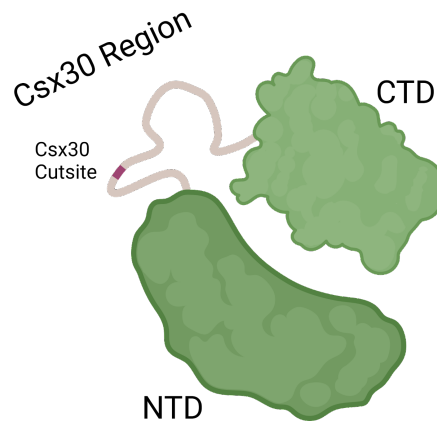


Figure 18. General gasdermin-Csx30 fusion protein with Csx30 cut site in the intrinsically disordered linker region. All fusions are tagged with a FLAG tag so that they can be detected via western blotting.

Currently, there exists no literature on the integration of the Craspase system into human gasdermins, making our fusion proteins (and the associated expression vectors designed for cloning) a completely novel design. As further detailed in Section 7.2, McGill iGEM plans to file a patent for these Csx30-integrated fusion proteins and their concerted use with the Craspase CRISPR-Cas system.

Bioinformatics Pipeline

Within NGS is the annotation and identification of genetic variants. Patients under the *PROTEUS* treatment will be required to provide DNA samples of their healthy and diseased tissue (hereafter dubbed normal and tumour) for sequencing. Reads are short sequences of inferred DNA base pairs retrieved from NGS. This information is often stored in the *FASTQ* file format, which has become the standard for many leading NGS platforms such as Illumina. From this point on, these raw reads will undergo several processes including (1) read mapping, (2) post-processing quality control, (3) variant calling, and (4) variant annotation and filtering. In short, these fragments are aligned against a human reference genome for comparison. Deviations present in the tumour sample and absent in the normal sample are flagged as candidate variants, which can then be cross-referenced with existing variation databases to gauge the degree of pathogenicity.

- ① Reads for normal and tumour samples retrieved from WES (*FASTQ*)

- ② Align sequence reads onto a reference genome (*BAM*)

- ③ Variant Calling (*VCF*)

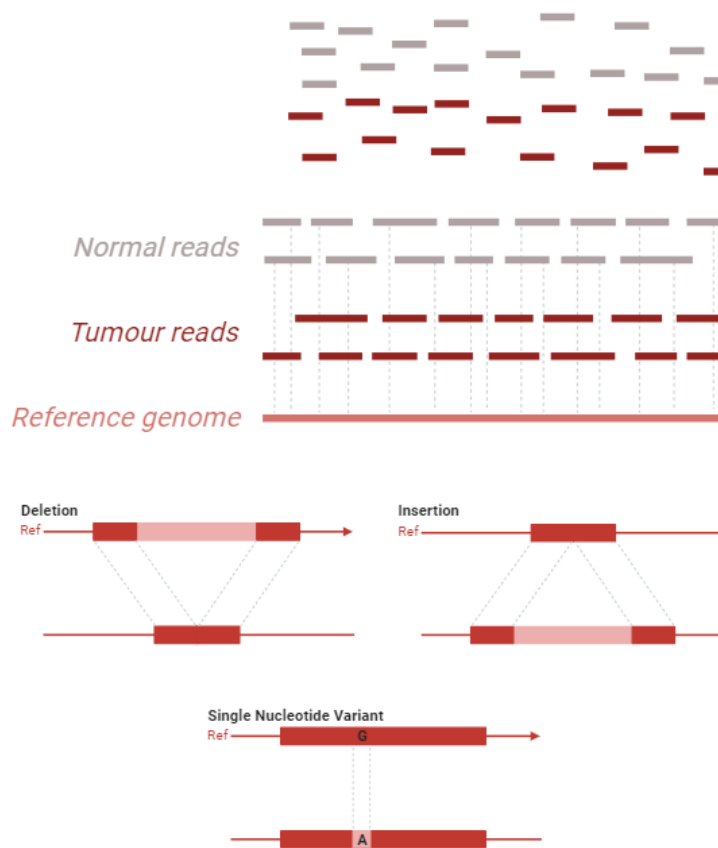


Figure 19. High-level visualization of the overarching bioinformatics pipeline.

To support our proof-of-concept, McGill iGEM developed a bioinformatics pipeline following recommendations from *GATK's Best Practices Workflow*. We retrieved *FASTQ* files of normal and tumour samples for PDAC from NCBI's Sequence Read Archive (SRA) public repository. For alignment, we used the Genome Reference Consortium Human 38 (GRCh38 or hg38), the most recent reference build released back in 2013. The table below outlines the main software we used and a brief description of their role.

Step	Software	Description
Read Mapping	<i>bwa: bwa-mem</i>	<i>bwa-mem</i> is one of three algorithms in the <i>bwa</i> package that maps DNA sequences against a reference genome. It uses the Burrow-Wheeler Transform algorithm to optimize space complexity.

Post-Processing Quality Control	<i>GATK: SortSam</i>	This tool sorts read alignments by coordinate or by query name. Downstream processes perform optimally on one or the other.
	<i>GATK: MarkDuplicates</i>	Duplicate reads often arise during PCR amplification and should not be mistakenly used as evidence for biological frequency. This program flags and removes them.
	<i>GATK: BaseRecalibrator</i>	The confidence of an assigned nucleotide is reported as a base quality score. This program builds an empirical model to predict the systematic errors that may arise during sequencing using a machine learning algorithm.
	<i>GATK: ApplyBQSR & AnalyzeCovariates</i>	These programs apply the results from <i>BaseRecalibrator</i> and plots the quality scores before and after.
Variant Calling	<i>MuTect2</i>	A variant calling software that specializes in detecting single nucleotide variants (SNVs) and short insertions and deletions (indels). It takes in a matched normal-tumour sample and outputs candidate variants in a <i>VariantCallFormat</i> (VCF) file.
	<i>GATK: FilterMutectCalls</i>	Raw VCF files are prone to report false-positive artifacts. This program was specifically developed to filter the VCF outputs from <i>Mutect2</i> . It applies a Bayesian model to predict the likelihood that a candidate variant

		is a true somatic mutation.
Variant Annotation and Filtering	<i>Variant Effect Predictor (VEP)</i>	A centralized annotation tool that aggregates variants across several public databases.
Other Miscellaneous Programs	<i>samtools</i>	A suite of programs that offers several utilities to manipulate high-throughput sequencing data.

Figure 20. Table of primary softwares used in our Dry Lab pipeline for identifying the most common oncogenic variants in pancreatic cancer patients following NGS.

Some clinical laboratories that perform NGS will have established their own bioinformatics pipeline, refined for their intended clinical use. Others will license a bioinformatics pipeline through a third party vendor and add customizations (Roy et al., 2018). In general, there is no single universal pipeline that can robustly handle all cases; appropriate variant callers must be chosen deliberately and rigorous validation must be performed to ensure clinical accuracy. Variant analysis in NGS pipelines is a vast undertaking. Different algorithms can produce divergent outcomes on the same sequencing data (Vestergaard et al., 2021) and requires interpretation by sufficiently qualified personnel (Shetty et al., 2021). As our pipeline was developed solely for the purposes of testing our proof-of-concept, we do not plan to license out this software to be used alongside *PROTEUS*. It was developed to specifically target single nucleotide variants (SNVs), corresponding to the KRAS G12V mutation in PDAC. The complexity of this task makes it difficult for our pipeline to accommodate the modularity of *PROTEUS*. As *PROTEUS* enters the market, we recommend the adoption of already existing bioinformatics pipelines that have been validated and approved for clinical use.

5.3 Unique Selling Proposition

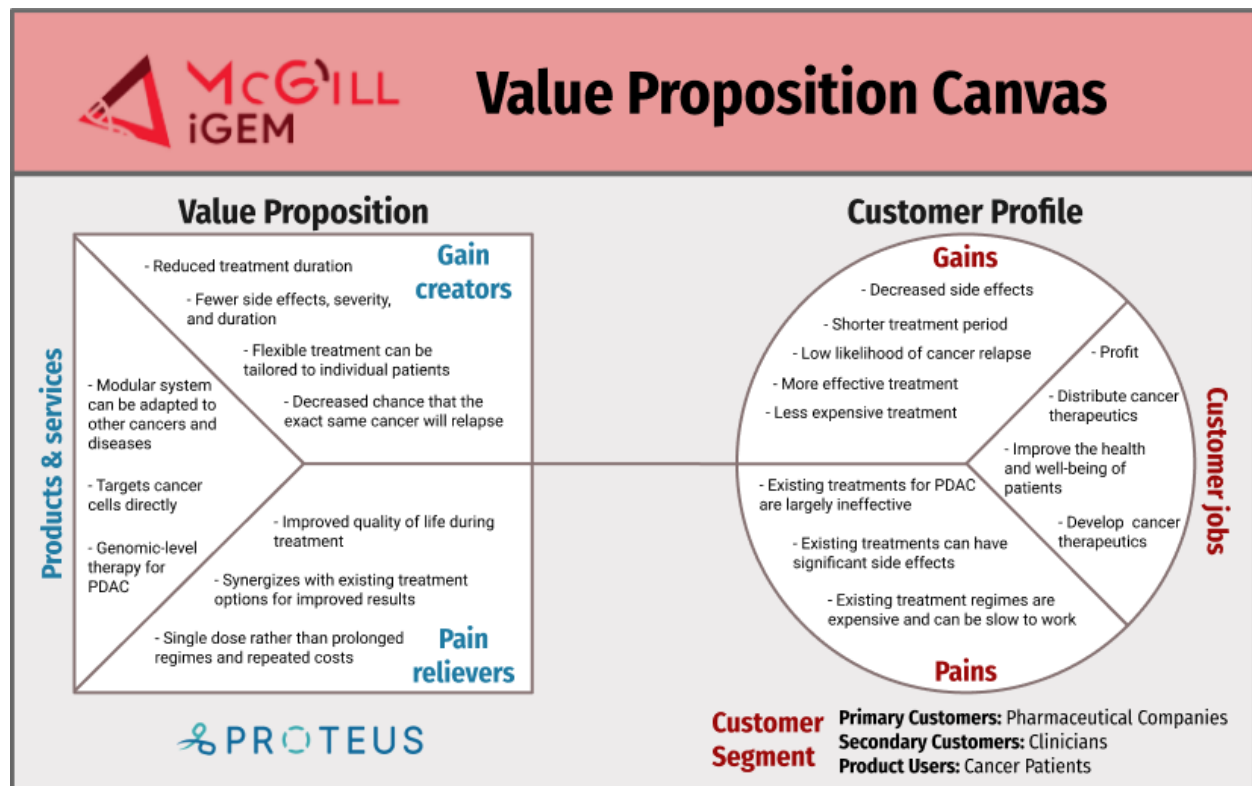


Figure 21. *PROTEUS*' unique selling proposition canvas, defining its key distinctive advantages against competitors, and potential opportunities for carving out a market share.

The most essential selling point of *PROTEUS* is that it targets cancer at a genetic level. This is something that other pancreatic cancer treatments fail to do, instead targeting *hallmarks* of cancer. For example, chemotherapy and radiation treatments both indiscriminately damage cells that are actively dividing (healthy and cancerous). Because cell division occurs much more frequently in cancerous cells compared to normal cells, it is true that these therapies mainly affect cancerous cells. When normal cells inevitably *do* divide, however, there is no way to distinguish them as healthy and avoid harming them. Side effects from the collateral damage of healthy cells typically persist throughout treatment and can last up to a few months after treatment ends (National Cancer Institute, 2022). *PROTEUS* mitigates this problem by taking advantage of the genomic difference between normal and tumour cells, targeting only those identified as cancerous and avoiding the aforementioned side effects. Since we are recommending *PROTEUS* to be used in combination with other treatment methods, possibly including chemotherapy, this does not guarantee the complete elimination of side effects. Instead, it would reduce the duration of a treatment regime, in turn reducing the duration of side effects and increasing the patient's quality of life.

Another quality of *PROTEUS* that sets it apart from currently existing treatments is that in addition to killing cancerous cells itself, it can also act similarly to an immunotherapy by

stimulating the body's immune response. The microenvironment of PDACs makes them a very difficult target for current therapies, especially immunotherapies; it is a hypoxic, hypo-vascularized, and immunosuppressive area, all characteristics that make it difficult for the body's natural defense system to fight off the cancer (Orth et al., 2019). The method that *PROTEUS* uses to kill cells– pyroptosis– results in the production of mature inflammatory cytokines, which boost the body's immune and inflammatory response (Bergsbaken, Fink, & Cookson, 2009). This turns the PDAC from a “cold” tumour, with minimal to no T cell infiltration, to one with a higher immune score (Ren et al., 2022). This change is beneficial regardless of the stage of treatment at which the patient begins using *PROTEUS*. If it happens early in the treatment regime, the tumour will be opened up and the effectiveness of other methods will increase. If it happens towards the end, the immune response will be able to clean up residual cancer cells that were not completely taken care of previously. This resultant synergy between therapeutics that comes from using *PROTEUS* is another key feature that increases its effectiveness and impact.

The value of our project has already started being recognized. Even before developing a working proof of concept, McGill iGEM received over \$150 000 in funding, including industry support, for our originality. In particular, we also received an award for our achievements in Engineering Inclusivity, Diversity, and Equity Advancement (E-IDEA). In addition to embodying these principles within our own team, they are also values that we have strived for in the development of *PROTEUS*. In line with the UN's 3rd and 10th Sustainable Development Goals, the promotion of good health and wellbeing and the reduction of inequality respectively, we have designed our treatment to be accessible to all people. Cancer does not discriminate, and the system we have created is flexible at the individual level, allowing it to be adapted to any patient regardless of their background.

PROTEUS addresses a particular gap in the landscape of pancreatic cancer treatments and its unique qualities will allow it to excel in that niche. The innovative nature of our project brings benefits that set it apart from any of the cancer therapeutics currently on the market, and these will only become more clear as its development progresses.

5.4 Intellectual Property Protection

As mentioned in Section 2.2, our team is planning to pursue patenting of our project; our future steps are to validate *in vivo* using mouse models given our success *in vitro* with yeast and mammalian cell lines.

Our team has worked extensively with McGill University Office of Innovation and Partnerships to prepare a Manuscript and Report of Invention demonstrating the novelty, utility and non-obviousness of our invention. Due to our support from McGill University, patent filing must be conducted through McGill's internal technology transfer office; our team is currently working with McGill's Technology Transfer Manager, Chris Corkery, to establish a patent for *PROTEUS*. As mentioned in Section 2.2, the benefits of filing with McGill include complete coverage of legal expenses associated with the upkeep of the patent, as well as support and

guidance on spinning off and scale-up (McGill Innovation and Partnerships, 2022; McGill Innovation and Partnerships, n.d.). As per McGill's Spinoff Handbook, McGill splits the revenues obtained from the license to the spinoff 60 to 40. That is, 60% of revenues go to the inventors and contributors named on the Report of Invention, and 40% goes to the university (McGill Innovation and Partnerships, 2022). Determining who is eligible to be considered an inventor or contributor was a difficult task for the team given the collaborative, open-source nature of iGEM, as well as our team's commitment to focusing on the group's success rather than individual recognition. This will be expanded upon in the Intellectual Property Strategy Section of the report.

The Technology Transfer Process

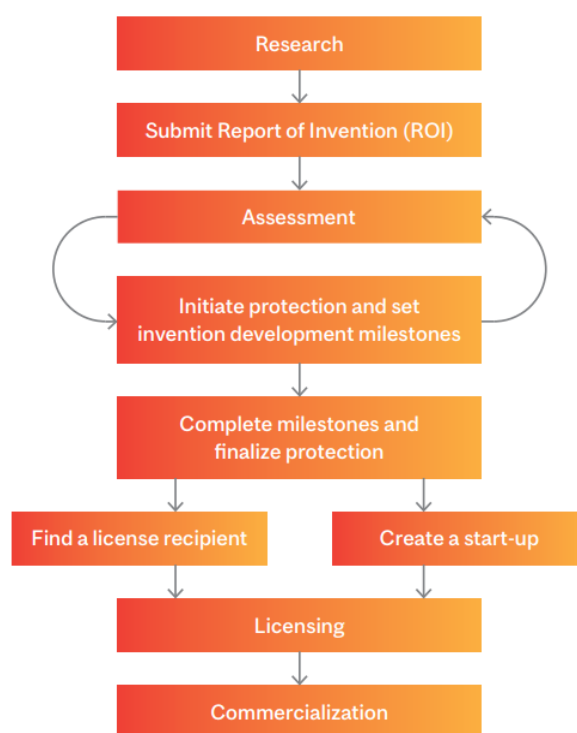


Figure 22. McGill University's standard technology transfer process from their *Guide for Faculty Inventors*. After ideating and developing a proof of concept, one can submit a report of invention (ROI) to McGill's technology transfer office. Afterwards, the office assesses the feasibility and patentability of the idea before initiating IP protection. Afterwards, the creator(s) can either find a license recipient to exit, or they can spin off to commercialize the project.

One consideration of spinning off is that, as per McGill's Policy on Inventions and Software, upon being granted license to spin off, the company would be expected to repay all

patent costs incurred by the university (McGill University, 2017). The timeline for this payback is negotiable on a license-by-license basis (McGill Innovation and Partnerships, n.d.; McGill Innovation and Partnerships, 2022), though it does represent a substantial liability in the first few years of the company that could have a severe impact on our balance sheet. Planning the payback effectively would be of utmost importance for the early financial stability of the company, as well as the appearance of our financial statements. After finalizing the licensing arrangements, the spinoff would also become liable for the entirety of its own patent expenses. Other licensing terms include a small annual royalty and equity in the spinoff (again both negotiable, though generally on the order of 5% or less) (McGill Innovation and Partnerships, 2022). Fortunately, after the spinoff incorporates, it is the sole beneficiary of the exclusive license created by patenting the invention through McGill, less McGill's minor equity claim and royalties.

Another consideration of filing with McGill University is the built-in measures to protect our IP assets against infringement, misappropriation, and unauthorized use of the technology by competitors and third parties. Given McGill's initial protection and complete payment for the legal expenses of upkeeping a patent, our team benefits from the inherent protection against misuse of the invention by others. While our team will do our due diligence to search for misuses or infringements, there will be a dedicated effort by McGill to ensure protection of the invention until the McGill iGEM team licenses the technology or abandons the patent.

A more comprehensive discussion of our project's IP strategy, including details on specific IP assets, protection mechanisms, and licensing opportunities, will be provided later in this report in the Intellectual Property Strategy section. Advance to this section for more information on IP.

5.5 Regulatory Pathway and Approval Process

In order to proceed with the regulatory and approval process of a new cancer therapy, we must first submit a provisional patent with proof of concept. In order to be considered for a patent, our team must be meticulous with regards to public disclosures. Luckily, in Canada and the US, there is a 12-month grace period that applies to one's own public disclosure; during this grace, any public disclosure on our end cannot be used to argue against the novelty of the technology during the patent approval process (Government of Canada, 2023b). This means that our team can discuss our technology at the iGEM Grand Jamboree so long as we file a patent application within 12 months of the competition.

It is important to outline the clinical trial application (CTA) and necessary steps in North America specifically to bring *PROTEUS* to light as a legally validated therapeutic. We've decided to outline the regulatory pathway for the US and Canada given this is our initial market for commercialization. After several discussions for human practices, we noticed that most therapeutic companies in Canada tend to go to the US to conduct their clinical trials and sell their product because the FDA is much more rapid for technology approvals. Additionally, the Canadian Health Agency tends to follow the decisions made by the FDA, so after approval

through the FDA, Canadian approval tends to follow suit. For this reason, we decided that if we were to pursue *PROTEUS*, we could raise capital in both Canada and the US. However, it would be much easier to get approval, as well as draft the CTAs in the United States.

There are several stages of testing, evaluation and review needed, prior to the approval of a new cancer therapy. This process involves multiple regulatory agencies and can be rather time consuming. The timeline of generating any product market launch must ensure safety and efficiency of the drug, prior to human administration. In the United States, the U.S Food and Drug Administration (FDA) has primary oversight on oncoming drugs, waiting for market approval. In Canada, Health Canada has primary oversight of new drug approvals, following a similar, multi-step process as in the United States. The first step in the drug approval process involves laboratory research and proof of concept, meaning a foundation of experimental evidence. Ideally, if experimental evidence supports the progression of a new therapy, trials will move into animal studies. Animal studies are a promising step following original proof of concept and can be used to assess the drugs safety, side effects, dosage and efficiency. Animal studies also help researchers understand risks of potential toxicity levels, as well as proper use.

Ideally, *PROTEUS* will test efficiency using syngeneic mouse models which are fully immunocompetent and often used to study tumor immunity and immunotherapy responses (Zeng et al., 2022). Mouse models would best represent human immune systems throughout the duration of animal trials given the similarity between many anatomical, cellular and molecular traits (Walrath et al., 2010). Syngeneic mouse models will also indicate optimal dosage levels and immediate side-effects following *PROTEUS* administration. If animal studies and further preclinical testing continue to generate promising results, the creator of the therapy must submit an Investigational New Drug (IND) application to the regulatory agency for further approval. There are three IND types; an investigator, emergency use, and treatment.

In the United States, the regulatory agency is the FDA, whereas in Canada, the IND will be reviewed by Health Canada Therapeutic Products Directorate (TPD), assessing therapeutics safety, efficiency and quality. If the review is positive, Health Canada will issue a Notice of Compliance (NOD), which means the therapeutic is now approved for sale in Canada. Once the therapeutic is approved, its use is monitored for post-marketing surveillance (clinical trial phase four) and Health Canada as well as the pharmaceutical company work together, assessing relevant post-market data. The Canadian approval process by Health Canada typically takes between one and two years.

For the US, an investigator IND is submitted by a physician who both initiates and conducts an investigation, as well as dictates the direction of the drug that is to be investigated. It is also possible that a physician or researcher can submit an IND for granted permission to study an unapproved drug, or an approved product for a new purpose (FDA, 2022a). Emergency use IND enables the FDA to authorize use of an experimental drug in an emergency situation, without following the normal timeline for IND submission. Emergency IND may also be used for patients who do not meet the criteria of an existing study protocol, or

non-existent study protocol (FDA, 2022a). A treatment IND is submitted under the circumstance of a promising experimental drug, with promising evidence in clinical testing. Treatment INDs are typically submitted for drugs with promise of treating critical illness' while FDA review takes place (FDA, 2022a).

Depending on the outcome of the review, the FDA will respond with either a clinical "hold" or "clearance". A clinical hold response indicates safety concerns and/or deficiencies in the submitted IND application (FDA, 2015). The hold will delay the proposed investigation. Thus, no subjects will be given the investigational drug, and no new subjects may be recruited to the study (FDA, 2015). In contrast, clearance indicates an acceptable application, and may include specific recommendations when moving forward to clinical trials.

In short, the IND outlines the investigational therapy's proposed use in moving forward to clinical trials, information regarding its chemical composition, manufacturing process, and experimental/preclinical data. Moreover, once an IND application is approved through the FDA, the investigational drug can move forward to clinical trials, further evaluating the therapy's safety and optimal dosage. IND review teams consist of medical officers, pharmacologists, pharmacokineticists, microbiologists, etc. (FDA, 2018). The team ensures that various field specialists are responsible to address potential concerns in different areas of the new drug.

Moving forward, if the investigational drug generates a positive outcome from the first three phases of clinical trials, the drug developer may submit a New Drug Application (NDA) to the FDA. The NDA will include detailed data on preclinical research and outcomes, clinical trials, the drugs manufacturing process, ingredients, and proposed labeling. The NDA serves as a vehicle in which drug sponsors propose approval of a new pharmaceutical, heading towards market launch, and must provide enough information to permit FDA reviewers to reach a conclusion (FDA, 2022b).

Once the drug is approved for marketing, it will be labelled with information such as approved use and dosage, and precautions such as side effects. Phase four of the clinical trial begins following successful NDA, hence beginning the phase of post-market surveillance. During this period, it is encouraged for users to report unexpected side effects or outcomes resulting from the new drug. The timeline of approval processes vary between country and regulatory agency, but similar principles apply to ensure safety and liability among the drug's users as well as the creator.

There are many potential setbacks during the first three phases of clinical trials, as roughly seventy percent of drugs surpass phase one, which lasts several months in testing for optimal safety and dosage (FDA, 2018). Phase two lasts several months up to two years, focusing on drug efficiency and side effects, with only one third of drugs moving forward to phase three (FDA, 2018). Additionally, phase three normally lasts between one to four years, assessing efficacy and monitoring of adverse effects from the product (FDA, 2018). Post-market surveillance is considered phase four of the clinical trial process, involving several thousands of volunteers who possess the disease/condition (FDA, 2018).

Any federal approval process can take between one and two years, depending on the composition and intended use of the product. The Canadian government is trying to implement a new “Agile Licensing” framework in an attempt to expedite the pre-market regulatory process of pharmaceuticals (Herder et al., 2023). This is part of the government's strategy to offer Canadians quality and affordable medicine. Adoption of Agile Licensing would allow drug marketing to occur nearly half a year sooner than the current systems implementation (Herder et al., 2023). Additionally, fewer pre-market clinical trials would be required providing firms continue studying their drug’s effectiveness once they are in use (Herder et al., 2023).

Moreover, with the lengthy approval and clinical trial process, time and funding have potential to hinder the process of bringing any new drug or therapeutic to market. Additionally, given *PROTEUS* targets cancer at a genetic level, using our therapy would require each patient to undergo exomic sequencing, which may cost up to \$3594 USD per person (Gordon et al., 2020). Furthermore, with any cancer immunotherapy, there is moderate risk of off-target effects. In order to monitor off-target effects and maintain patient well-being, it may be useful to keep patients under hospital supervision for several weeks following *PROTEUS*’ administration to ensure any adverse side-effects will be immediately attenuated. This brings further associated personnel and rooming costs.

There are strategies for harmonizing regulatory submissions and processes across multiple markets, such as leveraging the International Council for Harmonisation of Technical Requirements for Pharmaceuticals for Human Use (ICH) guidelines or participating in Mutual Recognition Agreements (MRAs) to streamline the approval process and expediting market entry. The International Council for Harmonisation of Technical Requirements for Pharmaceuticals for Human Use (ICH) is an international nonprofit association which brings regulatory authorities and the pharmaceutical industry together, harmonizing scientific and technical aspects of drug registration (FDA, 2020). With increased harmonization comes safer, more effective and higher quality drugs, registered in an efficient manner (FDA, 2020). Furthermore, international harmonization results in reduced time to bring a product to market, reduced patent burden through prevention of redundant duplication of clinical trials and post-market clinical evaluations, as well as reduced need for animal testing (without compromising safety and effectiveness of the drug) (FDA, 2020).

Additionally, there are also Mutual Recognition Agreements (MRAs) which promote trade in goods between the European Union and other countries, while facilitating market access (European Commission, n.d.). These are bilateral agreements that aim to benefit industry by providing easier access to conformity assessment (European Commission, n.d.). Conformity assessment bodies are mutual recognition agreements which state the non-member country will accept conformity assessment results (e.g. testing or certification) performed by the other party's designated conformity assessment bodies (European Commission, n.d.). Thus, showing compliance with the first party's requirements and vice versa. MRAs must also include relevant lists of designated laboratories, inspection bodies in both the EU and the other country (European Commission, n.d.).

5.6 *PROTEUS* Life Cycle

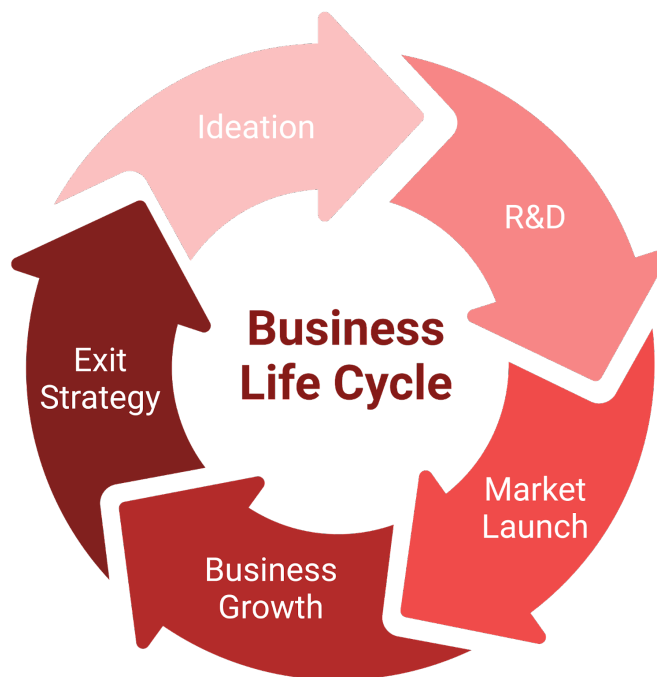


Figure 23. Overview of the business life cycle for *PROTEUS*, from ideation to exit.

PROTEUS was initially developed as a result of the market gap in pancreatic cancer treatment. Based on the 12% 5-year net survival rate and the market need illustrated in the Market Research Section of this report, it is clear that there is a significant need for a safe, effective, and consistent treatment, and one that is personalized to each patient. This section discusses the idealized life cycle of *PROTEUS* as a genetic-level pancreatic cancer therapeutic.

The idea was initially conceptualized in the ideation phase of our team's research cycle. Each year, our team splits into groups of three to four to brainstorm synthetic biology ideas that have potential to change the world and to be monetized. Through rigorous literature review, each group develops a theoretically valid project idea that solves a specific world problem. In the ideation phase, we encourage groups to consider which SDGs can be targeted and how the product will prove lucrative commercially. From here, each idea is crafted into a presentation, after which the entire team votes their top three favourite ideas. At this step, the entire team conducts more literature review to rule out any infeasible projects. Whichever best balances feasibility, intrigue, and real-world impact is selected as the project for the year. In this case, we selected *PROTEUS* back in October 2022.

With regard to research and development (R&D), *PROTEUS* has followed the classic synthetic biology design build test learn (DBTL) cycle, iterating over our wet lab and dry lab activities throughout 2023. The early months of the idea were greatly spent continuing ideation by conducting relevant literature review to confirm the feasibility of the chosen project. By

January 2023, our team started preliminary work on laying the foundations for the dry lab NGS pipeline. Later in April 2023, our team began wet lab work to validate expression of all vectors as well as cell death in our chassis. The NGS pipeline was greatly completed by July 2023, while wet lab validation in yeast and mammalian lines was not complete until October 2023.

Post-Jamboree, our team aims to more extensively validate the system to a level considered generally acceptable by the scientific community. We aim to eventually publish a research paper discussing the novel cancer therapeutic, as well as the BioBricks we have created. However, we plan to patent our product before publishing; if we publish a paper first and then file a patent application, we might encounter problems with the ‘novelty’ criterion, which can ultimately lead to the patent application being rejected (Tietze, 2023). Post-publication and patenting, our focus will shift to commercialization. This section of the business strategy involves product development and creation of a viable prototype – understanding the market audience, branding and marketing, and setting up mechanisms to be used for sales and distribution. Additionally, as detailed in the Regulatory Pathway and Approval Process Section, the path to market launch in North America includes submitting sound CTAs and seeking approval of the therapeutic by the FDA and Health Canada. We would likely pursue approval by the FDA first, as opposed to other national approvals, given that it is more rapid and opens the door to approval by other countries (such as Canada). To avoid the pitfalls of regulatory hurdles, we will work with regulatory agencies in the early stages of our development, as evidenced by our extensive iHP meetings, to provide comprehensive safety analyses. We are also working closely with ethicists to ensure that our project has taken common ethical analysis issues into concern, and that we have ethical principles developed early in the process. Additionally, our wide net of contacts established early on in our project will allow us to remain aware of the latest scientific advancements and cutting-edge technology to avoid technological hurdles.

One must also consider the business growth model in creating a successful startup. This is expanded upon in the Business Model section of this report. The first step in creating a successful business model is market penetration. This includes concretely identifying the problem our product solves and ensuring that it adequately addresses a significant problem such that our customer base exists and has need for *PROTEUS* as a therapy. With existing therapies for pancreatic cancer being largely unsuccessful, as reflected by the 5-year survival rate, there is a significant need on the patient's side for a novel therapy that can be used as a treatment for pancreatic cancer. Due to the nature of the disease, cancer is an ever-present challenge in our society, meaning that our product will simultaneously avoid obsolescence in the long-term unless a more cost-effective yet thorough cure is developed. The important part of this section for our therapy is attracting new customers and scaling operations to meet demand.

After building a successful customer base and improving upon feedback in the initial stages, we will pursue a monetization strategy. This will be the initial way of generating revenue from the business and using that revenue to continue to grow. We must consider pricing

models and balancing accessibility for those globally, all while generating profit. Depending on our initial patenting strategy and our pursuit of licensing, we may also be able to generate revenue from intellectual property licensing.

Finally, we will be pursuing an exit strategy. Many types of exit strategies exist, including acquisitions, initial public offerings (IPOs), management buyouts (MBOs), or liquidation. Depending on the progress of *PROTEUS* through the monetization process, we will pursue the exit strategy that is most financially feasible and profitable.

We have focused on discussing any potential setbacks with shareholders and have begun exploring patient input in collaboration with Kay Kays, an advocate for pancreatic cancer patients. This has allowed us to proceed with an ethical and patient-focused process of developing *PROTEUS*. Through many integrated human practices meetings, we have done a significant amount of market research and believe that the perceived need for a therapeutic like this exists both among providers and patients. Our objectives in market launch must be made clear. We aim to provide a novel genetic level therapy to target a mutation responsible for the deadliest common cancer in terms of survival rate. With many therapies targeting more niche populations (i.e. orphan drugs), and ours even taking cancer treatment down to the individual, genetic level, we know that one-size-fits-all therapies are a less viable market access strategy. However, with the onset of personalized medicine, the therapeutics industry is changing rapidly. In determining our approaches to pricing, contracting, access, and affordability, we will continue to consult these stakeholders to keep up with the changing market and keep *PROTEUS* a relevant therapeutic modality.

The development timeline for CRISPR-Cas-based therapeutics like ours can span decades with variation depending on the complexity of the therapy and outcomes in trials. Generally, it can take over a decade or more between initiation of the project toward general clinical use. However, as personalized medicine and gene therapies become more common, the timeline for approvals may become more rapid.

Potential extensions to our therapy may include modifications in the Craspase system to target mutations other than KRAS, allowing our therapy to expand beyond pancreatic cancer. However, this would involve future series of preclinical research, clinical trials, approvals, and scale-up, similar to the initial therapeutic. Realistically, we will push the KRAS therapeutic to market before considering extensions.

By employing user feedback from stakeholder groups and performance metrics, we will be able to analyze the success of *PROTEUS* in a clinical setting and monitor the response from patients and providers. Clinical feedback will be gathered from patients during the trial phases which will allow us to monitor potential safety concerns and adjust treatment delivery. Similarly, once the therapy is used more extensively, we will collect and analyze data on patient outcomes to ensure the best results are possible in the long term.

It is also important to continue on the path that we have been taking. We must keep communication open with stakeholders that we have been in contact with to address key concerns. It is also essential to continue to conduct research to monitor changes in the need

for *PROTEUS* as a therapeutic strategy. This will help us adjust pricing and access concerns. Similarly, we will continue to conduct regular literature reviews and collaborate with leading scientific institutions to scientifically develop *PROTEUS*.

Within the exit strategy, we have a variety of options which can be pursued at different times within the *PROTEUS* lifecycle. One of these strategies involves the sale of technology, which would be our preferred plan. This allows us to change the timeline of *PROTEUS* depending on the market conditions and project priorities. We can implement this strategy at any point in the project, which gives us a significant amount of flexibility, and can potentially pursue a merge or acquisition or MBO strategy. This would require negotiation and regulatory approval in the execution process. Another potential strategy would be to pivot *PROTEUS* toward a different use, depending on market conditions. If market conditions expressly change, rendering *PROTEUS* obsolete, we can use the basic technology of the therapy to target a mutation which exists in other types of cancer. This would require adjustments to our stakeholder discussion and clinical trial design, as well as the regulatory strategy.

6. Product Development

6.1 Value Proposition

McGill iGEM's primary asset under development in this project is the *PROTEUS* proof of concept in *S. cerevisiae*. We aim to provide validation of our system in the form of preclinical data from Phases 1 and 2 of our experimentation, as outlined in Section 6.2. A value chain analysis was performed and Figure 24 below highlights the progression of activities that McGill iGEM will take in order to deliver *PROTEUS* as a therapeutic option for cancer patients.

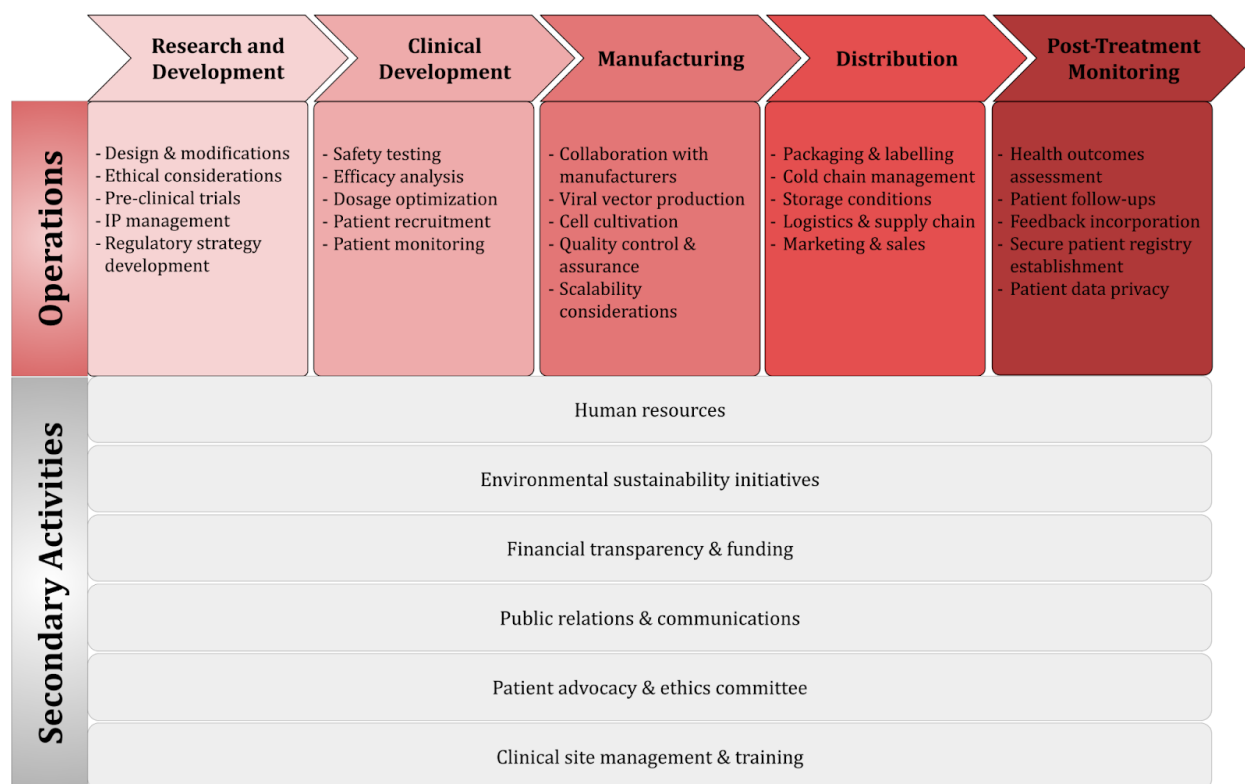


Figure 24. Value chain graphic for *PROTEUS* displays all business activities required throughout the development of the therapeutic, including operations and secondary activities.

6.1.1 Research and Development

The first activity in the development of *PROTEUS* is the Research and Development (R&D) stage. In this stage, we are focusing on the establishment of the *PROTEUS* proof of concept in both yeast strains and mammalian cell lines. Throughout this development, we converse with ethicists about the various intricacies of the project to ensure our final product will be ethically sound. After we finalize our proof of concept, our next step will be to test our system in pre-clinical trials where the aim is to validate our system in mouse models. Rigorous pre-clinical testing is crucial to evaluate the safety, efficacy, and potential side effects of the therapy. This phase encapsulates numerous iterations that ultimately optimize the specificity of our system not only to target cancerous cells, but also to minimize the harm to healthy tissues. Intellectual Property (IP) management is also integral at this stage in our business; further details highlighting our IP strategy are discussed in Section 7 of this document. While pre-clinical trials are ensuing, the team will also consult with lawyers, business professionals, and regulatory strategists to form a regulatory strategy, which establishes policies, procedures, and actions that will help us bring *PROTEUS* to market. See the Regulatory Pathway and Approval Process Section of this document for further information on our regulatory process. The formation of a regulatory strategy marks the end of the R&D stage.

6.1.2 Clinical Development

Following R&D, McGill iGEM will enter into the Clinical Development stage, where we will transition our product from laboratory experimentation to running clinical trials with real cancer patients. Clinical trials will be divided into three phases and further details of each phase are highlighted in the Development Timeline, Milestones, and Resource Requirements Section of this business plan. The first phase will assess the safety of the therapeutic in a small sample of patients. Phase II trials will then expand the diversity of the patient sample group, and further evaluate the efficacy of the treatment and optimize the dosage amount. This phase is crucial to moderate the effects of the therapeutic on both cancerous cells and the patient's immune response. The final phase, Phase III, will once again expand the patient population, and will definitively establish the safety and efficacy of this treatment. In all three phases of clinical trials, patients will be recruited in a randomized manner, thus ensuring the most diverse sample group receiving treatment. All the patients involved will receive close monitoring from clinicians and healthcare providers who will assess the impact of the treatment on both their tumour and overall well-being.

6.1.3 Manufacturing

After clinical trials prove the efficacy of *PROTEUS*, the manufacturing stage will ensue, marking the production of *PROTEUS* as a scalable and tangible product. On our end, this stage requires ample communication between us and our associating manufacturers. Specifically, manufacturing a gene therapy requires sophisticated facilities equipped with demanding quality control measures. A significant part of the manufacturing stage is the production of the viral vectors, which are the systems that deliver the therapeutic system's vectors to target cancerous cells. Proper manufacturing practices of these vectors requires strict adherence to regulatory guidance to ensure safety, consistency, and no contamination, which is something that McGill iGEM will closely monitor throughout. Therefore, we will be implementing quality assurance and quality control measures at each step of the manufacturing process, from sourcing of raw materials to distribution of the final product (more information on quality control is provided in the Quality Control and Testing Procedures Section). This allows us to facilitate a reproducible, scalable product ensuring that patients receive a consistent and reliable therapy.

6.1.4 Distribution

Following Manufacturing, the next business activity that would occur is the Distribution stage, which is the process in which our finalized product is transported from the manufacturing facilities to hospitals where it will be administered. To ensure timely and safe transport, McGill iGEM will develop a diligently planned logistics network, highlighting key considerations including labelling requirements, temperature surveillance, and handling controls. Specifically, cold chain management will be key to regulate as the *PROTEUS* therapy requires strict temperature controls to maintain its efficacy and stability (Marshall et al., 2016). For instance, our therapeutic will be delivered in a gutless adenovirus, which needs to be

stored in sub-zero Celsius conditions, typically -80°C (VectorBuilder Inc., 2023). Thus, McGill iGEM must ensure that our therapy is delivered in specialized freezers with temperature monitoring to maintain these cold conditions and ensure the product's potency.

6.1.5 Post-Treatment Monitoring

The Post-Treatment Monitoring stage represents our commitment to ensure the ongoing health and well-being of patients receiving the *PROTEUS* treatment. Our team will establish a patient-centered surveillance system that requires a continuous monitoring effort by the administering healthcare professionals, whereby they employ frequent blood sample tests and imaging techniques, and genetic analysis to monitor whether the cancerous mutation maintains prevalence. As each patient will respond differently, the continuous genomic data collection will allow us to make specific adjustments to the patient's dosage, providing us with the opportunity to make our treatment optimal for each individual. McGill iGEM will also ensure the privacy of a patient's genetic information. We will incorporate robust data encryption and access controls, thus complying with strict data protection regulations set out by the Canadian Personal Information Protection and Electronic Documents Act (PIPEDA), the U.S. Health Insurance Portability and Accountability Act (HIPAA), and the E.U. General Data Protection Regulation (GDPR) (Grynwajc, 2020). Additionally, quality of life assessments and patient-reported outcomes allows us to foster an environment of open communication between ourselves, patients, and healthcare providers, allowing patients the freedom to voice their concerns.

6.1.6 Secondary Activities

The secondary activities of a value chain are the supporting activities not involved in the day-to-day operations, but must still be considered as they are crucial to the efficiency and success of the primary operation stages. For starters, human resources (HR) is instrumental for ensuring safety regulations are being properly implemented and followed, daily tasks are being monitored, and also for proficient talent acquisition at the various stages of the project. HR will also ensure that clinicians are receiving the proper handling and storage skills that our therapy requires, by way of establishing some of our own employees on-site in hospitals where *PROTEUS* is being administered. Environmental sustainability initiatives will be monitored closely throughout the implementation of *PROTEUS*, and McGill iGEM will ensure that all procedures are being carried out in the most environmentally friendly way and that all applicable SDGs are being maintained. McGill iGEM will also maintain complete transparency with our partners and investors regarding our financial state, and diligently manage our resources and financial reporting throughout operation stages. Finally, we aim to implement patient advocacy groups and support networks for clinical trial patients and patients receiving our finalized treatment.

6.2 Development Timeline, Milestones, and Resource Requirements

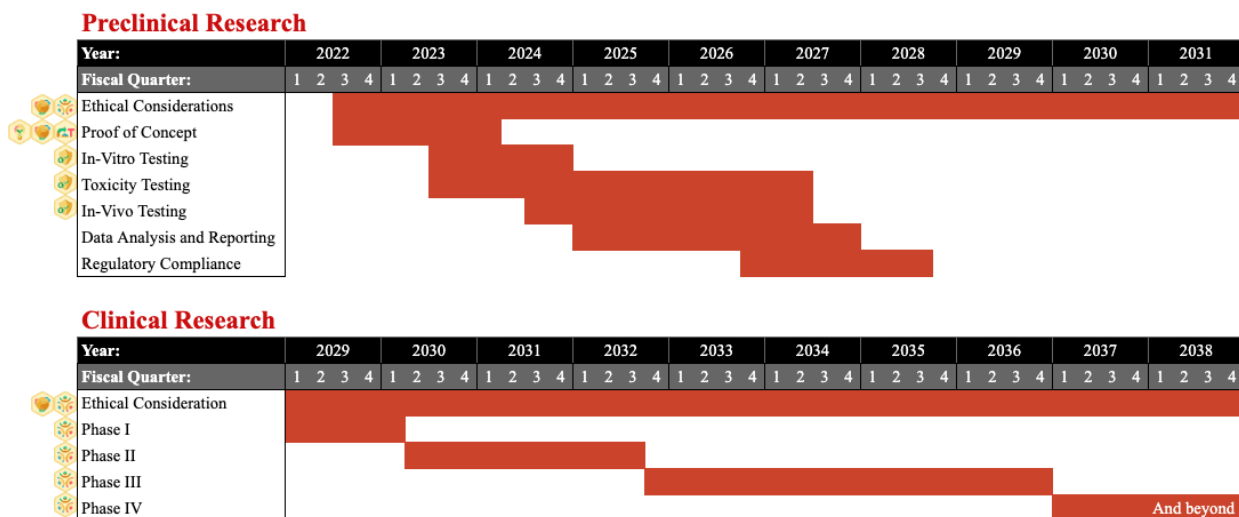


Figure 25. McGill iGEM's 2022-2038 Gantt chart for development and clinical trials of *PROTEUS*. Figures are estimated based on average cancer therapeutic development expenses (Prasad & Mailankody, 2017).

6.2.1 Key Milestones, Roadblocks, Resources

Throughout the process of developing cancer therapeutics, there are many important milestones that must be hit to ensure the project is bound for completion. Typically, it takes 6 to 15 years to develop a cancer therapeutic, which reinforces the need to complete the following goals on time (Prasad & Mailankody, 2017). In addition, considerations must be made for any roadblocks that would delay the development of the therapeutic.

Proof of Concept

Timeline: 1-2 Years

The Proof of Concept represents a critical stage that determines whether the research project should move forward with further experimentation. A significant challenge that comes with developing our CRISPR-based therapeutic is expressing all of the components of the CRISPR system to cleave Csx30. Thus, a majority of the time would be spent on engineering the right plasmid and inserting all of the appropriate plasmids into yeast and mammalian cells.

Preclinical Testing

Timeline: 3-4 Years

After a more robust proof of concept is established, more data on the efficiency and efficacy of the therapeutic must be gathered. This operation requires collaboration between the dry lab and the wet lab. One notable area of investigation is off-target effects, which are relevant to a CRISPR system that activates based on binding to RNA with a mismatch mutation. In addition, it is important to address the unanswered questions raised by the work of

Strecker et al., such as understanding the effect of modifying single base pairs in the spacer region of the gRNA.

Another important aspect of preclinical testing is toxicity and dosage (U.S. Food and Drug Administration, 2018b). Proteus must be tested in in-vivo models using its intended delivery method to assess whether it poses unintentional harm to the test subject (U.S. Food and Drug Administration, 2018b). In addition, in-vivo models would also further our understanding of how the therapeutic interacts with the entire body and allow us to determine any potential side effects.

The major concerns of Preclinical Testing are funding and expertise. In-vivo and toxicity testing involves extensive research in state-of-the-art facilities following regulations outlined in 21 CFR Part 58.1: Good Laboratory Practice for Nonclinical Laboratory Studies (U.S. Food and Drug Administration, 2018b). In addition to having personnel with proper training to conduct in-vivo research, we would also require the expertise of leading scientists, regulatory specialists, and industry professionals to build a portfolio for the IND application.

Approval of Investigational New Drug (IND) Application

Timeline: 4-6 months

With the data collected from in-vitro and in-vivo, the team would put together a Investigational New Drug Application to jump start the clinical trials phase.

Clinical Trials Phase I to III

Timeline

- **Phase I: Several Months - 1 Year (American Society of Clinical Oncology, 2019)**
- **Phase II: 2 years (American Society of Clinical Oncology, 2019)**
- **Phase III: Several Years (American Society of Clinical Oncology, 2019)**

During Phase I of clinical trials, the therapeutic is tested on 20 to 80 volunteers with the specific cancer that our therapeutic targets (U.S. Food and Drug Administration, 2018b). It gives researchers a first look at the effect of drugs based on dosage and the best way of administering the drug (U.S. Food and Drug Administration, 2018b).

During Phase II of clinical trials, the number of participants increases to a few hundred (U.S. Food and Drug Administration, 2018b). Phase II studies allow researchers to gather additional safety data in order to refine research investigations for Phase III (U.S. Food and Drug Administration, 2018b). Moreover, the therapeutic is tested alongside the standard treatment in randomized groups to demonstrate the efficacy of the therapeutic (American Society of Clinical Oncology, 2019).

Phase III of clinical trials involves 300 to 3000 participants (U.S. Food and Drug Administration, 2018b). The participants come from various backgrounds in order to test the drug's efficacy to a specific population (U.S. Food and Drug Administration, 2018b). The length

of Phase III clinical studies is also longer, which allows researchers to gather long-term data on safety and side-effects (U.S. Food and Drug Administration, 2018b).

The challenge that comes with all three Clinical Trials is failure to make it to the next stage. Many factors can lead to failure: failure to follow FDA guidance, problems with patient recruitment and retention, and issues with safety (Fogel, 2018). All failures come at a financial cost, and can potentially impact the ability to improve a therapeutic that can re-enter clinical trials. Therefore, it is important to select the correct study site and address patient concerns, minimizing patient burden and maximizing patient appreciation to ensure the success of clinical trials (Fogel, 2018).

The personnel are also important for the success of clinical trials. It starts at the beginning, with the research and development team investigating the effectiveness of the drug compared to standard treatment and any potential off-targeting effect that may lead to unsuccessful clinical trials. Building a connection with patients, doctors, and healthcare organizations is crucial for the recruitment and retention of patients (Fogel, 2018). It involves hiring patient representatives and study support staff that seek to minimize patient stress and lack of knowledge on their own treatment (Fogel, 2018). Moreover, hiring researchers who have held successful trials in the past would increase the chance for the therapeutic to pass all three stages of clinical trials and move forward with Manufacturing Scale-Up and the final approval (Fogel, 2018).

Manufacturing Scale-up

Timeline: 1-2 Years

In-between Phase II and Phase III trials, important consideration for manufacturing scale-up must be made. This is the moment in time in which business professionals, supply chain experts, and marketing gather to discuss the commercialization of the product once it is approved. Coordination between research and development teams, regulatory authorities, and manufacturing experts is essential to streamline the process and ensure that the product meets safety and efficacy standards. The goal is to create a strategy for optimizing production without losing product purity and finding an ideal market for the distribution of the therapeutic.

Approval of New Drug Application (NDA):

Timeline: 6 months - 10 months (U.S. Food and Drug Administration, 2018b)

The Approval of New Drug Application requires data collected from preclinical and clinical research to demonstrate the drug's safety and efficacy (U.S. Food and Drug Administration, 2018b). If approved, the drug is required to be labeled with the appropriate prescribing information and it is ready for commercialization (U.S. Food and Drug Administration, 2018b).

6.2.2 Estimate of Total Expense and Coverage of the Expense

The average cost to develop a cancer drug is \$648.0 million; the revenue generated has a median of \$1658.4 million (Prasad & Mailankody, 2017). On average, Phase I trial costs \$3.4 million, phase II trial costs \$8.6 million, and Phase III trial costs \$21.4 million (Martin et al., 2017). Note that the cost of clinical trials are averages, meaning that to develop a cancer therapeutic, it could be more expensive to run clinical trials because it is more regulated. Based on the estimations above, one can assume that the bulk of the cost originates from research and development.

In order to cover the expense of developing a cancer therapeutic, the organization requires funding from various sources. Currently, the research being done at McGill iGEM is funded by external grants, institutions of McGill University, and collaboration with scientists at the Goodman Cancer Institute. As the research project scales up in the future, the organization would seek out funding from the government, venture capital, and investors to accommodate the high cost of synthesizing a cancer therapeutic.

6.3 Regulatory and Safety Considerations

The biotechnology sector plays a significant role in the development of new drugs and therapies. Hence, this sectors' components are under tight regulatory authority of organizations such as Health Canada and the FDA. There are regulations which apply depending on the nature of the product (i.e. gene therapies, monoclonal antibodies, biosimilars, etc.).

In the United states, the Orphan Drug Act provides incentives for the development of drugs and biologics intended to treat rare diseases. Companies may receive orphan drug designation which comprises benefits such as: market inclusivity, tax credits and fee waivers to encourage the development of new products. Following the approval of biotechnology products, they are subject to ongoing monitoring for safety and effectiveness. Manufacturers are also required to report adverse events and product defects to regulatory authorities. Regulatory agencies worldwide collaborate through organizations like the Internal Council for Harmonisation of Technical Requirements for Pharmaceuticals for Human Use (ICH) to harmonize standards and streamline the drug approval process. Thus, resulting in safe, effective and higher quality medications.

There are strict regulatory and approval processes which follow any cancer therapeutic, prior to being deemed safe for administration. *PROTEUS*, currently specific to pancreatic cancer, would follow the approval process outlined in the Regulatory Pathway and Approval Process Section, however the modularity associated with the Craspase system will theoretically allow specific treatment for a wide range of RNA diseases. Firstly, the new drug must exhibit experimental proof of concept ahead of successful animal trials. Animal model organisms are chosen based on specific selection criteria such as: the incidence of specific diseases which may complicate experimental results; features of the animal, such as response to microflora; and the ability of the animal to accurately model the drug's mechanism of action (Davidson, Lindsey, and Davis, 1987). Pre-clinical testing is required prior to submission of an

IND application. The IND is submitted to the country's regulatory agency. Clinical trials begin, subdivided into four phases, following approval of the IND Application. The preclinical trial data strives to indicate proof of concept, while also adhering to the regulatory and approval process, ensuring promising results are established prior to moving forward to clinical testing.

Progression of the drug throughout the clinical trial process involves selectively larger demographics of volunteers, all of which possess the disease/condition of interest. The final phase of the clinical trial process is termed post-market surveillance, ongoing for multiple years. The newly approved product is under constant monitoring via the country's regulatory agency. Hence, adverse and long-term side effects can be closely monitored using a widespread demographic of patients.

Commercialization strategies can be hindered due to time constraints of the regulatory and approval process, as well as adequate funding. Companies must allocate funds for clinical trials, toxicology studies, manufacturing, and professionals in regulatory affairs in order to sustain financial resources. In terms of human resources, regulatory affair professionals play a critical role in managing compliance with regulatory requirements. Additionally, regulatory agencies require adherence to Good Manufacturing Practices which can drive the severity of quality control measures.

There are a variety of potential safety considerations regarding the market launch of any new product in the Biotechnology sector, and a variety of solutions which may be proposed to solve the issue. For example, inadequate clinical trial design or suboptimal sample size/niche demographic may not detect rare side effects of the therapy. Thus, in order to overcome this risk, long-term broader clinical trials can be conducted using diverse patient populations. Moreover, post-marketing surveillance is needed to monitor the risk of such rare side effects. If/when adverse side effects or toxicity occur from the product, they can be strictly monitored throughout the duration of clinical trials, ensuring that product safety information/labeling is consistently updated. In addition, phase four clinical studies are in place to ensure that healthcare providers report adverse events from the drug through pharmacovigilance.

Unwanted off-target effects and/or potential interactions with other medications or substances may also pose an issue during new drug development. To address this, studies which assess drug-therapy interactions should be conducted, as well as the addition of appropriate warnings and precautions in product labeling. Another consideration regarding the biotechnology industry is off-label use. Healthcare providers may prescribe drugs for unapproved indications. Evidence-based prescriptions are the most effective way to communicate potential risks and appropriate usage, in tandem with clear product labeling. Big pharma products must develop comprehensive risk communication plans to develop product-consumer responsibility, while also encouraging a simple process for consumers to report adverse effects directly to regulatory agencies.

There are variations in regulatory standards and requirements across countries, which can be overcome through the promotion of International Collaboration and Harmonization of

Regulatory Standards. Through harmonization between research and big pharma corporations, there can be standards to ensure consistent safety evaluations around the globe.

Another setback which may arise throughout the process of product approval is data manipulation and/or fraud among clinical trial results. In order to overcome this risk, strict regulatory agencies such as the GDPR (Europe) or Government of Canada, can provide strict data monitoring and auditing procedures. Additionally, there is the potential setback of the public having limited access to complete safety and efficiency data. Thus, it is important to promote transparency through public access to clinical trial results, even among reports of adverse patient side effects, assuming patient privacy is maintained.

Addressing these safety considerations requires a multifaceted approach involving pharmaceutical companies, regulatory agencies, healthcare providers, and patients. Continuous vigilance, robust post-marketing surveillance, and open communication are crucial elements in ensuring the safety of newly approved drugs.

In order to manage risks and potential side effects of *PROTEUS* administration, McGill iGEM suggests constant patient monitoring up to several weeks following the primary dosage. McGill iGEM understands that this will serve as a costly act of precaution, and we hope that following animal trials we will collect enough data to determine optimal dosage and potential adverse side effects, such that we can reduce patient monitoring times all while respecting proper safety parameters.

Post-marketing surveillance strategies are also in place following drug approval which ensure pharmaceutical companies continue to monitor the drug's safety and efficiency, especially while the drug is introduced to a broader patient demographic. Post-marketing surveillance serves as a way to identify rare or long-term side effects and continues for a number of years post-approval. Pharmaceutical companies also develop Risk Management Plans (RMPs) which outline strategies for monitoring, mitigating, and communicating risks associated with the drug through its lifecycle.

Throughout the duration of real world market launch, strategies such as pharmacovigilance, labeling and communication, and drug utilization reviews are in place, which identify potential risks, misuse, or off-label use. Moreover, Risk Evaluation and Mitigation Strategies (REMS) aid to restrict drug access or provide special training to healthcare providers tailored to the specific risks for new-drug treatment.

During implementation of any new drug following market approval, users are also highly encouraged to stay proactive and report new information/side effects from taking the drug, while also being well informed about the potential risks and benefits, prior to initial use. Minimizing the risk associated with new therapeutics approvals involves multiple stakeholders and requires a collaborative effort between the pharmaceutical industry and consumers, so various clinical and post-market surveillance measures are in place to reduce the risk of premature approvals.

6.4 Quality Control and Testing Procedures

During product development and creation of an eventual proof of business, McGill iGEM aims to provide the highest quality of care available in the pancreatic cancer therapeutics space. Taking the proper precautions is essential for customer safety and satisfaction. *PROTEUS* has significant promise as a therapeutic, but comes with inherent risks given its ability to kill host cells based on recognition of a certain transcriptomic sequence.

For each patient that is eligible for using *PROTEUS* as a treatment option, we will act according to the following systematic approach to treatment:

1. Patient cells are extracted, both healthy and oncogenic cells
2. Cells are lysed, and DNA is extracted. Whole exome sequencing (WES) is performed to create a read for each lysed cell.
3. Running the reads through our next-generation sequencing (NGS) pipeline, we can determine the best oncogenic target transcripts.
4. We use our gRNA program to produce the most efficient gRNAs with minimal off-target effects based on the patient's WES results.
5. We test the gRNAs *in vitro* on patient-derived cells to determine the effects on both healthy and cancerous cells.
6. We provide a minimal dosage to the patient using our viral vector delivery system and monitor the effects *in vivo*.
7. We provide the suggested dosage, specifically targeting the tumour. The patient is monitored hourly for several days after the dosage to ensure comfort and no complications. Monitoring of general tumour shrinkage and patient health is recommended for several months following the first dose. Secondary or tertiary doses may be required.

The figure below provides an overview of the workflow:

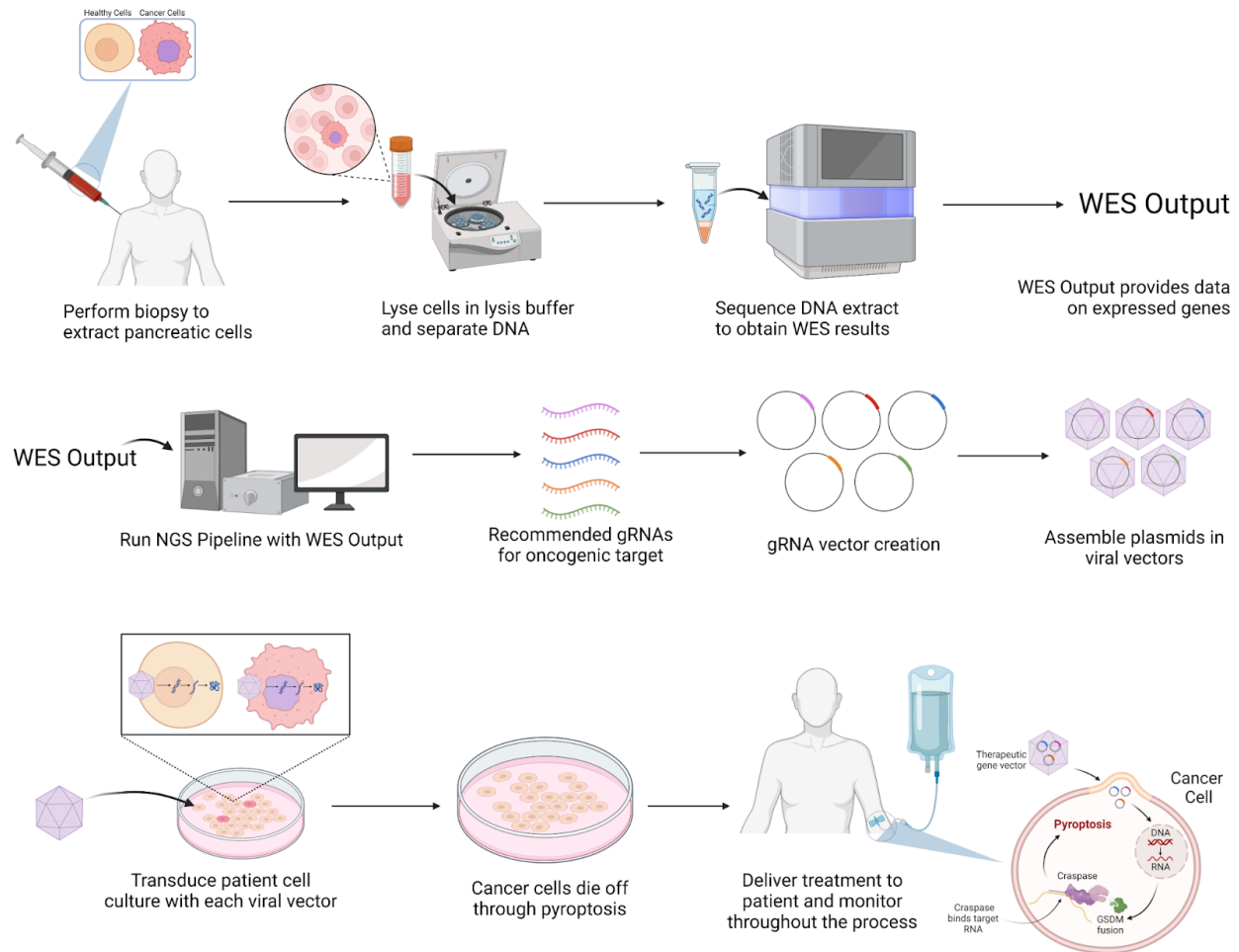


Figure 26. Overall workflow for *PROTEUS*, from patient biopsy to final full-dose delivery. This is an idealized workflow that does not consider failure; in the case of poor quality or deleterious results, the workflow is restarted to ensure safe, effective treatment for each and every patient.

At each step of the workflow, several testing procedures are enforced to ensure the quality and reliability of the product as well as the safety for each and every patient. In the cell extraction step, we will recommend endoscopic ultrasound-guided fine needle core biopsy (EUS-FNB), which has been shown to be an effective method for diagnosing pancreatic malignant lesions; EUS-FNB provides well preserved tissue for histologic grading and subsequent molecular biological analysis (Yang et al., 2016). Most importantly, EUS-FNA is widely accepted as safe and effective; it is considered a minimally invasive biopsic protocol with an exiguous complication rate of approximately 1-2% (Yang et al., 2016; Adler et al., 2005; Kedia, Gaidhane, & Kahaleh, 2013). For DNA extraction and WES, there are minimal direct safety concerns for the patient; however, it is exceedingly important to ensure that WES is performed on a large sample size of both cancerous cells and healthy cells to effectively differentiate their phenotypes with statistical confidence. Both subsets will then be compared with a reference genome to perform variant calling. After developing a list of the most common

variant alleles in the cancer cells (and absent in healthy cells) through the NGS pipeline, a medical practitioner can select which cancer genes should be targeted for the *PROTEUS* treatment. Instead of simply selecting the most common mutations, we recommend the support of medical professionals to ensure a balance of safety and efficiency in determining the optimal target mRNA(s). Specifically, the recommendations of both a clinical geneticist and a clinical immunologist would be optimal. These physicians may provide insight in the mRNA selection process to prevent over-inflammation, inefficient targeting of cancer, or off-target effects. Based on the mRNA target selected, there could potentially be high variance in the effects observed in the patient; hence, we consider the advice of credible medical professionals essential at this stage of the workflow.

After target selection, we run the target transcript sequences through our gRNA program. This program acts as an *in silico* model, outputting gRNAs with the most efficient binding properties to the target. It searches the output of the patient's WES and compares the host genome to the most efficient gRNAs. The system has a built-in safety metric; if the gRNA exceeds a certain level of off-target activity based on its efficiency, the gRNA will be rejected, and the system will scan the next most efficient gRNA for its safety. This system has exceedingly conservative requirements to ensure the safety of the patient; any gRNAs with relatively high efficiency binding to other sites throughout the patient's transcriptome are immediately discarded. One parameter we use to filter gRNAs is the extent of complementarity between the gRNA and potential secondary targets; any gRNAs with less than three mismatches with another region in the patient's transcriptome are conventionally rejected. Even if the gRNA has minimal complementarity with off-target loci, it is nonetheless thoroughly scrutinized in terms of hybridization minimum free energy, RNA secondary structures, GC content, and other factors. gRNAs that pass these filters move on to *in vitro* testing in patient-derived cells. We transform patient cells – both healthy and cancerous – with the system and create growth curves by measuring cell viabilities at short increments over several days. By comparing cancer and healthy cell viabilities, we are able to more accurately determine how the gRNAs will behave *in vivo* on a patient-by-patient basis.

If at any point in the *in silico* to *in vitro* to *in vivo* workflow we find potential risks for the patient, we immediately return to *in silico* modelling to ensure patient safety. We go above and beyond taking minimal precautions, focusing on patient outcomes first and foremost. After finding acceptable results *in vitro*, we provide a minimal dose using our AAV delivery system both to assess the patient's tolerance to the AAV, as well as to see the effects on tumour viability and any unforeseen consequences for healthy tissue. Depending on the extent of pyroptosis in healthy tissue, we will decide from the minimal dose whether the suggested dose will be safe and comfortable for the patient. These parameters will be determined based on results from human trials elucidating the ideal regimen to administer based on the patient's specific circumstances and needs.

The final step of the workflow is administering the full dosage. At this point, the proper safety precautions have been taken to minimize harm to the patient all while delivering an

effective, genetic level therapeutic against the patient's cancer. We recommend maintaining constant monitoring of the patient in the first few days after delivering the treatment, followed by routine care and monitoring for the months following. Based on the patient's disease progression following the primary dose, as well as monitoring of their immune response and potential health complications, a secondary or tertiary dose may be recommended to expedite patient recovery or increase the strength of the treatment for especially aggressive cancer cases.

This workflow will provide efficient, consistent results for patients eligible for treatment using *PROTEUS*. To ensure the highest quality products, we aim to seek review through ISO 9001:2015 Quality Management Systems and adhere to the guidelines provided by the Government of Canada for Good Manufacturing Practices (GMP) (ISO, 2015; Government of Canada, 2023a). Naturally, the cancer therapeutic space is already tightly monitored for safety and quality; regardless, our team aims to enforce frequent quality inspections and maintain exhaustive documentation on a patient-by-patient case. Post-seed funding, our team aims to have a separate division for quality control. This is further discussed in Section 3 of the report. This division would ensure we meet the criteria to be certified under Quality Systems ISO 13485 by a third party auditing organisation akin to recognized registrars of the Canadian Medical Devices Conformity Assessment System (CMDCAS) (Government of Canada, 2019).

One potential bottleneck that arises in the workflow is the time constraint of providing care on a patient-by-patient basis. Naturally, given each patient will have unique genetic drivers behind their cancers, it becomes difficult to create a standardized, generalizable approach to cancer treatment. One option is to create a 'library' of sorts, where the most frequent driver mutations are kept on file; we could create standard gRNAs for these mutations and have them readily available in the case that they are requested by a physician's office for a patient. This would save time for our most common orders, and would provide more consistency and assurance that the system will yield its regular results.

As mentioned, we plan to keep record of each treatment to ensure proper documentation in the case of regulatory scrutiny as well as for patient safety and medical histories. This will also allow us to develop our library of the most common orders by tallying the frequency of each mutation targeted. Our record-keeping practices through our Administration and Accounting division will have policies in place to maintain the privacy and dignity of all patients who choose *PROTEUS*. We plan to have a policy of no third-party disclosure of patient data.

7. Intellectual Property Strategy

7.1 Overview of Relevant IP

The Craspase system is a patented technology, even though it is an exceedingly novel discovery. The inventors on the patent are Sam Pelle Bertram van Beljouw (S.P.B.v.B.) and Stan Johan Jozef Brouns (S.J.J.B.), a PhD candidate and Full Professor of Microbiology respectively,

both affiliated with Delft University of Technology. In the founding paper, which was produced in collaboration of researchers from TU Delft and Cornell University, it is mentioned that the Cornell cohort has additionally filed a provisional patent “related to this research” (Hu et al., 2022). The patent owned by S.P.B.v.B. and S.J.J.B. is an international patent application through the World Intellectual Property Organization (WIPO). The publication number is WO2022255865A1 (van Beljouw, 2021; PatentScope (WIPO), 2022). This patent covers “uses of gRAMP and Craspase” according to the founding paper (Hu et al., 2022). Thus, using Craspase for *PROTEUS* means that we will have to license the technology from the inventors. Potential synthetic biology or bioengineering applications of Craspase are not elucidated in the paper, nor protected by the patent; the paper focused on Craspase’s discovery and characterization, while the patent merely covers gRAMP and Craspase as protein machines (and similar iterations of the proteins in terms of amino acid sequence alignment) (van Beljouw, 2021).

In addition to the TU Delft patent and the Cornell provisional, there are two more provisionals from other researchers on different applications of the Craspase technology. Strecker et al.’s paper, “RNA-activated protein cleavage with a CRISPR-associated endopeptidase”, mentions US patent application 63/337,386 by Jonathan Strecker (J.S.), F. Esra Demircioglu (F.E.D.), and Feng Zhang (F.Z.), which will cover “programmable protease compositions” once publicly available (Strecker et al., 2022). The paper which mentions this provisional primarily focuses on Craspase’s potential applications for RNA sensing; more specifically, as a proof of concept of the RNA sensing capacity, the researchers generated a fluorescently labeled engineered variant of Csx30 and demonstrated its ability to detect RNA in vitro down to 250 fM without nucleic acid amplification (Strecker et al., 2022). They also demonstrated RNA detection capacity *in vivo* using separate experiments involving HEK293T cells and Mouse Neuro-2A cells. Further discussion of these RNA sensing technologies can be found in the “RNA-sensing applications with DiCASP” section of Strecker et al.’s paper (Strecker et al., 2022).

While J.S., F.E.D., and F.Z.’s final patent submission is not available publicly to date, this is the most threatening to McGill iGEM in terms of its potential coverage. Claiming any “programmable protease compositions” suggests protection of any and all manipulations of the system which make use of the TPR-CHAT protease activity in a programmable way. By definition, *PROTEUS* is a system that uses TPR-CHAT in a programmable manner to induce pyroptosis in pancreatic cancer cells which express certain oncogenic target transcripts. Depending on the contents of the patent, we may also need to license use of Craspase as a programmable protease from J.S., F.E.D., and F.Z., though only when we extend our business to the US (given this was a US patent application, not an international application). Regardless, if *PROTEUS* sees any preliminary success in Canada, the inventors will likely act quickly to file a similar patent for protection in Canada. At this point, we would not be able to avoid licensing the use of the technology (assuming the patent is indeed approved for coverage of all “programmable protease compositions”, which may or may not be the case).

The final provisional is alluded to in Kato et al.'s "RNA-triggered protein cleavage and cell growth arrest by the type III-E CRISPR nuclease-protease". The competing interests section simply notes that "[t]he authors have filed a patent application related to this work" (Kato et al., 2022). While the work briefly mentions the potential for cellular RNA sensing and RNA/protein technologies, this publication has a less threatening associated patent application than J.S., F.E.D., and F.Z.'s given the secondary focus of the *applications* of the technology in biotech. The authors of this paper "developed a molecular RNA sensor for transcripts in mammalian cells", which is likely the topic of their patent given the novelty and potential utility (Kato et al., 2022). The sensor they developed placed the Csx30 protein sequence between citrine (a fluorescent protein) and a degron. Thus, after transfecting the model cell line (HEK 293FT in this case) with all the Craspase components as well as a target mRNA, the system would cleave Csx30, separating citrine from the degron. If no cleavage had occurred, the degron would cause the whole fusion (with Csx30 and citrine) to be degraded, preventing a fluorescent signal. However, the paper found a significant level of fluorescence between experimental tests and negatives, suggesting successful *in vitro* cleavage of the fusion. More details can be found in the "Engineering Csx29 and Csx30 for programmable RNA sensing in mammalian cells" section of the paper (Kato et al., 2022). Based on the technology developed in Kato et al., our team does not feel in danger of infringing on the likely patent; we do not plan to use Craspase for RNA sensing, so we are not too concerned with a patent covering this application of the system.

Based on the four relevant patents and three relevant publications discussed in this section, our team recognizes our potential liability with two of the patent applications. We will minimally need to license use of Craspase from S.P.B.v.B. and S.J.J.B., and we may need to consider J.S., F.E.D., and F.Z.'s rather vague but threateningly broad claim of a patent application for "programmable protease compositions". Regardless, these patents and patent applications do not seem to create any issues for our own patent application; the novelty of our fusion proteins in tandem use with the Craspase system will likely be sufficient to prove utility, novelty, and non-obviousness at the time of provisional patent application in Canada and the US.

7.2 Strategy for Protecting IP and Application Filing

As discussed thoroughly in other sections of the report, we have taken all necessary precautions to prevent pre-Jamboree public disclosure. Given that we cannot avoid public disclosure at the Grand Jamboree, our team plans to take advantage of the grace period provided in many countries. Many provide a 12-month grace, while others a 6-month. We would prioritize patent applications in the countries with the shortest graces (Germany, the UK, France, China, etc.) then continue to those with longer graces (the US, Canada, Japan, Brazil, the Philippines, etc.), always prioritizing our primary markets (Canada, the US, and Europe primarily, followed by East Asia, Australia/New Zealand, and potentially the Arab Gulf States). Countries that do not provide a grace would become inaccessible for patent protection

following the Grand Jamboree; thus patent coverage in these countries must be assumed to be impossible. A country-by-country overview of patent laws (especially related to grace periods) is provided by the WIPO (WIPO, 2023).

Given the relevant existing patents and patent applications issued by other groups, McGill iGEM must be meticulous in deciding what to consider in our eventual patent application; in particular, given the extensive patent claims for Craspase and Craspase-related applications (especially in RNA sensing), we will likely focus our IP towards our novel gasdermin-Csx30 fusion proteins, and their use in tandem with the Craspase CRISPR-Cas system. We likely will not focus our patent application on pancreatic cancer given the inherent modularity for the system to target and effectively subdue any cancer with highly homogeneous driver mutations. Thus, we aim to focus our patent on the use of Craspase to cleave fusion proteins that go on to induce pyroptosis in the host cell (assuming *in vivo* usage). More generally, we would like to patent fusion proteins that integrate Csx30 into a flexible, intrinsically disordered linker region between two native proteins or native protein domains (where the domains are from different proteins or the same original protein). For instance, similar to the technology in Strecker et. al, our patent would cover integrating Csx30 between a membrane anchor protein and a Cre recombinase for the goal of spurring Cre recombinase to perform a downstream excision of a LoxP site elsewhere in the cell.

7.2.1 Patenting in Canada

Canada is one country that luckily provides a 12-month grace post-public disclosure. The US is the same case (these facts played largely into our interest in focusing on Canada and the US for the early stages of the venture). The specific process for filing a patent in Canada is outlined in the infographic timeline provided below:

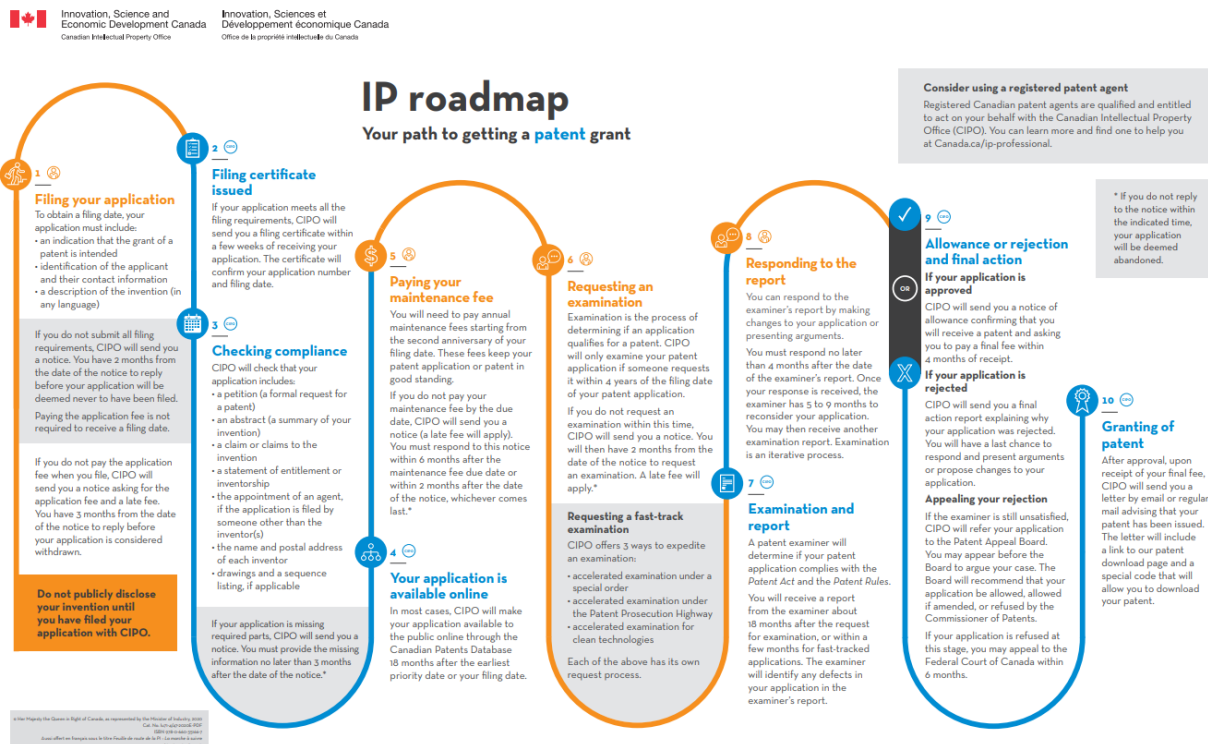


Figure 27. The standard IP roadmap in Canada, provided by the Canadian Intellectual Property Office (CIPO) (Innovation, Science and Economic Development Canada, 2020).

The timeline demonstrates the key deliverables that must be generated within the year following public disclosure at the Grand Jamboree. The initial patent application must include many details, such as the inventors, a detailed description of the technology (an abstract and technical drawings to support the technology), and several legal formalities (a petition for the patent, a claim to invention, statement of entitlement, etc.). Following approval of the application and certification of compliance by the CIPO, most inventors need to ensure payment of maintenance fees. Luckily, in our case, our team is affiliated with McGill University, which has its own internal policies for university inventors. As mentioned in the Intellectual Property and Technological Assets Section, this comes with the benefit of interest-free financial coverage of the legal expenses associated with maintaining the patent up until the date that we license the technology from McGill (at which point there is a payback period to the university for patent expenses incurred). Within four years of our initial filing, we must also undergo an examination by an officer of the CIPO. If successful, our patent is granted, and we have 20 years of protection on *PROTEUS*. If the examining officer makes suggestions for reconstruction of the patent application (which in most cases they do), we have four months to make the necessary changes before submitting for re-evaluation. If after reconsideration, the officer still deems the technology unfit for patent protection, then we will be officially unsuccessful; otherwise, we will be able to assume that *PROTEUS* would be patentable as early as late 2025, or approximately 2 years following the 2023 Grand Jamboree (for lack of certainty, we can estimate it as November 2025).

7.2.2 Patenting in the US

The US follows a very similar route to patenting as in Canada. Uniquely, the US has three types of patents: utility, design, and plant. Utility patents are for anyone who invents “a new and useful” invention (USPTO, n.d.). Design patents are for the invention of an “ornamental design for an article of manufacture” (USPTO, n.d.). Finally, there are plant patents for “anyone inventing or discovering and asexually reproducing any distinct and new variety of plant” (USPTO, n.d.). Evidently, the best fit for *PROTEUS* is the utility patent. We must decide whether to submit a provisional or nonprovisional patent as well in the US. The provisional only provides 12 months of protection before being abandoned. Therefore, our team must file a nonprovisional regardless; it is only a matter of whether we will directly file a nonprovisional, or first a provisional to buy time. Given the time restraints for covering our IP following exposure at the Grand Jamboree, our team will likely file a provisional in the US first. This exempts us from examination until we file for a nonprovisional (USPTO, n.d.). Realistically, we can expect a provisional patent by January 2024, not long after the competition. The buffer of a few months is to account for the time required to assemble a provisional on our end, as well as the turnaround time for the United States Patent and Trademark Office (USPTO) to settle the provisional.

Immediately after finalizing a provisional, our team would begin working on a nonprovisional. This patent application is more rigorous, requiring the following:

- A specification (description and claims)
- Technical drawings
- An oath or declaration to inventorship
- Filing, search, and examination fees

Following submission of the nonprovisional, there is an examination by an officer of the USPTO, similar to the process in Canada with the CIPO. The USPTO describes the examination as follows:

“The examination consists of a study for compliance with legal requirements, along with a search through U.S. patents, publications of patent applications, foreign patent documents, and available literature. This is to see if the claimed invention is new, useful, and non-obvious, and if the application meets patent statute requirements and rules of practice. If the examiner gives a favorable decision, a patent is granted.”

More information on the specific process of inspection is provided by the USPTO in their online services (USPTO, n.d.). After examination, the examiner will either issue an approval, or demand changes be made to the claims, description, or legal documentation. The examiner searches for prior art and will be scrupulous in ensuring we formally state relevant past works and patents, and that we thoroughly differentiate our technologies from prior art in order to be

patentable. Based on the standard timeline in the US, we estimate that we could expect nonprovisional protection by mid-to-late 2025, though likely just before Canadian protection is issued (we estimate September 2025).

7.3 Collaboration Opportunities and Industry Partners

Although our team is determined to maintain privacy with regards to our technology until the Grand Jamboree, we would be interested in seeking industry partnerships post-Jamboree. Given the foreseeable bottleneck in production of our therapeutic based on our team's size, skillset, and limited infrastructure, it would be imperative to seek partnerships for manufacturing. Perhaps industry collaborations would even be feasible, depending on the successes of patent applications. For example, our team has talked with Cameron Black, the Vice President at Repare Therapeutics. Both Repare's technologies and our own are kept concealed in discussions, though given the similar spaces of our products as well as our proximity (both Repare and McGill iGEM are based in Montreal), a collaboration opportunity may be feasible and mutually beneficial. While their technology and our own would likely stay separate in terms of inventorship and ownership, one possible opportunity would be to develop a future technology in tandem, or develop a partnership for funding rounds.

The more pressing consideration for industry partnership is manufacturing partners. Our team does not have the bandwidth to produce the service at the scale we have discussed in this report. Even after major hiring rounds following seed and post-seed funding, our team would not be large enough to compete with key industry players, nor make the mark that we predict we can make. Thus, it is necessary for us to develop industry relations with large oncology firms or CRISPR-Cas therapeutic ventures. One key player is CRISPR Therapeutics. We had a discussion with Demetrios Kalaitzidis from CRISPR Therapeutics on his suggestions for *PROTEUS*' process development and scale-up to clinical use. Given CRISPR Therapeutics' current success and market share in the space, a partnership for scale-up on the manufacturing side would be mutually beneficial assuming *PROTEUS*' success in patenting and clinical trials. Beyond CRISPR Therapeutics, there may be opportunities to partner with some of the largest oncological pharmaceutical companies to scale up manufacturing. Companies such as J&J, Celgene, Pfizer, or BMS would be ideal given that they are primarily based in North America and have a substantial stake in the cancer therapeutics space. If *PROTEUS* was perceived to have the potential to cause a major shift in the space, these companies would be sure to offer lucrative partnerships or acquisition offers. In either case, we would weigh the benefits and drawbacks to make the most optimal decision for profit margins, venture risk limitation, and rapidity in commercialization.

8. Business Model

8.1 Lean Business Model

McGill iGEM's Lean Business Model Canvas is on the following page:

McGill iGEM Lean Business Canvas

Problem - Pancreatic cancer is the deadliest common cancer worldwide - No current treatments are effective nor curative - The disease continues to grow in incidence rates	Solution <i>PROTEUS</i> is a novel, genetic-level therapeutic - It targets cancer mRNA, allowing us to kill cancer cells without harming healthy ones - The treatment induces pyroptosis in cancer cells, alerting the immune system of the cancer	Unique Value Proposition Current treatment modalities cannot target cancer at the genetic level, causing major off-target effects that harm and weaken the patient. <i>PROTEUS</i> targets cancer at its root and activates an immune response, helping the body to mount its own response to the cancer.	Distinct Advantage <i>PROTEUS</i> uses a novel CRISPR-Cas system, making it difficult to reproduce without extensive knowledge on the system and mechanism. We plan to patent the technology, allowing us to protect our intellectual property against big pharmaceutical companies that could rapidly reproduce the tech.	Customer Segments - Elderly (pancreatic cancer is more common with age) - Affluent (CRISPR-Cas therapeutics are costly) - Smokers and those with high BMI (strongest indicators of pancreatic cancer)
Existing Alternatives - Chemotherapies: 5-FU, Abraxane®, Gemzar®, onivyde®, and combination therapy from each - Radiation therapy - Surgical resection - Palliative care Notably, none of these treatments, besides early stage surgery, are highly effective.	Key Metrics - After contacting cancer support groups, we decided we want to get cancer survivors' inputs on the therapeutic and how we can tailor it to help them - We also aim to conduct extensive surveys to integrate patient perspectives	High-Level Concept <i>PROTEUS</i> uses a CRISPR-Cas system to sense cancerous mRNAs. Once detected, the system cuts a fusion protein that goes on to kill the cell. If no cancer is detected, the cell is left alone. The system is delivered using an adenoviral vector, enabling for host cells to constitutively express the CRISPR-Cas system.	Channels McGill iGEM aims to partner with large pharmaceutical companies as wholesalers of our technology. We would ship the product to them, then individual physicians would request the therapeutic, after which our wholesalers would dispense the product to the physician to administer.	Early Adopters - As a CRISPR-Cas therapy, people would need to trust gene therapies - Countries with more rapid approval processes (US especially with the FDA) - Patients with life threatening disease progression would be more likely to adopt the technology
Cost Structure Fixed Costs: employee salaries, overhead on rental spaces and basic laboratory equipment & consumables, early stage R&D, patent maintenance fees Variable costs: manufacturing costs, legal costs (it will highly depend on the stage of development), debt financing interest, marketing costs, clinical trial expenses Customer Acquisition Costs: marketing & advertisement, clinical trial cohort recruitment, market and customer research Distribution Costs: shipping and freight, maintenance and storage, communication with wholesalers and hospitals	Revenue Structure Revenue Sources: sale of product, licensing royalties, wholesale fees, capital investments from VCs, angels, etc. Revenue Model: license our scientific platform to more established biopharma companies while scaling up our own manufacturing capacity until we surpass the production bottleneck given our limited size and established infrastructure. Life Time Value: one-time use; assume average revenue of \$1 million / customer Exit Strategy: depending on market conditions, we may pursue an acquisition from a more established firm, or we may have an IPO if we experience great success.			

8.2 Revenue Generation Strategies and Eventual Exit

The strategic landscape for biotech companies in the biopharma sector is a complex one. Among the most crucial decisions is the choice of business model, which in turn influences who the company partners with and how it scales. The primary business models exist: technology partnering, asset-creation/out-licensing, and product development and commercialization; however, many firms operate as some combination of these three.

In the technology partnering model, biotech firms license their scientific platforms to more established biopharma companies. This approach has the advantage of providing immediate revenue streams, thereby alleviating some of the financial pressures that young companies often face. Importantly, this model also minimizes risk, as the responsibility for developing and commercializing pharmaceutical products lies primarily with the partnering biopharma company. Entering a technology partnership can have ripple effects beyond immediate financial relief; it serves to validate the biotech's platform in the eyes of the wider biopharma and investor communities. Companies that engage in early partnerships with well-known biopharma corporations see a significant boost in their credibility. This credibility is not just symbolic; it enhances the young company's bargaining power in future negotiations, be they for subsequent partnerships or funding rounds.

On the other side of the spectrum is the asset creation and out-licensing model. Here, biotechs use their proprietary technology to develop their pharmaceutical assets. This approach is resource-intensive upfront, requiring both time and financial investment. However, the payoff can be considerable. Once these assets reach a scientific proof-of-concept, they can be licensed to larger biopharma companies for further development and commercialization. This not only generates revenue through upfront fees and milestone payments but also significantly de-risks the later, often more expensive, stages of asset development. It is similar to planting a tree, nurturing it until it is sturdy enough, and then handing it over for others to grow it to its full potential. In a crowded market, developing your own pharmaceutical assets can be a game-changer. It provides a means to stand out and assert technological and innovative superiority, particularly if the developed asset addresses an unmet medical need or outperforms existing treatments. Partnerships in this model often revolve around complementing capabilities, as the biotech company can focus on early-stage development while the biopharma partner can use its greater resources and expertise for late-stage trials and commercialization.

A third model, and by far the most ambitious, is one of complete in-house product development and commercialization. This model is a complete in-house project in which the young firm inherits all the risks but also all of the rewards of successful development. Companies adopting this model intend to take pharmaceutical assets from conceptual stages right through to market. The financial risk here is much higher, as is the burden of development and regulatory compliance. But, if successful, the returns are unparalleled, as the company retains full ownership of the product. Naturally, this model requires the most significant influx of capital, either from investors or from a steady revenue stream generated from previous deals, other sources or a combination. The complexity of taking a product to market also means that most companies cannot go it alone. Strategic alliances, often with big pharma, are almost a necessity, providing the expertise in areas where the biotech may lack the necessary experience, such as late-stage clinical development, particularly phase III trials, manufacturing, or market access strategies.

As companies grow and evolve, many find themselves adopting a hybrid approach that combines elements of the above models. For example, a company might initially engage in technology partnerships to generate quick revenue and validate its platform while also investing in asset creation for the longer term. This dual approach maximizes both immediate revenue and the potential for higher returns down the line. As companies mature, they often find themselves in a position to transition into full product development and commercialization, taking greater control over their assets while reaping larger financial rewards. It is also very common for young biotech companies, even those that have gone public, to be acquired after a successful Phase II trial because of the enormous capital costs of conducting a successful Phase III trial.

To summarize, the choice of business model for a biotech company in the biopharma sector sets the stage for a myriad of other strategic decisions, from partnership formation to scaling. Technology partnering offers a quicker, less risky path to revenue, while asset creation and out-licensing promise higher returns but at greater initial costs and risks. Product development and commercialization offer the highest returns and the most control but require the most resources and carry the highest risk. Therefore, it's not uncommon for companies to employ a blend of these strategies as they grow, navigating their way through the challenging yet exciting landscape of biotech innovation. With regard to *PROTEUS*, we believe that it is too early to determine which commercialization strategy is optimal, but it is important to understand the landscape of options.

With regard to planning an exit strategy, executives in early-stage biotech companies are faced with many potential options. Successful early-stage biotechs are either acquired or go public via IPO, whereas failed commercial startups will generally go bankrupt. M&As and IPOs have alternated as the preferred exit route over the past several decades. Post-interest rate hikes in 2022, the more common route has definitely moved away from public markets and towards acquisitions.

Financial metrics obviously matter significantly when contemplating an exit. This includes not just revenue streams, profit/loss, and growth margins, but also venture capital funding and other forms of financial backing. However, a company's valuation at the time of exit is not solely influenced by its balance sheet. Whether a company has a single promising asset or a diversified portfolio can affect its attractiveness to potential acquirers or investors. The richness and maturity of the R&D portfolio also weigh heavily and are almost always the exclusive factor driving an early-stage biotech company's valuation.

It's equally crucial to consider market applicability and potential scalability of the biotech solutions in development. Companies targeting therapeutic areas with larger commercial markets, such as oncology, diabetes, or autoimmune diseases, often find themselves more appealing acquisition targets. Moreover, the presence of a lead candidate that has demonstrated proof-of-concept in (preferably) Phase II or Phase I clinical trials is often a strong indicator of not just safety, but also efficacy, making the company a more attractive prospect for acquisition.

Beyond these factors, the importance of strategic partnerships cannot be overstated. Collaborations with academic institutions can lend credibility and depth to R&D efforts, while partnerships with other biotech and pharma companies can provide the necessary capital and resources to bring a product to market more rapidly. These alliances often serve as a *de facto* validation of a company's potential, making it more appealing to potential acquirers.

Data indicates that the upfront deal values tend to rise significantly between Phase II and Phase III. However, the total deal value remains relatively consistent across these phases. Essentially, this means that if a company is willing to take on the additional risk of advancing its asset to a later clinical stage, it might command a higher upfront payment upon exit without significantly altering the total deal value. This underscores the importance of a company's tolerance for risk in shaping its exit strategy.

The ability to exit is also crucial to securing venture capital funding; VCs have a variety of investing styles, but many will prefer to maximize their internal rate of return (IRR), which favours shorter time horizons. Thus, depending on the specifics of VC funding, if an investor has control of the company board, it may not be up to the management whether or not an exit, in this case an acquisition, will occur.

In summary, navigating a successful exit in the volatile and high-stakes world of biotech requires a multi-faceted approach. Executives must balance financial metrics, R&D maturity, market potential, and strategic partnerships, all while carefully timing their exit to maximize both upfront and total deal values. It's a delicate act, necessitating a nuanced understanding of both the scientific and commercial landscapes, as well as a willingness to manage risk strategically.

8.3 Pricing Strategy and Cost Structure

In many markets, the price of drugs is regulated by the government on behalf of the citizens. The exception to this is the United States, where prices are almost entirely unregulated. This gives firms much more price-setting potential and is a main contributor to the US being the world's largest drug market. The US pharmaceutical industry sees substantial revenue growth, partially due to the lack of regulation around drug pricing. Companies can raise the prices of their drugs beyond inflation rates, even if the demand is not high. Most of a pharmaceutical company's revenue comes from repeatedly increasing the prices of existing drugs rather than new drug sales, which has led in part to US medical inflation rapidly outpacing the rest of the consumer price index. In this report, we focus on drug pricing in the United States because that is where a young biotech company has the most freedom to set prices and the jurisdiction where the firm will more than likely capture most of its revenue.

It is important to note that recent legislation– most notably the Inflation Reduction Act of 2022– aims to negotiate the prices of high-cost prescription drugs for Medicare beneficiaries. Starting in 2026, the act will target 10 high-cost drugs and expand to 20 by 2029. Companies refusing to negotiate could face a 95% tax penalty. The act also caps Medicare beneficiaries' out-of-pocket drug costs at \$2,000 starting in 2025 and monthly insulin costs at \$35.

Once *PROTEUS* is a commercial therapeutic, the following factors will need to be considered when setting its price.

1. The uniqueness of the therapeutic: If a firm is developing a therapeutic or drug for a condition that has limited treatment options, the company may be able to set a higher price when the drug eventually goes to market.
2. Competition: Understanding the landscape of existing treatments will be crucial. If the market is saturated, the young biotech may need to price its drug competitively. If the drug offers additional benefits over existing ones, it might justify a higher price. Pancreatic cancer has many drugs indicated for its treatment, however, patient outcomes remain suboptimal.
3. Effectiveness: The more effective the drug or therapeutic, the higher the potential price. If the biotech's therapeutic can prevent expensive surgeries or other medical treatments, or if it can significantly extend or improve quality of life, these factors could justify a higher price point. If *PROTEUS* can offer superb patient outcomes, then a higher price for the therapeutic is justified.
4. Research and Development Costs: R&D is often the most expensive part of therapeutic development, especially for a young biotech without revenue from other IP. These costs will need to be recouped through the therapeutic's pricing. Investors and stakeholders will be keenly interested in how these costs balance against projected revenue. The goal is to minimize the cost of research and development to be able to maximize return on investment.
5. Regulatory Environment: Being aware of regulations and regulatory hurdles like the Inflation Reduction Act of 2022 can help a young biotech plan for potential negotiations

with Medicare or other governmental bodies in the future. Additionally, a significant sum will be spent during the R&D phase on compliance with the FDA, which needs to be faceted into the pharmaceutical's price.

6. Revenue Goals: Like any business, the ultimate aim would be to maximize revenue. Understanding the dynamics of pricing — too high could limit prescriptions, too low could undervalue the drug — will be key to financial viability.
7. Customer Out-of-Pocket Costs: Depending on the patient population, consideration of what customers can afford may also be a factor, especially in light of potential caps on out-of-pocket expenses for certain categories of drugs and therapeutics.
8. Generic Competition: Once the patent on the therapeutic expires, generic versions could become much cheaper. Revenue models must take into account this eventual drop in pricing power.

Understanding these elements could be key for any young biotech company, aiding in attracting investors by providing a detailed roadmap of how they intend to price their drug or therapeutic for maximum profitability and market penetration.

Pricing a therapeutic incorrectly can have a significant impact on its market success. Pharmaceutical companies raise prices primarily to maximize revenue, and the lack of competition for patented therapeutics allows them to do so freely. Generic drugs are cheaper because they do not incur the original R&D costs. Insurance companies and pharmacies determine the final amount a customer pays for a therapy, though the US government is not involved in setting the initial prices.

9. Financial Projections

9.1 Revenue, cost, and profit projections from 2025 to 2035

We have decided to use a 10-year model for projecting our finances for *PROTEUS*. As a CRISPR-Cas cancer therapeutic, it is expected that developing a minimally viable product and proof of business may take over a decade given the regulatory impediments to commercializing. Our financial projections make assumptions about the year-by-year progress of our commercialization:

- Years 1-4 represent the initial R&D & development phase, resulting in operating losses.
- Years 5-8 involve increased expenses as clinical trials progress.
- Year 9 marks a turning point with regulatory approval and initial revenue.
- Year 10 sees significant revenue from commercialization and additional financing activities. Following Year 10, or after 2035, we expect to see major opportunities for exit, either through acquisition offers by large pharmaceutical companies, or through a management buyout given the scale of the company after 2035.

Year	Milestones
1	Technology Development
2	Preclinical Testing
3	Regulatory Affairs (IND prep)
4	Preclinical Data Collection
5	Phase I Clinical Trials
6	Phase I Results Analysis
7	Phase II Clinical Trials
8	Phase II Results Analysis
9	Phase III Clinical Trials
10	Regulatory Approval, Commercialization, Market Entry

Figure 28. Year-by-year milestones over our 10-year timeline for *PROTEUS*' approval.

Capital investments from VCs and angel investors are assumed to occur in the early years, primarily in Years 1-4. R&D expenses include costs associated with technology development, preclinical testing, and clinical trials. Legal expenses account for regulatory compliance, patent filings, and legal support. Salaries encompass employee compensation across various functions within the company. Manufacturing expenses include production, shipping, and administering costs for the CRISPR-Cas therapeutic. Interest income and interest expenses are minimal until Year 9, when revenue is generated, and financing activities increase.

The financial projections generated provide a detailed view of *PROTEUS*' financial performance, indicating the initial investment requirements, the transition to profitability, and the potential for substantial revenue as the therapeutic reaches the market. These projections are based on estimations and assumptions and should be periodically updated to reflect actual performance and changing market conditions.

To support the development timeline and financial projections, McGill iGEM plans to raise capital through multiple funding rounds, including seed, series A, series B, and series C rounds.

Detailed financial models for each round will be developed as the company progresses.

A breakdown of the salary figures in the projections is provided below:

- Salaries (HR): Covers human resources staff responsible for recruitment, employee benefits, and HR administration.
- Salaries (R&D): Encompasses research and development team members, including scientists and researchers.
- Salaries (Biz Dev): Accounts for business development personnel focused on partnerships and collaborations.
- Salaries (Legal): Represents the expenses associated with a small in-house legal team responsible for regulatory compliance and patent filings and continued protection.
- Salaries (Manufacturing): Includes staff involved in the manufacturing, shipping, and administration of the therapeutic.
- Salaries (Upper Management): Covers top executives and management personnel overseeing various divisions.

These financial projections provide a detailed overview of *PROTEUS*' financial performance, with a comprehensive breakdown of salaries across different divisions within the company.

On the following page is our Projected Financial Statements for *PROTEUS*, including an income statement estimate, balance sheet estimate, and cash flows statement estimate. As mentioned, these projections are based on estimations and assumptions and should be periodically updated to reflect actual performance and changing market conditions.

Projected 10-Year Income Statements 2025-2035 (in Millions CAD)

End of Calendar Year	Revenue	Cost of Goods Sold (COGS)	Gross Profit	R&D Expenses	Legal Expenses	Salaries (HR)	Salaries (R&D)	Salaries (Biz Dev)	Salaries (Legal)	Salaries (Manufacturing)	Salaries (Upper Management)	Other Operating Expenses	Operating Profit	Interest Income	Interest Expense	Net Profit
2025	0	0	0	0	5	2	1	2	1	0.5	1	2	4	-18	0	-18
2026	0	0	0	0	7	2	1.2	2.2	1.2	0.6	1.2	2.2	4.4	-22	0	-22
2027	0	0	0	0	8	3	1.4	2.5	1.4	0.7	1.4	2.5	4.8	-26	0	-26
2028	0	0	0	0	10	3	1.6	2.8	1.6	0.8	1.6	2.8	5.2	-30	0	-30
2029	0	0	0	0	15	4	2	3.5	2	1	2	3.5	7	-43	0	-43
2030	0	0	0	0	15	4	2.2	3.8	2.2	1.1	2.2	3.8	7.6	-47	0	-47
2031	0	0	0	0	20	5	2.5	4.5	1.3	1.3	2.5	4.5	8.5	-53	0	-53
2032	0	0	0	0	20	5	2.7	4.8	1.4	1.4	2.7	4.8	9.2	-58	0	-58
2033	50	20	30	30	30	6	3	5.5	1.5	3	3	5.5	10.5	14.5	2	15.5
2034	150	40	110	30	30	6	3.2	5.8	1.6	3.2	3.2	5.8	11.6	98.4	2	99.4

Projected 10-Year Balance Sheet 2025-35 (in Millions CAD)

End of Calendar Year	Assets	Liabilities	Equity
2025	0	0	4
2026	0	0	9
2027	0	0	-9
2028	0	0	14
2029	0	0	-14
2030	0	0	-20
2031	0	0	25
2032	0	0	-25
2033	0	0	31
2034	0	0	-31
2035	0	0	38
2036	0	0	-38
2037	0	0	44
2038	0	0	-44
2039	67	0	44
2040	217	0	44
2041	173	0	44

Projected 10-Year Cash Flows 2025-35 (in Millions CAD)

End of Calendar Year	Operating Activities	Investing Activities	Financing Activities	Net Cash Flow
2025	-18	0	0	-18
2026	-22	0	0	-22
2027	-26	0	0	-26
2028	-30	0	0	-30
2029	-43	0	0	-43
2030	-47	0	0	-47
2031	-53	0	0	-53
2032	-58	0	0	-58
2033	15.5	45	-10	50.5
2034	99.4	50	-150	-0.6

A graphical representation of the major data points is provided below; the graph includes figures for projected revenues, expenses, and resulting profits, specifically considering revenues from sale of goods. Equity financing is not factored into the analysis given its uncertainty in timing and scale of investment. This factor might positively impact both our financial statements and the appearance of the graphic below:

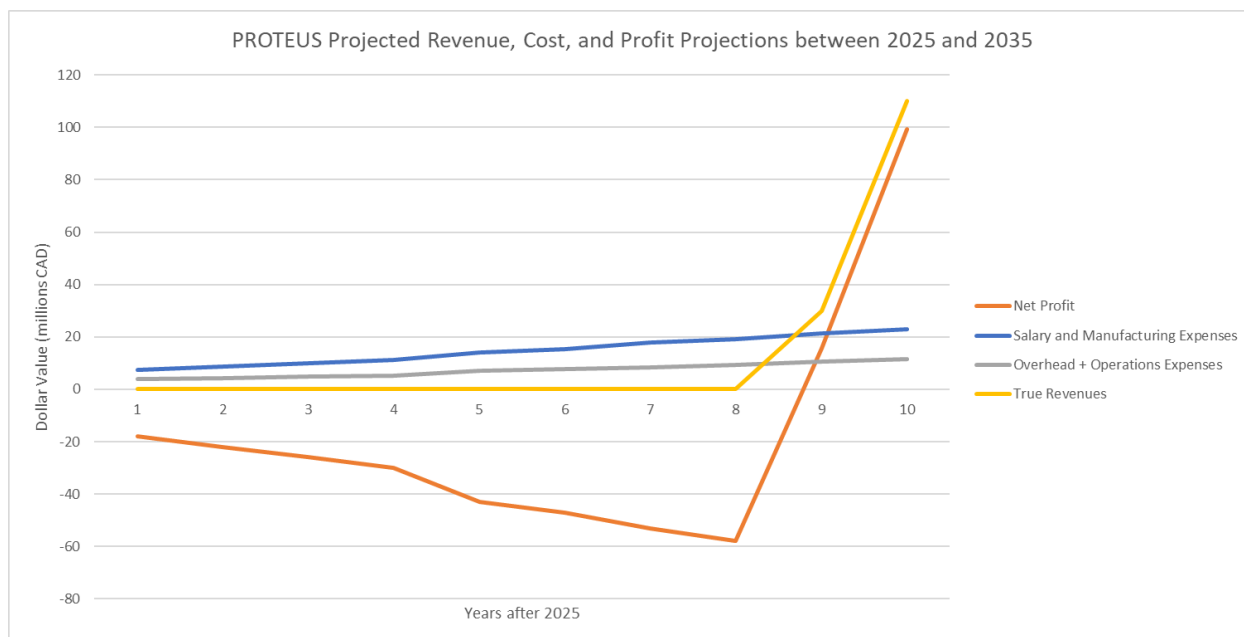


Figure 29. Projected revenues, expenses, and net profits between 2025 and 2035 in millions CAD, not considering financing options such as capital investment from VCs/angels.

9.2 Funding requirements and potential sources

The cost of drug development is a complex and expensive process that involves multiple stages, from initial discovery to clinical trials and eventual market approval. Companies may spend anywhere from tens to hundreds of millions of dollars during this process. These figures can vary widely depending on whether they include only out-of-pocket expenses, capitalized costs, or both. Studies have shown that the average cost of developing a drug can range from \$350 million to as high as \$5.5 billion, depending on the company and number of drugs developed (Herper, 2013). The costliest stage is usually the drug discovery phase, where the focus is on understanding the underlying causes of diseases and developing chemical compounds to treat them. The process can take 3-20 years and can cost billions of dollars. Following this, companies must obtain an IND application from the FDA before proceeding to pre-clinical and clinical trials. Clinical trials are mandated by the FDA, with a single phase potentially costing upwards of \$100 million. Adding to the financial burden is the high rate of failure. Only about 10% of drugs that enter pre-clinical trials make it to human testing, and many of these ultimately fail as well (Sun, 2022). Therefore, successful drugs have to cover the costs of those that fail, substantially inflating the total expenditure. Given the high costs, pharmaceutical companies use their research and development expenses as a basis for

setting drug prices. This profit-driven approach often results in high retail prices for new drugs. In this context, there are efforts to explore alternative models involving collaboration between universities, governments, and the industry to optimize resources. Overall, understanding the full scope of drug development costs is crucial for pharmaceutical companies in setting pricing strategies and projecting profits.

10. Conclusion

This report has comprehensively examined all major aspects of developing *PROTEUS* from its current state to the stage of market launch. In truth, it has provided a developed strategy for the entire life cycle of the product and the firm, from ideation to research development, to clinical trials, to business development, to exit strategy. Based on the research in this document, our team now possesses an advanced, easy-to-follow strategy to move forward with our product. One of the greatest foibles of biotechnological startups is their inability to transfer their technology to the real world; over 90% of startups fail to break even (Patel, 2015), and only a select few of the successful firms receive funding in the form of capital investment (Eve, 2020). In order to succeed where others – with equally innovative, applicable technologies – have failed, our team realized the importance of developing a top-notch business plan. We have discussed in great detail the path to success by developing: a highly efficient corporate organizational structure, as well as a skill gap analysis to determine what expertise we currently have and currently lack; a market research plan to determine our target market and manufacturing and partnerships plan; SWOT and PEST analyses to determine the current ‘playing field’; in-depth project description and project development plans; an IP strategy to protect our technologies and to avoid infringing on existing proprietary materials; business models to guide our future business development; and financial projections to give us a rough estimate of future cash flows, revenues, expenses, and market trends to look out for.

Based on this report, McGill iGEM 2022-2023 plans to move forward with *PROTEUS* in hopes of eventually realizing its use in the real world. From the start, our goal has been to help those in need; this was a driving factor in choosing pancreatic cancer, given its abysmally low survival rate compared to other diseases, and even to other cancers. We are excited for the opportunity to share our innovation with the world, and hopefully to have a real impact on the lives of millions of patients worldwide. Only through sound firm development will we achieve this goal given the cutthroat environment with biotechnology startups paired with that of oncological therapeutics. This report has laid the foundations for us to break into the space and make our mark for the betterment of humanity.

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